Who Killed the Green Fairy? Temperance, Language, and Wine in the Swiss Absinthe Ban of 1908

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Abstract

On July 5, 1908, Swiss voters wrote a ban on absinthe into their federal constitution, the first national prohibition of the drink enacted by popular vote. The temperance movement led the campaign. Deputies and historians also pointed to the wine industry, which stood to gain, but no one has measured its part. We measure it with each canton's share of the country's estimated wine revenue, across 25 cantons and at both stages of the initiative, the petition and the vote. At the petition, the Protestant share, our measure of organized temperance, predicts signatures per hundred eligible voters, and no relationship with the wine share is detectable. At the vote, language masks wine: most of the country's wine was grown in French-speaking cantons, and those cantons gave the ban less support. In the primary specification, which holds language fixed among other controls, a one-percentage-point higher wine share is associated with a 0.677-percentage-point higher yes share, significant at five percent under four small-sample procedures. On the other federal ballots we study, the wine share predicts support only once, in the 1906 vote on the federal food law, an act that protected wine. Because direct democracy separates setting the agenda from deciding it, the case measures both partners of the coalition that theories of interest-group regulation describe: temperance carried the petition, and the wine interest, which never took the podium, registered at the ballot box.

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1 Introduction

In 1910 Gantner's caricature for the Swiss satirical revue *Le Guguss* laid the green fairy out dead. A blue cross is driven into her chest, a black-robed figure wearing the emblem of the Blue Cross temperance society stands over her with a Bible, and the wreaths of Neuchâtel and Geneva, the only two cantons to vote against the ban, lie at her feet (Gantner, 1910, reproduced in Montagne, 2013, p. 509). The ban's opponents had named their killer.

The Swiss wrote a ban on absinthe into their federal constitution in 1908 (Swissvotes, 1908), the first national prohibition of the drink enacted by popular vote, in the middle of the first wave of absinthe bans, which ran from 1906 to 1915 (Royaume de Belgique, 1906; Arnold, 1989, p. 116; Lachenmeier et al., 2006, p. 367). The majority was itself a measure of the moment: 63.5 percent overall, and a local majority in all but two cantons (Swissvotes, 1908; Schweizerischer Bundesrat, 1908). Both chambers carried it against the Federal Council's own recommendation to reject (Schweizerischer Bundesrat, 1907b, p. 351).

Breadth of support, however, is not identity of authorship. That a majority said yes tells us little about which of the coalition's members—the temperance churches, the medical establishment, or the drink trades that competed with absinthe—moved votes, where, and by how much.

To contemporaries, little about the outcome required explanation. Absinthe was cheap, by the campaign's own reckoning (Studer, 2024, p. 126), and it carried a diagnosis of its own: absinthism, a pathology held to be distinct from ordinary drunkenness and attributed to the essence of wormwood.[1] Nor was the instrument novel: the Confederation had curbed the rivals of the "natural" drinks since the alcohol monopoly of 1885–87 (Swissvotes, 1885, 1887; Section 2.2), and the pro-ban press set the initiative in that line of standing federal restrictions (Journal de Genève, 1908, July 3, p. 1). The public-health premise does not bear the weight the conventional account places on it, and the era itself knew the case was contested (Section 2.3). Modern analysis finds nothing in pre-ban absinthe, at obtainable doses, that distinguishes it from chronic alcohol abuse (Lachenmeier et al., 2008, p. 3073; Padosch et al., 2006, p. 11).

A yes-or-no vote buries a fact both camps acknowledged: the stakes ran beyond absinthe to other drinks. The Swiss Alcohol Board's own expert predicted that other spirituous beverages "would take the place of the prohibited absinthe" (Milliet, 1907, p. 365). On the temperance side, the abstainers' paper *Freiheit*, organ of the Anti-Alcohol League, had

[1] By the campaign years, it was attributed more narrowly to its ketone thujone.

conceded in its No. 11 of 1905, before the federal initiative was launched, that absinthe was "by no means the most harmful drink for us Swiss" and that "there are worse things to fight," while urging its readers to fix on "what we can already achieve now" (Trechsel, 1990, p. 153).

A ban on one drink, which both camps called a small step, does not by itself explain majorities in all but two cantons. The support ran against the drink's own geography: absinthe was made and drunk mostly in French-speaking Switzerland, yet the ban won its widest margins in German-speaking cantons and its narrowest in French-speaking ones, while French-speaking cantons signed the petition at a somewhat higher rate, 23.9 against 20.4 signatures per hundred eligible voters (Figure 2, panels C and D). No existing account weighs the economic stakes against that cultural divide. Theories of interest-group regulation suggest where to look (Stigler, 1971; Peltzman, 1976): a moral movement and an economic interest can back the same rule for different reasons (Yandle, 1983).

We study the two canton-level records the initiative left—the petition that put the ban on the ballot and the vote itself—across the twenty-five cantons. We estimate a cross-section by least squares with the French-speaking share held fixed, and we check each estimate with four small-sample inference procedures, among them a Freedman–Lane permutation test. Signing was voluntary, so the petition, measured in signatures per hundred eligible voters, records organized mobilization; the vote records the choice of the electorate at large (Section 3.1.1). The petition was filed with 167,814 valid signatures (Schweizerischer Bundesrat, 1907b, p. 353), the largest signature count then recorded (Trechsel, 1990, p. 152). The cantons differ in wine production, confession, language, and exposure to absinthe manufacture.

Once language is held fixed among the other controls, a canton's stake in the national wine harvest predicts its support for the ban: a one-percentage-point higher share of national wine revenue goes with a 0.677-percentage-point higher yes-share (standard error 0.216), and the estimate holds under each of the four inference procedures. Holding language fixed is not a specification chosen for convenience: language overlaps both the wine geography and the absinthe geography, the raw cross-section reverses sign without it, and the cultural divide therefore masks the within-bloc relation rather than explaining it away. Prior work treats culture as a qualitative explanation for the ban's support. Here, we quantify it and the economic stakes alike.

The petition's results differ. Signatures per hundred eligible voters rise with the Protestant share, our measure of standing temperance organization (0.170, standard error 0.056), while the wine share is not detectable. That difference in what we can detect at the

two stages is the central refinement our cross-canton evidence brings to existing readings of the 1908 vote. The viewpoint that the vote was mostly a cultural, moral, or health-related decision is partly right, but it ignores how closely wine production overlaps the linguistic cleavage, and how closely both overlap with absinthe production.

We then ask what distinguishes this vote from other federal ballots of the period. Under the same specification the confidence interval of the wine coefficient includes zero on each of six ballots whose subjects had nothing to do with drink. Of the three earlier ballots that engaged wine, two show no detectable relationship; the third, the food act of 1905, confirmed by referendum in June 1906, gives a positive coefficient just under the ten percent line on the shared specification (permutation p =.083). The Act's own legislative record points at the wine interest directly: the wine merchants' associations had advocated for an outright ban on artificial wine (Schweizerische Bundesversammlung, 1905a, pp. 393, 1050–1051), as Section 5.3 sets out.

A purely cultural account leaves one feature of the record unexplained. The absinthe-producing cantons gave the ban less support in the raw returns, and the only two cantons to reject it, Neuchâtel and Geneva, were producing cantons, as Heimberg (2000, p. 105, n. 12) had already noted, and indeed the country's largest producers (Swissvotes, 1908; Milliet, 1907, pp. 360–361). The Council's December 1907 message carries an annex by E. W. Milliet, director of the Swiss Alcohol Board, listing the forty houses that bought monopoly spirit, with each one's purchases and exports (Milliet, 1907, pp. 360–361); we check that roster house by house (Section 3.1.4; Appendix A.3). With five producer cantons, four of them French-speaking, the producer indicator is indistinguishable from zero, and we do not read that as the absence of resistance. Those raw returns are not proof, only the reason a third factor had to be measured rather than asserted.

The role of the wine interest has been asserted in passing, doubted in passing, and left open by careful authors for want of evidence; to our knowledge no one has estimated it. The existing literature offers three readings of the 1908 vote. The first treats wine-industry interests as a material contributing factor. Cahannes (1981, pp. 44–45), in the closest existing analysis of the Swiss case, concludes that the initiative passed partly because absinthe competed with white wine and the winegrowers therefore supported it (Section 2.4). Padosch et al. (2006, p. 4) name the winegrowers among those calling for the ban in France, and Prestwich (1979, p. 319) characterizes the banned drink as "less a villain than a convenient victim" of a coalition that combined hygienic concern with material interest. The second reading treats the temperance movement and the cultural cleavage as effectively sufficient: much of the medical and cultural-history literature takes the Lanfray murders of

1905 (Studer, 2024, p. 178) and the Protestant revival mobilization (Trechsel, 1990, pp. 29–30) as the sufficient explanation. The Federal Council, by contrast, dismissed the petition as "a hurried, little-thought-through legislative attempt by the bearers of the temperance movement" (Schweizerischer Bundesrat, 1907b, p. 351). The third treats the two as jointly necessary, with the relative weight of each unsettled in the absence of quantitative evidence. Cahannes (1981, p. 44) is a careful proponent of it, qualifying her own wine attribution with the observation that "the motives underlying this initiative were without doubt a concern not only for public order but also for the physical and mental health of the population," and Studer (2024, p. 18) treats the campaign as similarly multi-causal. Our analysis builds on the third reading.

Two modern-era treatments read the geography of the result from the raw returns. Heimberg (2000, p. 99) holds that the prohibition "did not really touch major economic interests" and was decided by German-speaking regions little affected by absinthe consumption, and Auderset (2021, pp. 152–153) reads the returns the same way. Both readings are fair on the raw returns and neither was offered as a test. Studer's history names the pressure of the French wine lobby among the causes of the 1915 ban but treats the Swiss vote narratively, with no canton-level analysis (Studer, 2024, p. 18, ch. 7). Elsewhere the wine calculation appears as authorial supposition, the growers having "deduced that absinthe could easily be sacrificed" (Adams, 2004, p. 197).

Popular votes on prohibition elsewhere in the same era give the nearest comparisons. In Canada's plebiscite of 1898, Dostie and Dupré (2012, pp. 511–512) find no effect of beer and spirit production. In the American statewide votes of 1907–1919 that Lewis (2008, pp. 389, 393–394) studies—on laws that, as he describes them, prohibited wine along with the other drinks, all but four sparing sacramental wine—brewing predicts the dry share and grape-growing does not. Ours was a vote on one spirit that left wine on sale. To our knowledge this is the first quantitative analysis of the 1908 Swiss absinthe vote, and it joins a short list: outside the United States, Dostie and Dupré (2016, pp. 493–494) could find only one earlier empirical study of a prohibition referendum, their own on the Canadian vote, and added a second on New Zealand's votes of 1911–1922.

We confine the analysis to what the evidence can answer, and we allege no explicit collusion between wine interests and the temperance movement. Where every written account is an interested account, testimony cannot arbitrate among motives: the government that recommended rejection also sold absinthe's makers their spirit, through its own alcohol monopoly (Sections 2.2 and 2.5), and the opposition charged, on one faction's stated ambitions, that the ban was a first step toward total prohibition (Gantner, 1908).

It's superficial to argue merely that markets play a role when historical accounts witness that the prices of spirits and wine are considerations. The tractable questions are how market interests were involved, and by extension, whether their involvement is legible in the cross-canton pattern, granting from the outset that the temperance movement led the campaign and that the cultural cleavage is real. We reject two statements, each of which ignores the other: that cultural attitudes and popular consensus drove support for the ban and markets played no important part, and that markets drove it and culture played none. A movement can lead a coalition whose success serves an interest that never took the podium. Writing the temperance movement's own history, Trechsel (1990, p. 153) records the order of events: the abstainers "climbed aboard as a free rider on a movement that started from a winegrowing village." What we measure is the political weight of a producer interest at the ballot. We make no claim that wine sales rose once absinthe was gone.

The political economy of regulation supplies the mechanism such a reading needs. Regulation tends to reward organized, concentrated interests at the expense of diffuse ones (Stigler, 1971; Peltzman, 1976). On that logic, a prohibition that reaches one drink and spares the rest is a prize for the rest. Room (1984, fol. 3) puts the mechanism plainly: drawing a line between a bad drink and a residual good one "splits the industry interests and potentially avoids a head-on attack on the most powerful." In a public-health vote the best-known form of that mechanism is the coalition that Yandle (1983, p. 13) called bootleggers and Baptists: a moral movement and an economic interest back the same rule for different reasons, both gain from it, and regulators accept it because it is easy to administer. Such coalitions succeed best, he argues, where a broad public concern, alcohol among his examples, lets a rule move resources from one group to another (Yandle, 1983, p. 14). We read the Swiss case as that coalition in its looser form: the two sides shared a goal, and we find no evidence that the economic interest financed the campaign (Section 6). For France, Prestwich (1979, p. 309) documents a wine industry long resentful of absinthe as a newer competitor, which had used political influence to secure wine's tax-favored standing as a hygienic drink (Prestwich, 1997, p. 1258; Studer, 2024, p. 5), while cautioning that it was a beneficiary rather than the campaign's prime mover. Switzerland has a precedent of its own: when the Confederation rebuilt its alcohol taxes in 1885–87, the Federal Council meant the abolition of the cantonal entry duties to cheapen wine, cider and beer (Schweizerischer Bundesrat, 1884, p. 473), as Section 2.2 sets out. The food laws of 1897 to 1906 extended that preference to wine against its imitations, and the absinthe article of 1908 removed one more rival of the fermented drinks. To a wine canton the protection was attractive: the Valais government itself conceded the price disadvantage of

its own wine (Schweizerischer Bundesrat, 1907b, p. 345).

Section 2 supplies the half-century of record behind the vote, since the differential treatment of the drink classes is older than the initiative. Sections 3–5 construct the measures and test them.

2 Background

Two questions run through this section: why wine would be involved in a health vote about one spirit, and what distinguishes that vote from the period's other legislation aimed at wine's competitors. Wine had faced cheap substitutes since the vine blights of the 1860s through 1880s (Nye, 2007, p. 58), two decades before the ban (Section 2.1). The state was no bystander: in 1885–87 it rebuilt the taxation of drink around alcohol content and gave the fermented drinks a constitutional shield (Section 2.2), and the food laws of 1897 to 1906 extended that protection to wine against its imitations (Section 5). The case against absinthe was medical and disputed among physicians at the time (Section 2.3). What set the ban apart was temperance backing on a scale no earlier measure had, alongside open talk of the vineyard in the chambers (Section 2.4), and an article that banned the distilled liqueur while sparing vermouth, the wine flavored with the same herb (Section 2.5).

2.1 Wine and its substitutes

Two blights struck Europe's vineyards in succession. Oidium, a powdery mildew, came first; phylloxera, an insect of the aphid family indigenous to America, followed it (Nye, 2007, p. 58). Phylloxera first appeared in southern France in 1863, and by 1890 it had destroyed 40 percent of French vineyards; output recovered only as growers replanted on resistant American roots (Banerjee et al., 2010, pp. 714–717). It reached Switzerland in 1874 (Furrer, 1889, s.v. Reblaus, p. 646). We do not try to settle how far Swiss vineyards were themselves hit; the cantonal area record for these decades is coarse, and nothing in what follows turns on it.

Swiss wine production was meanwhile trending downward. The Federal Council's November 1884 message reckoned that Swiss wine yields since 1876 had stood on average a quarter to a third below the mean and had barely reached a million hectoliters (Schweizerischer Bundesrat, 1884, p. 381). The Council drew the implications for the Swiss drinker:

> "If one further considers that in France, the principal wine country, wine

> production has fallen considerably as a consequence of the devastation by the phylloxera; that in consequence the price of wine has risen while its quality has fallen; and moreover that our purchasing power has also diminished through the growing industrial crisis, is it then still a wonder that our population, accustomed in the good years to a heightened enjoyment of life, has gone over to other drinks?" (Schweizerischer Bundesrat, 1884, pp. 381–382)

The opening that cheap spirits filled is the federal executive's own reading, not our reconstruction.

This is a textbook setting for substitution effects: the price of wine rises, and demand shifts toward the substitutes, cheap spirits among them. Wine and beer, the Council wrote, "as a daily fortification, have been made almost inaccessible to the worker, while a schnapps injurious to health is everywhere to be had at derisory prices," and it called this the "economic, sanitary and moral fruits" of the small-distillery industry, which we take up in Section 2.2 (Schweizerischer Bundesrat, 1884, p. 456; Appendix A.1).

The French record puts numbers to the price path (Figure 1: Swiss producer prices peaked in the early 1870s, stayed high into the mid-1880s, then fell by more than half in the late 1880s). The wholesale price of industrial alcohol in France fell from 166 francs per hectoliter in 1854 to 28 francs in 1901, "despite increases in taxes on alcoholic consumption" (Marrus, 1974, pp. 128–129; Appendix A.1), and once phylloxera had deprived French makers of the wine alcohol absinthe had been built on, the drink could be bought "for ten centimes, less than a glass of wine" (Prestwich, 1979, p. 302).

A naive free-market story omits a critical detail: none of this played out on neutral ground. The federal government had been legislating on exactly this margin for two decades before the absinthe vote, and each time it wrote the law, wine, beer, and cider came out on the protected side.

2.2 The federal alcohol monopoly

The Council's 1884 message stated the public-health justification. "The chief evil is the quality of our schnapps," it wrote; the task was to do away with that drink and put less harmful ones in its place (Schweizerischer Bundesrat, 1884, p. 455). Its general doctrine said the same: since alcoholic drinks could not be condemned outright, the more harmful ones should be displaced by the less dangerous, or their effects reduced by legislation (Schweizerischer Bundesrat, 1884, p. 401).

The tax system the reformers inherited worked against that substitution. The cantonal

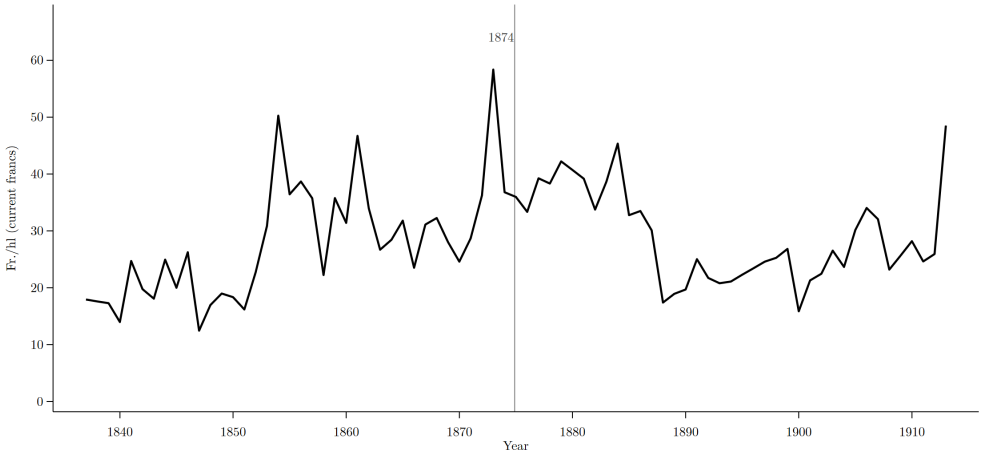


Figure 1: Swiss wine prices, 1837–1913, shown for context. Nominal producer prices in current francs per hectoliter: what growers received, not what drinkers paid. The vertical line marks November 23, 1874, when phylloxera was found in the vine greenhouses at Pregny, near Geneva, in the year the insect first appeared in Switzerland. Source: HSSO Table H.2a, column X (Historische Statistik der Schweiz Online (HSSO), 2012e). Every value in that table is an estimate after Ritzmann rather than an observed price.

taxes were levied on the volume of drink (Dubler, 2003), and a charge on volume falls more lightly on strong drink per unit of alcohol unless the tariff offsets it. Before 1887, no Swiss tariff taxed drink by alcohol content, and the cantons' tax structures differed from one another. Brandy, which on that principle would have paid some six times the duty on ordinary wine, paid on average only about twice as much, the Council reckoned (Schweizerischer Bundesrat, 1884, p. 443).[2] Industrial alcohol production was climbing at the time (Marrus, 1974, p. 128). While price generally rises with alcohol content, the tax ran the other direction, and demand shifted toward cheap, high-proof schnapps. The Federal Council attempted to reorient this pricing structure around alcohol content.

The correction came through Article 32bis, adopted on October 25, 1885, and the law of December 23, 1886, confirmed at the polls on May 15, 1887, whose monopoly articles took effect that July (Schweizerischer Bundesrat, 1886, p. 421; Swissvotes, 1887; Schweizerischer Bundesrat, 1887, p. 675). The Council had said what the law was for: dearer and purer spirits for consumption, an end to the abuses of distilling and sale, revenue enough to relieve the drinks that were not distilled, and the fullest regard for the agricultural and industrial interests bound up with distilling (Schweizerischer Bundesrat, 1886, pp. 423–424; Appendix A.1). Cheap fermented drink was an objective in its own right. The planned abolition of the cantonal entry duties was meant "quite specifically" to cheapen wine, cider and beer, so that in a country rich in wine and fruit "a part of the population should not be dependent on schnapps," and "in good fruit years," the Council promised, "cider is to

[2]Nor was the problem peculiar to Switzerland (see Nye, 2007, pp. 34–36).

be had throughout the whole country at cheap prices" (Schweizerischer Bundesrat, 1884, p. 473). Revenue entered the design as a means and a requirement rather than a measure, and at least a tenth of what reached the cantons was reserved for combating alcoholism (Schweizerischer Bundesrat, 1886, pp. 423–424; Schweizerische Bundesversammlung, 1885, pp. 478–479).

The settlement was an agricultural bargain. A farmers' petition of March 1886 welcomed a dearer spirit "provided that not merely fiscal but also sanitary and moral purposes are pursued thereby," and provided that the tax did not make the agricultural distillery impossible (Schweizerischer Bundesrat, 1886, p. 456). The same settlement relieved wine of the entry duties, left the distilling of wine and fruit outside federal law, on the Council's reasoning that such spirits were too dear to be much abused (Schweizerischer Bundesrat, 1886, pp. 422, 440), and barred special cantonal taxes on fermented drinks (Schweizerischer Bundesrat, 1886, p. 422; Appendix A.1); these passages establish what wine gained within a broader agricultural bargain, not who secured each provision.

France made the same public-health claim for its drink taxes as Switzerland did, by different means. Both cast hard liquor as the danger and favored fermented drink over cheap spirits, and both later protected wine against its imitations (Section 5). They differed in how the rules worked and in which competitor they hit hardest: France's came to turn on cheap and artificial wine, Switzerland's on cheap spirits. France's law of 1897 capped town taxes on wine and the other "hygienic" drinks and raised the tax on spirits, and Franck et al. (2014, pp. 78, 81) credit it to "progressives" who wanted "to improve citizens' health."[3] Switzerland's settlement of 1885–87 had already relieved wine of the cantonal entry duties and put spirits other than wine and fruit brandies under a federal monopoly. The wine interests differed too. France's spanned regions: after 1901 the small, traditional producers of table wine in regions like Burgundy and Gironde and, "more importantly," those of the South felt threatened by cheap wine (Franck et al., 2014, p. 87). Switzerland's was more concentrated: Vaud and Valais alone held about half of national wine revenue in 1907 (Appendix Table A.3). The 1908 debates tie the wine interest to the four French-speaking cantons whose governments backed the initiative (Section 2.4), and Fazy, for the ban, said there was only one raisin-wine factory in the country, "it seems" (Schweizerische Bundesversammlung, 1908, National Council, April 7, 1908, p. 96).[4] The Swiss government had itself conceded that the public-health claim was unproven, finding it

[3]The law took effect in January 1901 and capped the town taxes on wine, cider, mead and beer, those on wine by size of town; its implementation was settled under a minister who relied for much of his support on the wine producers of the South (Franck et al., 2014, p. 81).

[4]We do not test this difference: our wine measure is built from the yearbook's harvest totals, which combine white, red and mixed wine.

"impossible to affirm with any certainty" that fermented drinks had appreciably restricted the consumption of spirits.[5] Whether French policy shaped Switzerland's later measures or both answered the same pressures, the French record is context for the Swiss case, not evidence of Swiss motives.

With the law of December 1886, Parliament enacted a monopoly and created a federal body to run it, the Swiss Alcohol Board (*Eidgenössische Alkoholverwaltung*), established in 1887 under the Alcohol Act (Federal Office for Customs and Border Security, n.d.). The Confederation took the exclusive right to make and import spirit from every material federal law did not exempt—in practice, the Council said, chiefly potatoes and grain—and bought domestic spirit under contracts tendered in lots of at least 150 hectoliters of absolute alcohol (Schweizerische Bundesversammlung, 1886, pp. 1309–1310; Schweizerischer Bundesrat, 1886, p. 479).

Absinthe entered this settlement as a customer rather than a target. The 1886 message singled out the export trade in liqueur and absinthe, an industry working predominantly from foreign fine spirit (Schweizerischer Bundesrat, 1886, p. 468; Appendix A.1). After 1887 that input came through the monopoly, and the Board later wrote that absinthe liqueur was made wholly from the drinking spirit the Confederation supplied (Milliet, 1907, p. 358). In 1897 the monopoly's domestic sales of spirit per head came to 35.2 kilograms in the Val-de-Travers, against 2.2 for Switzerland as a whole, figures the Board said described its traffic, not a statistic of consumption, though a rough guide to where drinking was concentrated (Schweizerischer Bundesrat, 1898a, p. 89). Its export accounts carried absinthe on a line of its own from the monopoly's first year (Schweizerischer Bundesrat, 1900, p. 1013; Schweizerischer Bundesrat, 1889a, p. 174). The Board did not claim to see all of the trade: its register of makers omitted those working by the cold method, and purchases through intermediaries went unrecorded (Milliet, 1907, pp. 359, 361). That record made absinthe a feasible object of regulation; by itself it does not explain the ban. Subsequent sections study the explanatory gap.

Outside the contracts, the monopoly had little ability to monitor production or detect evasion. Its control service was built around the contract distilleries: the decree of 1890 named ten controllers and drew their nine districts as lists of those distilleries (Schweizerischer Bundesrat, 1890b, pp. 161–164). Beyond them the Board relied on reports. Officials, gendarmes and police were bound to report every offense they found (Schweizerischer Bundesrat, 1888, p. 877), and each collected fine was split in thirds among

[5] The Federal Council's message on the revision of Article 32bis that the voters rejected in 1903 (#63), read to the National Council by Federal Councilor Ruchet in 1908 (Schweizerische Bundesversammlung, 1908, National Council, April 6, 1908, p. 85).

the informer, the canton and the commune where the offense took place (Schweizerische Bundesversammlung, 1886, p. 1313). Few reports came. Over 1887–98 the Board issued about twenty-two penal orders a year, informers were paid Fr. 3,682.35 in 1890 but Fr. 73.40 in 1898, and by 1896 the complaints came, with few exceptions, from the Board's own employees (Schweizerischer Bundesrat, 1899a, pp. 301–302, 460–462; Nationalrätliche Alkoholkommission, 1897, p. 1018; Appendix A.1). Nor could its accounts show whether harmful drinking as a whole had fallen: parliament's committee credited the surplus of 1897 in part to monopoly spirit that poor fruit and wine harvests had pushed drinkers toward (Nationalrätliche Alkoholkommission, 1898, pp. 191, 193; Appendix A.1). The free distilleries that worked fruit and wine outside federal law were never counted, so the Board could not measure how much spirit escaped it. Absinthe's principal factory trade stood on the unusually visible side of that line: concentrated, identifiable and recorded in its purchases and exports, it offered a feasible target on which responsibility for a wider drinking problem could later be placed.

2.3 The public case: science and temperance

The stated case for singling out absinthe was medical, and older than the campaign. Valentin Magnan built his account of absinthism on a single case he treated at Bicêtre, published in 1864 (Luauté, 2007, pp. 518, 523). By the 1890s, Prestwich writes, his work on absinthism "was considered classic" (Prestwich, 1979, p. 305). The federal government's experts endorsed the diagnosis: as Auderset summarizes their report, absinthe's "poison properties" produce a clearly distinct form of alcoholism (Gaule et al., 1907, pp. 5, 11–13, as reported by Auderset, 2021, pp. 149–150, n. 57). The *Journal de Genève* reported Dr Pallard, at a meeting in Vernier, invoking "the unanimous opinion of the physicians and of the experts appointed by the Federal Council" (Journal de Genève, 1908, July 3, p. 4).

Contemporary scientists *did* disagree. Most asylum doctors stood behind Magnan, but dissent came from established clinicians, from experimenters and from a government inquiry (Luauté, 2007, p. 525).[6] According to Luauté, in 1907 the French government ordered an inquiry into Pontarlier, a heavy absinthe-drinking district; its prosecutor, Couleru, found the disorders blamed on absinthe less marked there than where other alcohol was drunk heavily, and crime halved between 1876 and 1906 while absinthe consumption rose (Luauté, 2007, pp. 526–527). Though these accounts come from France, a Swiss rapporteur turned

[6]Professor Joffroy, 1900: "the proportion of epileptics is minuscule compared to the colossal number of absinthe drinkers" (Luauté, 2007, pp. 524, 530). Lalou's 1903 experiments, reported in the 1905 *Traité de l'alcoolisme*: commercial absinthe owed its acute effects "almost exclusively" to the alcohol (Luauté, 2007, p. 524).

the same comparison on wine, while granting absinthe's dangers. In the National Council in 1908, the physician Müller (Bern), who spoke for the commission majority against the ban, held that all alcoholic drinks do harm in excess, "the Vaud wine just as much as the marc brandies." He had seen almost more delirium tremens from that wine than from the federal schnapps (Schweizerische Bundesversammlung, 1908, National Council, April 2, 1908, pp. 27, 30).

As Padosch and colleagues review the laboratory record, the seizures in the period's animal experiments were produced with pure wormwood essence rather than the drink, and *The Lancet* questioned as early as 1869 whether such findings carried over to drinkers (Padosch et al., 2006, p. 7). Modern chemical analysis of thirteen vintage pre-ban absinthes found nothing besides ethanol that could explain the syndrome (Lachenmeier et al., 2008, p. 3073), and a review of the literature concludes that "the so-called absinthism cannot exactly be distinguished from chronic alcoholism" (Padosch et al., 2006, p. 11).

Swiss temperance began with societies that "required from their members total abstinence from distilled beverages only, and moderation in the use of wine" (Cahannes, 1981, p. 42). Trechsel writes that the Blue Cross's campaign against absinthe was "internally almost uncontested, probably because it was directed against distilled drinks" (Trechsel, 1990, p. 46). Temperance societies and wine producers drew on the same hierarchy of drinks from different sides, which made their interests appear aligned without requiring any coordination. We treat that hierarchy as context, and our design does not test it.

2.4 The campaign and the recorded interests

On August 28, 1905, at Commugny, in canton Vaud, Jean Lanfray killed his wife and two daughters, and the killings, which public opinion blamed on his drinking of absinthe, became the catalyst for a cantonal petition against its sale (Studer, 2024, pp. 178, 181). Blaming absinthe rather than alcoholism was dubious. A newspaper report of the defense psychiatrist's evidence had him drinking up to four liters of wine a day, besides many absinthes and marc (Studer, 2024, p. 180). Federal Councilor Ruchet, speaking against the initiative, later called the crime "an honest pretext" (Schweizerische Bundesversammlung, 1908, National Council, April 6, 1908, p. 83). Vaud banned the retail sale of absinthe in May 1906 and Geneva in February 1907, and cantonal votes confirmed both laws (Schweizerischer Bundesrat, 1907b, pp. 344, 346; Heimberg, 2000, p. 98; Trechsel, 1990, p. 151). The café-keepers forced the Vaud vote by referendum (Heimberg, 2000, p. 98). Secretan, for the ban, told the National Council that the café-keepers, who said they earned little on absinthe, had at first "seemed to want to stay neutral," until the manufacturers "summoned them to

make common cause with them" (Schweizerische Bundesversammlung, 1908, National Council, April 3, 1908, p. 73). Trechsel records that the abstainers, who in principle stood with the ban movement, remarked "not uncritically" that the Vaud initiative had found large majorities above all in the winegrowing regions (Trechsel, 1990, p. 151).

The proposal for a national ban came from the Neuchâtel and Jura branches of the *Ligue patriotique contre l'alcoolisme*, a moderation league founded by the Blue Cross. After a congress at Bern on December 17, 1905, an initiative committee formed, and Robert Hercod's abstinence secretariat took over "the coordination of the signature collection, as later also of the voting campaign" (Trechsel, 1990, pp. 33, 151–152). Hercod later judged: "The abstainers, to be sure, did the real work, but the non-abstainers at least helped us morally, immensely" (Trechsel, 1990, p. 153). The ban's opponents charged that it was a first step toward total prohibition: an anti-ban poster for the vote printed, as the words of a "Dr Herco of Lausanne" from a report it cited by page, that the ban would have "only a quite weak effect on alcoholism in general," that the aim for a more distant future was "the abolition of all alcoholic drinks, wine and beer included," and that "it is a beginning" (Gantner, 1908).

The 1908 deliberations record the wine stake in the participants' own words. Müller observed that only the governments of Vaud, Geneva, Fribourg and Valais had declared for the initiative, "although competition with wine seems to play a certain role in this" (Schweizerische Bundesversammlung, 1908, National Council, April 2, 1908, p. 27). Martin, the majority's French rapporteur and deputy for the absinthe-making Val-de-Travers, found it "rather curious" that absinthe drinking "must have developed above all" in the cantons that produce white wine (Schweizerische Bundesversammlung, 1908, National Council, April 2, 1908, pp. 30, 34). Both remarks come from the ban's opponents. Müller's remark attributes to the ban's side a competition with wine, which is motive whether it was realized or only *perceived*. Brüstlein, for the ban, told the National Council that "in the canton of Vaud the movement is fortunately helped by the fact that the wine-growers see in absinthe a dangerous competitor and would gladly be rid of it, also from a selfish standpoint" (Schweizerische Bundesversammlung, 1908, National Council, April 3, 1908, p. 69). Writing in 1981, Cahannes concedes that the initiative's "passage was also due to the fact that absinthe, particularly popular in the French part of the country, competed with white wine, and the initiative was therefore supported by the winegrowers" (Cahannes, 1981, pp. 44–45). Gobat, for the ban, charged that the Federal Council "has declared itself ready to protect wine, vineyard and winegrowers against raisin wine, an innocent beverage, but it refuses to protect them against a poison which likewise competes with wine" (Schweizerische Bundesversammlung, 1908, National Council, April 6, 1908, p. 81).

Section 5 documents the food-law record. Federal Councilor Ruchet, defending the Council, restated the 1885 program: relief "... of the wholesome drinks, wine, cider, beer" and the "raising of the price of alcohol" (Schweizerische Bundesversammlung, 1908, National Council, April 6, 1908, p. 84). Fazy, already quoted on raisin wine, named the mechanism in the same speech: "when it is a matter of protecting wine, of protecting anything whatever, everyone agrees and nobody invokes freedom of industry." He then declined it: "we do not have as our pure and simple mandate the suppression of competition with wines or with this or that product" (Schweizerische Bundesversammlung, 1908, National Council, April 7, 1908, pp. 96–97). The deputies recognized a wine interest and disputed it, but the debate cannot measure its weight at the ballot, which Sections 3–5 estimate.

2.5 What was enacted

Whereas the cantonal laws had banned retail sale, Article 32ter prohibited the "manufacture, import, transport, sale and storage for the purpose of sale" of absinthe and its imitations (Schweizerischer Bundesrat, 1907b, pp. 353–354). Auderset notes that Geneva, which already had a cantonal ban, rejected the federal initiative (Auderset, 2021, p. 152). The article left "all other absinthe-containing beverages which constitute a public danger" to later federal legislation (Schweizerischer Bundesrat, 1907b, pp. 353–354). The Board counted "Chartreuse and above all so-called vermouth wine" among them, and left vermouth out of its reckoning of the ban's effects as "only optionally foreseen" and too little measured (Milliet, 1907, pp. 358, 364 n. 1). The Neuchâtel government had put the same question from the makers' side: most Val-de-Travers houses also made liqueurs flavored with fennel, anise, hyssop or lemon balm and no wormwood, and "there exist other liqueurs, such as Chartreuse, which contain absinthe; will these also fall under the prohibition?" (Schweizerischer Bundesrat, 1907b, pp. 347–348). Among absinthe-containing drinks, the ban fell only on the distilled side of the period's own line between "wholesome" fermented and distilled drink, with the notable asymmetry regarding Chartreuse and vermouth: a distilled liqueur was deferred alongside a wine (Schweizerische Bundesversammlung, 1908, National Council, April 6, 1908, p. 84; Milliet, 1907, p. 358; Schweizerischer Bundesrat, 1907b, p. 348). The government that recommended rejection had a fiscal stake, which we do not claim drove its recommendation: its Alcohol Board put the lost monopoly profit at up to about 873,000 francs a year if no other monopoly drink took absinthe's place. The Board said the loss would fall especially on the cantonal treasuries, and thought it would not arise in full (Milliet, 1907, pp. 364–366).

3 Data

The analysis uses a canton-level cross-section of the $N = 25$ cantons that existed under 1908 boundaries.[7] The outcome is the canton-level share of votes in favor of federal vote #68, the absinthe-ban constitutional amendment of July 5, 1908. Four groups of right-hand-side variables enter: measures of wine-industry presence, which carry the wine-market interest; the absinthe industry's own purchases of monopoly spirit, taken from the Milliet 1907 firm-level annex and aggregated to canton shares of the national total; cultural and political controls for language and religion; and population density.

Each arm, the vote and the petition, is anchored to the event it describes. The vote arm uses the 1907 harvest and canton population, and the petition arm those of 1906. The ballot fell in mid-year, so the vote arm takes the last complete harvest and the prior year's population (Appendix A.2). Language and religion, and the land area behind density, come from the 1900 census, for which there are no intercensal estimates; the federal census cycle is decennial (1900, 1910).

3.1 Data sources

3.1.1 Vote and petition outcomes

The outcome variables fall into three groups: the canton yes-share on the absinthe ban itself, vote #68, which carries the primary results; the canton's signature rate on the 1906 petition for the ban, per hundred eligible voters; and the yes-share on other federal votes of the preceding decade, which serve as falsification and corroborative comparisons.

The vote-share dataset draws on publicly available primary sources: the yes-shares for every referendum come from Swissvotes (Année Politique Suisse, University of Bern), the canton-level archive of every Swiss federal vote since 1848 (Swissvotes, 2026). We cross-validate the canton-level outcome totals against the Federal Chancellery's official archive (Schweizerische Bundeskanzlei, 1908a,b); Appendix A.4 gives the official record's page.

Five cantons required their citizens to vote in this period (Appendix A.4). Turnout there measures compulsion as much as interest, so it cannot stand in for mobilization. The petition stage gives the cleaner contrast: the same electorate faced the same question seventeen months earlier, and signing was voluntary in every canton.

[7] Bern absorbs the future Jura, which separated from Bern only in 1979 and is therefore not a distinct unit during our period.

Swiss federal votes in this period take three legal forms. A constitutional amendment, whether drafted by the Federal Assembly (a mandatory referendum) or by citizens (a popular initiative), needs a double majority, of voters nationally and of cantons, because the constitution is the compact among the cantons; an ordinary law put to an optional referendum needs only the popular majority (Appendix A.4).

The absinthe ban was a popular initiative, so its outcome was determined canton by canton as well as nationally, which is why we analyze it at the canton level and why the votes we use as falsifications are drawn from the double-majority forms.

The dependent variable in our primary specifications is the canton-level yes-share on vote #68,

$$Y_{i,68} \;=\; \texttt{pct_yes_68}_i \;=\; \frac{\text{yes votes}_i}{\text{valid votes}_i} \times 100,$$

for $i = 1, \ldots, 25$ cantons, drawn from Swissvotes (anr = 68; Swissvotes 1908).

The cantons enter unweighted. Under the double majority each casts a single vote, the six half-cantons a half each, but reproducing that arithmetic in the regression would distort the interpretation of the coefficient rather than sharpen it. The question is whether wine production predicts local support for the ban. Weighting by population or by ballots cast would answer a different question, whether it predicts the national result.

The specification is estimated on nine further federal votes alongside the absinthe ban. Six are falsification tests, on which no wine relationship is expected; three are corroborative, on which a positive one is. Three criteria narrow the set. We take only votes held before the ban, so that no comparison is contaminated by behavior the ban itself changed. We restrict the falsification arm to votes decided under the double majority. And we require the falsification votes to carry no wine-related subject matter. The corroborative votes meet the double-majority criterion with one exception, referendum #65, which is included because it is a wine-salient vote named explicitly in the deliberations on the absinthe ban. Table A.1 lists all ten.

Every referendum uses the vote-share construction given above for vote #68.

3.1.2 Petition signatures

The initiative reached the ballot through a signature drive, and that drive leaves a canton-level record of its own, seventeen months before the vote. We measure it as signatures

submitted per hundred eligible voters,

$$P_i \;=\; \frac{\text{signatures submitted}_i}{\text{eligible voters}_i} \times 100,$$

for the same $i = 1, \ldots, 25$ cantons, with eligible voters taken from each canton's roll for the federal vote of June 10, 1906 (#65).

The numerator comes from the twenty-five-canton table at printed p. 984 of the Federal Council's report of February 22, 1907, which gives for each canton the signatures submitted, those declared valid, and those rejected, closing at 169,377, 167,814 and 1,563 nationally (Schweizerischer Bundesrat, 1907a, p. 984). We take the submitted count. The canton figures reconcile: each of the three columns sums exactly to its own printed national total (Appendix A.4).

The denominator wants a count of those entitled to sign, and no such count survives. We use the cantonal electoral roll recorded for the federal vote of June 10, 1906, the federal ballot nearest the drive in its own year. A population denominator is the obvious alternative and it is the worse one: women, minors and non-citizens could not sign as electors, so scaling by population would import cross-cantonal differences in age structure and foreign residence into a measure meant to record who signed.

The same 1906 ballot also supplies a corroborative outcome (§5.3); its roll and its yes-share never meet in one specification. Appendix Table A.20 reports the petition-arm variables.

3.1.3 Wine harvests

The central variable of interest is canton-level wine-industry presence, standing for the producer interest that a ban on a competing drink would serve. We digitized the canton wine-harvest table series of the *Statistisches Jahrbuch der Schweiz* to measure it. The vote arm uses the 1907 harvest (BFS, 1909, printed p. 51) and the petition arm the 1906 harvest (BFS, 1908b, printed p. 55). Table A.2 gives the volume and printed page behind every referendum in the set.

The table reports two quantities for each canton: the total harvest in hectoliters and its total value in francs. We take both as printed and convert each to a share of the national total,

$$X_{2i} \;=\; \frac{Q_i}{\sum_j Q_j} \times 100, \qquad\qquad X_{3i} \;=\; \frac{V_i}{\sum_j V_j} \times 100,$$

where Q_i is canton i's total harvest in hectoliters and V_i its value in francs. We report both,

and for brevity show wine revenue in the main paper where possible and volume in the appendix; §4.2 sets out why revenue is preferred.

The denominator is national by construction. These variables measure the share of Swiss wine production sitting in canton i; they do not measure how large wine looms within canton i's own economy. The vote was federal, and what carried weight in it was a canton's standing in the national industry.

V_i values an entire harvest at the average prices returned by the cantonal governments and statistical bureaus, so it is a valuation rather than a receipt; we call it wine revenue for brevity (Appendix A.2).

We code cantons with unreported wine production as zero, except where the publications indicate incomplete reporting; where a vintage is incompletely recorded, we substitute the yearbook's footnoted estimates.

3.1.4 Absinthe producers

The absinthe-industry measures come from the annex that E.W. Milliet, Director of the Swiss Alcohol Board, prepared for the Federal Council's December 1907 *Botschaft* on the popular initiative (Schweizerischer Bundesrat, 1907b, pp. 341–366), with the firm-level table at pp. 360–361 (Milliet, 1907). It lists forty houses in eight cantons, each with its purchases of spirit at 95° ABV from the federal monopoly over 1902–1906; the other seventeen cantons have no listed house and enter every measure at zero. The proxy is coarse, but it is unique: Switzerland did not report liquor production by product until the 1930s.

The register is a list of monopoly customers, not a census of absinthe distilleries. The Board itself warned that the purchase figures give no reliable indication of the scale of absinthe production, because the listed houses also made other products from the same spirit (Milliet, 1907, p. 361). We therefore verify the register house by house and strike, from both numerator and denominator, every house the sources document as producing primarily or exclusively another drink; a documented absinthe producer with a mixed line stays in (Appendix A.3). Two series bracket the one judgment call. The loose series strikes five houses; the floor series, carried in the main tables, also strikes one Fribourg wholesaler with no production of its own.

The continuous measure, Abs_i, is the canton's share of the retained national total in percent. Under the floor series Neuchâtel holds 62.4 percent, Geneva 19.8, Vaud 10.9, Basel-Stadt 6.0 and Valais 1.0, and the three other register cantons hold zero. A discrete counterpart, $\mathbf{1}\{\text{Abs}_i > 0\}$, marks the five cantons with at least one attested absinthe

producer. The loose versions keep the Fribourg house; both are reported in Appendix Table A.8.

Beyond the stills the industry's footprint was narrow and lay in the same places. Wormwood was grown in the Val-de-Travers beside the distilleries (Trechsel, 1997), and the compensation paid after the ban went to the growers of 33 hectares of herbs as well as to forty distillers and their workers (Berthoud, 1969, p. 656), so the producer measure carries the growers with it. Downstream stood the cafés, whose keepers forced the Vaud referendum (§2.4); the tavern employment share of Table 4 measures that trade. Nor was absinthe made in one place and drunk in another. The Board put exports at 64,111 of the 809,850 kilograms of spirit a year that went into absinthe in 1902–1906, a production figure it thought more likely too high than too low (Milliet, 1907, pp. 361–362); the rest was drunk at home, in the French-speaking cantons (§3.2.1). Basel-Stadt is the exception, a German-speaking canton whose three houses worked "principally for export" (Schweizerischer Bundesrat, 1907b, p. 347).

3.2 Cultural and political controls

3.2.1 The French-speaking share

The first cultural control is the canton-level French-speaking share,

$$\text{French share}_i = \frac{\text{French speakers}_i}{\text{Resident population}_i} \times 100,$$

constructed from HSSO B.32 (1900 federal census, native-language classification; Historische Statistik der Schweiz Online (HSSO) 2012d) over the resident population of HSSO B.1a (Historische Statistik der Schweiz Online (HSSO), 2012a).

The denominator is the whole resident population rather than French and German speakers alone. A two-language denominator would omit the language most of Ticino speaks (Appendix A.4). The five Romande cantons cluster at the top, with Vaud, Geneva, Neuchâtel, Fribourg and Valais all above 60 percent.

The French-speaking share marks two things. The first is where absinthe was drunk. Asked in 1907 how the initiative bore on their cantons, the governments of the German-speaking cantons and of Ticino answered that consumption among their populations was "wholly insignificant," and the Federal Council concluded that a ban on sale was worth discussing for five cantons only, Fribourg, Vaud, Valais, Neuchâtel and Geneva (Schweizerischer Bundesrat, 1907b, pp. 343–344): the five in which French is the majority

language. No canton-level record of consumption can stand in its place, since the monopoly's sales described its traffic and not consumption (Schweizerischer Bundesrat, 1898a, p. 89) and some spirit never passed through its books (§2.2), so recorded spirit is likely to understate what was drunk. The second is the standing difference between French- and German-speaking Switzerland over federal measures in general. We therefore read its coefficient as co-explanatory rather than treat it as a nuisance control.

It does not measure wine. French-speaking Switzerland grows much of the country's wine, so the two shares are correlated across cantons (Appendix Table A.10), but the wine variable is a producer measure, a canton's share of the value of the national harvest, and the overlap is incomplete (Figure 2, panel A; Appendix Table A.3). Fribourg is two-thirds French-speaking and grows almost no wine; Zürich, Schaffhausen, Aargau and Ticino, with almost no French speakers, hold nearly thirty percent of national wine revenue. Nor were the two drinks consumed in different places: absinthe was drunk where wine was the customary drink, as its cheaper substitute, which is what the Valais government said in wishing the initiative success (Schweizerischer Bundesrat, 1907b, p. 345). The French-speaking share thus stands for the consumers of the banned drink and the wine share for the producers of its nearest substitute. Panel B shows why each is read with the other held fixed: among the twenty cantons that are not French-plurality, support rises with the wine share with no control at all, though imprecisely (Appendix Table A.9), while the five French-speaking cantons, lower as a group, turn the pooled relation negative.

3.2.2 The Protestant share

The second cultural control is the canton-level Protestant share,
$\text{Protestant share}_i = \frac{\text{Protestants}_i}{\text{Protestants}_i + \text{Catholics}_i} \times 100$, constructed from HSSO B.27 (1900 federal census; Historische Statistik der Schweiz Online (HSSO) 2012c). The denominator is Protestants and Catholics rather than the resident population, since no canton is predominantly Jewish, Eastern Orthodox or unaffiliated (Appendix A.4).

The Protestant share measures religious composition. What it proxies is the density of standing temperance organization, and the two are joined by how the organizations actually spread rather than by assumption. Switzerland's oldest and largest abstinence body, the Blue Cross, was founded in 1877 and declared itself neutral between denominations in Article 2 of its 1884 statutes. It did not spread that way: it reached, in Trechsel's account, only Protestant areas, with the single exception of the Bernese Jura, and the founding of the Catholic Abstinent League in 1895 ended its cross-denominational ambitions (Trechsel,

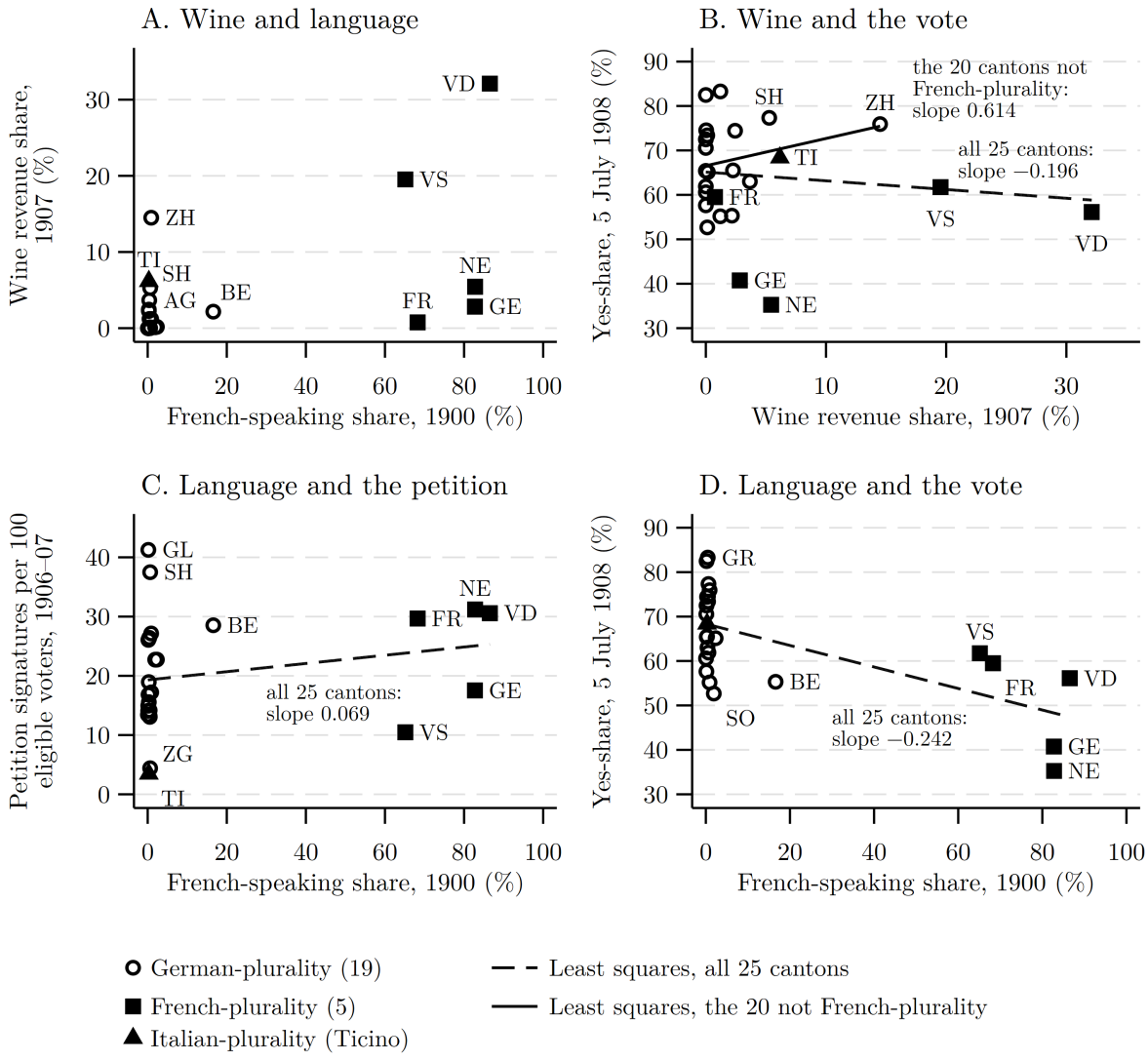


Figure 2: Wine, language, the petition and the vote, by canton. Panel A: share of national wine revenue (1907 harvest) against the French-speaking share (1900 census). Panel B: yes-share of July 5, 1908, against the wine revenue share; the dashed line is the least-squares fit through all twenty-five cantons, the solid line the fit through the twenty that are not French-plurality (Appendix Table A.9). Panel C: petition signatures per 100 eligible voters in the 1906–07 drive against the French-speaking share. Panel D: yes-share of July 5, 1908, against the French-speaking share. In panels C and D the dashed line is the least-squares fit through all twenty-five cantons, with slopes of 0.069 (HC3 standard error 0.062) and −0.242 (0.076).

1990, pp. 35–36).[8] The pattern is not one organization's idiosyncrasy (Appendix A.4).

Catholic temperance was not absent, and we do not claim it was (§5.2; Appendix A.4). The claim is one of timing and organizational depth. The Protestant body had been compounding its membership since 1882, while the Catholic League was assembled in 1895 out of three local societies founded in 1893 and 1894 (Trechsel, 1990, p. 98). Reviewing his own membership figures, Trechsel (1990, p. 132) concludes that the abstinence movement was to a large extent a Protestant one.

Religious affiliation is an indirect measure of the organizational capacity the argument runs on. Many Protestants did not support temperance, and some Catholic clergy did, particularly in cantons with large numbers of both. We keep the variable because the mechanism runs through it, and because omitting it would load the Protestant–Catholic divide onto the wine coefficient.

3.3 Population density

The structural control is the log of canton population density, $\ln D_i = \ln\left(\frac{\text{Population}_i}{\text{Land area}_i}\right)$, with population in persons and land area in square kilometers. The two arms differ in the numerator alone: the vote arm uses the 1907 canton population and the petition arm the 1906 population, both read from the Statistical Yearbook rather than interpolated. Land area is recovered exactly from the 1900 census, as population over published density (Historische Statistik der Schweiz Online (HSSO), 2012a,b; Appendix A.4). Density runs from 15.4 to 3,510 persons per square kilometer, a factor of more than two hundred, so it enters in logs.

3.4 Summary statistics

Table 1 reports summary statistics for the analysis dataset, each variable on the scale at which it enters the regressions, for all twenty-five cantons and for the German- and French-plurality cantons separately. Ticino, the one Italian-plurality canton, appears only in the all-canton columns.

[8] "Obwohl laut Statuten konfessionell neutral, verbreitete sich das Blaue Kreuz mit Ausnahme des 'Berner' Juras nur in protestantischen Gebieten. [...] Faktisch wurde so das Blaue Kreuz eine protestantische Organisation." Our translation.

Table 1: Descriptive statistics: canton-level analysis variables, 1908 cohort

	All cantons (N = 25)					German-plurality (N = 19)		French-plurality (N = 5)	
Variable	Mean	SD	Median	Min	Max	Mean	SD	Mean	SD
Yes-vote share, referendum #68	64.3	11.7	65.1	35.3	83.3	67.7	9.2	50.7	11.9
Petition signatures (per 100 eligible)	20.5	9.5	17.5	3.4	41.3	20.4	9.1	23.9	9.4
Wine revenue share, 1907	4.0	7.5	1.2	0.0	32.1	1.7	3.4	12.1	13.4
Wine volume share, 1907	4.0	6.9	1.0	0.0	25.6	1.8	3.7	10.9	11.1
Attested absinthe producer (0/1)	0.20	0.41	0.00	0.00	1.00	0.05	0.23	0.80	0.45
Absinthe-sector exposure, 1902–06	4.0	13.0	0.0	0.0	62.4	0.3	1.4	18.8	25.6
French share, 1900	16.5	31.3	0.5	0.1	86.5	1.4	3.7	77.1	9.7
Protestant share, 1900	43.1	35.0	48.2	1.3	90.2	44.1	34.4	47.5	39.3
Tavern employment share, 1905	2.09	1.01	1.94	0.75	5.18	1.91	0.64	2.18	1.31
Tavern and inn employment share, 1905	3.99	1.57	3.49	1.48	6.82	3.93	1.51	3.74	1.69
Population density, 1907 (per km^2)	264	687	108	15	3510	292	782	202	234
Log population density, 1907	4.67	1.12	4.68	2.73	8.16	4.68	1.15	4.77	1.20

Notes: Every variable is summarized on the scale at which it enters the regressions. Percentage term measures include: national shares (wine revenue, wine physical output, absinthe-sector exposure) defined as the canton's share of the Swiss total and sum to 100 across cantons. The French and Protestant shares are within-canton percentages, of the 1900 census population and of Protestants plus Catholics respectively. The tavern employment shares are within-canton percentages of male employment, net of cottage industry, in class 301 of the federal business census of August 9, 1905 — restaurants and taverns not attached to an inn — and in classes 300+301, which add inns, hotels and boarding houses (BFS, 1908a). The yes-share of votes (64.3) is unweighted; it treats half-cantons and full cantons with equal weight. Absinthe-sector exposure is the canton's share of national purchases of 95-degree spirit from the federal monopoly over 1902–1906 (Milliet 1907 annex) after the record-verified exclusion of non-absinthe firms; the indicator for attested-producer cantons marks Neuchâtel, Geneva, Vaud, Basel-Stadt and Valais.

4 Empirical approach

4.1 Specification

For canton i and referendum r, the estimating equation is

$$Y_{i,r} = \alpha + \beta_X X_i + \beta_{\text{Fr}}\text{Fr}_i + \beta_{\text{Abs}}\text{Abs}_i + \beta_{\text{Pr}}\text{Pr}_i + \gamma_D \ln D_i + \varepsilon_{i,r}, \tag{OLS}$$

where $Y_{i,r}$ is the yes-share in percentage points, X_i the wine measure of §3.1.3, and the remaining terms are the French-speaking share, absinthe-industry presence, the Protestant share and log population density.

Every specification we report is a variation of equation (OLS), excluding or substituting some selection of the regressor set. The columns of Table 2 enter the controls in a fixed order, holding the not-yet-entered coefficients at zero: the wine measure alone, then the French share, then the Protestant share, then absinthe-industry presence, then log population density. Two further columns repeat the last two with the absinthe conditioner in its continuous rather than its discrete form. The order is fixed in advance and does not vary across tables.[9]

The petition arm re-estimates equation (OLS) with the signature rate P_i of §3.1.2 in place of the yes-share, reading the harvest and population covariates at their 1906 values; the census shares keep their own dates. The coefficient of interest is the wine measure at the vote (Table 2) and the Protestant share at the petition (Table 4), where the absinthe regressor enters as the indicator for attested-producer cantons. The choice trades one bias for another: the continuous purchase share carries measurement error of unknown direction, since the Alcohol Board itself warned that its purchase figures give no reliable indication of the scale of absinthe production (§3.1.4), while the indicator's error is consistent, the same constant for every producer canton whatever its share of the market. One column of Table 4 adds the tavern employment share: tavern employment among men, net of cottage industry, since only men could sign or vote (see the table note).

Wine and tavern never enter a column jointly. Their correlation over the cantons is indistinguishable from zero (0.030 by Pearson, 0.163 by Spearman, and −0.029 on the broad tavern measure), and at $N = 25$ a joint column would cost a degree of freedom we cannot spare. Appendix Tables A.12 and A.18 show which cantons carry each coefficient and what deleting each one does to it.

[9] Two appendix tables depart from it: Appendix Table A.7 enters log population density before the absinthe measures, and Appendix Table A.19 enters cattle and the food-control regime ahead of the shared controls for referendum #65.

4.2 Revenue versus volume

We report the wine revenue share as the focal measure and the volume share beside it. The two differ by one term: wine revenue is volume scaled by the canton's average wine price relative to the national average, so the wine revenue share is the volume share weighted by local price levels.

That weighting is the reason to prefer it. If absinthe competed with wine on price, local prices put more weight on the cantons where absinthe was the closer substitute. And in the Stigler–Peltzman account what regulation transfers is wealth, so the stake that counts is the revenue at risk rather than physical output (Stigler, 1971, p. 12; Peltzman, 1976, pp. 212–215).

We add that price is an estimate and carries its own assumptions: that the price measure is locally accurate, that expected revenue approximates realized revenue, and that the within-canton price distribution does not make the average unrepresentative. None is testable here. The volume share, which is not price-scaled, is reported throughout for that reason, and the two measures agree in sign and closely in magnitude.

4.3 Estimator choice

The yes-share is bounded, and the Papke–Wooldridge fractional logit respects the bounds where least squares need not (Papke and Wooldridge, 1996, pp. 619, 621). These data sit away from the bounds: no fitted value falls outside them on any referendum, neither specification test rejects, and the average marginal effects of the fractional logit, our check on least squares, never differ in sign from the least-squares coefficients and on the 1908 vote are slightly smaller in magnitude in every column (Tables A.5, A.13 and A.4). We report both and take inference from least squares, which admits the finite-sample corrections below.

4.4 Collinearity diagnostics

Because the wine measure and the French-speaking share describe overlapping sets of cantons, the foot of Table 2 reports two collinearity diagnostics for each column: the centered variance inflation factor of the wine measure (O'Brien, 2007, n. 1), and the Belsley–Kuh–Welsch condition number, computed with the constant and without centering so that a dependency on the intercept stays visible (Belsley, 1991, pp. 36–39). Neither is a test. Their usual cutoffs are conventions, and a favorable value certifies nothing the

standard error does not already carry (Belsley, 1991, pp. 38, 43; O'Brien, 2007, p. 681 and n. 2). We use them for one question: whether the width of the intervals comes from near-dependence among the regressors or from the information twenty-five cantons carry.

One reading of the VIF does not carry over to this paper. Under the classical variance formula the VIF is the factor by which collinearity raises a coefficient's sampling variance relative to an orthogonal design, but that formula has no place for leverage, and the heteroskedasticity-robust (HC3) standard errors of the next subsection weight each squared residual by a function of its observation's leverage. What the VIF states here is a property of the design, the share of a regressor's variation the controls span, and not the width of any interval in the table; collinearity is one source of a large variance among several, and it is the standard error that carries the conclusion (Goldberger, 1991, §§23.1, 23.5).

4.5 Small-sample inference

Inference at twenty-five observations cannot rest on one asymptotic approximation, so we report four readings of every least-squares estimate in the main-text tables, each correcting a problem the others do not. None is an extrapolation: the smallest design in the simulations that established HC3 is twenty-five observations with four regressors (Long and Ervin, 2000), and the evidence behind the resampling procedures reaches still smaller samples (Anderson and Robinson, 2001; Roodman et al., 2019). The wine share is right-skewed: most cantons grew little or no wine, and Vaud alone held about a third of national wine revenue in 1907. Standard errors are HC3, the most conservative of the MacKinnon–White family (MacKinnon and White, 1985); at this size the uncorrected robust estimator rejects a true null more than twice as often as it should, and HC3 holds the distortion below two percentage points (Long and Ervin, 2000). HC3 corrects the variance but not the distribution the statistic is compared against, and with a skewed regressor that is where coverage fails (Imbens and Kolesár, 2016). The Bell–McCaffrey adjustment pairs an HC2 variance with degrees of freedom computed from the design, which fall well below $N - k$ when a coefficient rests on a few cantons; it can land on either side of HC3, and Imbens and Kolesár (2016) recommend it for exactly this case.

Two resampling readings build their reference distribution from the data instead of taking it from a table. A wild bootstrap-t with Rademacher weights and the null imposed flips each canton's residual in place, so each keeps its own variance (Roodman et al., 2019). A Freedman–Lane permutation test on the HC3 t statistic asks how often a statistic this large would arise if the residual yes-vote were shuffled across cantons (Freedman and Lane, 1983; Anderson and Robinson, 2001; DiCiccio and Romano, 2017); its validity under

heteroskedasticity is asymptotic, so at twenty-five it too is an approximation. No two of the four use the same reference distribution: HC3 uses the conventional t distribution, Bell–McCaffrey a t with degrees of freedom computed from the design, the wild bootstrap the distribution of its resampled statistics, and the permutation test that of its shuffled ones; it is their agreement that carries the inference.

5 Results

5.1 The vote

Table 2 reports the least-squares cascade for the July 5, 1908, vote. The raw association between a canton's share of national wine revenue and its yes-share, column (1), is negative and indistinguishable from zero. Unsurprisingly, without controls the wine coefficient mixes the wine relation with the language divide. Wine cantons are disproportionately French-speaking, and the French-plurality cantons supported the ban by 17.1 points less, on average, than the German-plurality ones (Figure 2, panel B). With the co-explanatory French share included, from column (2) onward, the wine coefficient is positive in every column and below ten percent on at least three of its four readings in every column but (7). The French share is negative throughout and significant in some columns only. The Protestant share and the producer indicator, columns (3) and (4), move the wine coefficient further, to 0.639 (0.207) and then 0.677 (0.216) in the preferred specification. The Protestant share enters negative in both columns and reaches five percent in neither.[10] Only five cantons had attested absinthe producers, four of them French-speaking (eight appear in Milliet's register), too few to identify an absinthe effect, so both absinthe measures serve as controls only.

Because the wine measure is a share of a national total, the coefficient prices each canton's stake in the national outcome rather than the intensity of viticulture within it. One percentage point of national wine revenue is associated with 0.677 points of yes-share, holding the French share, the Protestant share and producer status fixed, so between two cantons alike in those three respects, one at the mean share of four percent (Table 1) and one with no wine at all, the specification attributes about 2.7 points of yes-share to the

[10] Leaving the Protestant share out of column (4) lowers the wine coefficient to 0.601, with all four readings still below five percent (.023 to .040). In the purchase-share column (6), where the estimate is weaker to begin with, the same omission lowers it to 0.404, and none of the four readings clears ten percent (.104 to .164). Leaving the wine share out of column (4) instead lowers its R^2 from 0.58 to 0.46 and raises its AIC and BIC, which already charge for an extra regressor, by 4.3 and 3.0.

wine share. The conditioning carries the sign: the unconditional comparison in column (1) runs the other way, because the cantons with wine are the cantons that speak French. All four readings of the preferred column reject at five percent: .005 on HC3, .014 under Bell–McCaffrey, .032 on the wild bootstrap and .012 on the Freedman–Lane permutation test. The Bell–McCaffrey reading uses only 3.68 effective degrees of freedom instead of the nominal 20, a number that depends on the regressors alone, not on the votes (Bell and McCaffrey, 2002, p. 176). At so few degrees of freedom the test rejects a true null less often than the nominal five percent (Bell and McCaffrey, 2002, Table 3), so a result that clears it has passed a conservative test.

The remaining columns stress the estimate. Adding log population density in column (5) moves the coefficient to 0.559 (0.305), 0.55 standard errors below the preferred specification's value, while the density term itself is indistinguishable from zero and the interval widens; the four readings range from .072 to .147. Columns (6) and (7) replace the producer indicator with the continuous absinthe measure and give 0.514 (0.292) and, with density, 0.439 (0.315), with readings from .029 to .093 and from .105 to .180. No column after the first turns the sign, the French share stays between −0.240 and −0.316 throughout, and the volume measure gives the same picture, with a preferred-column coefficient of 0.644 and readings between .026 and .037 (Appendix Table A.6). Why the density columns widen the interval is a question about where the wine measure's variation sits among the cantons, and we take it up after the collinearity diagnostics.

The diagnostics at the foot of Table 2 give no sign that near-dependence sets the interval widths. In column (4) the condition number is 4.43, inside the range read as weak, and the wine measure's variance inflation factor is 1.58: about a third of its variation across cantons is predictable from the other regressors. The two large condition numbers, in the density columns (5) and (7), register a dependency between log density and the constant, which centering removes without changing any slope. What sets the widths is the number of cantons and where the wine variation sits among them.

Appendix Table A.12 reports where that variation sits. Four cantons have leverage above twice the average (Hoaglin and Welsch, 1978, pp. 18–19): Basel-Stadt (0.652), Vaud (0.631), Fribourg (0.534) and Valais (0.410). Leverage matters only when the canton's vote is also surprising given its regressors (Hoaglin and Welsch, 1978, p. 20), and two cantons combine the two: Fribourg and Basel-Stadt, at standardized influences of −0.448 and −0.423 (Belsley et al., 1980). They are the two cantons where language and producer status part ways, Fribourg French-speaking with no attested producers, and Basel-Stadt German-speaking with attested producers. Both work against the result: neither grows

Table 2: Wine revenue and the 1908 absinthe ban

	Raw	+ French share	+ Protestant share	+ producer (binary)	+ population density	+ purchase share	+ population density
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Wine revenue share	–0.196	0.547**	0.639***	0.677***	0.559*	0.514*	0.439
	(0.219)	(0.257)	(0.207)	(0.216)	(0.305)	(0.292)	(0.315)
French share		–0.316***	–0.315***	–0.271	–0.298	–0.249**	–0.240**
		(0.087)	(0.073)	(0.158)	(0.258)	(0.117)	(0.113)
Protestant share			–0.083*	–0.078	–0.048	–0.062	–0.030
			(0.046)	(0.046)	(0.080)	(0.051)	(0.068)
Absinthe (binary)†				–4.974	–0.514		
				(11.844)	(20.235)		
Absinthe (share)‡						–0.202	–0.189
						(0.305)	(0.199)
Log population density					–2.110		–2.084
					(3.757)		(2.654)
Constant	65.13	67.38	70.56	70.48	79.05	69.87	78.32
	(2.58)	(2.08)	(2.77)	(2.80)	(14.67)	(2.72)	(10.96)
R^2	0.02	0.51	0.56	0.58	0.60	0.59	0.62
Adjusted R^2	–0.03	0.46	0.50	0.49	0.49	0.51	0.52
AIC	196	181	180	181	182	181	181
BIC	199	185	185	187	189	187	188
Small-sample diagnostics for wine coefficient (p-values)							
HC3 robust	0.379	0.044	0.006	0.005	0.083	0.093	0.180
Bell–McCaffrey	0.385	0.072	0.025	0.014	0.078	0.078	0.139
Wild bootstrap-t	0.435	0.036	0.031	0.032	0.147	0.029	0.105
Freedman–Lane	0.325	0.059	0.016	0.012	0.072	0.079	0.134
Bell–McCaffrey effective D.O.F.	2.04	3.58	3.74	3.68	3.81	4.08	4.40
BKW condition no. (with constant)	1.68	2.48	3.56	4.43	16.54	4.79	13.19
VIF (wine revenue share)	1.00	1.47	1.53	1.58	1.85	1.81	1.89

Notes: Ordinary least squares estimates; $N = 25$ cantons. The dependent variable is the canton yes-share in the July 5, 1908 referendum. Wine is the canton's share of national wine revenue (×100), 1907 harvest — the last complete harvest before the vote. *, **, and *** mark $p < 0.10$, 0.05, and 0.01, on heteroskedasticity-robust (HC3) p-values; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell–McCaffrey, wild bootstrap-t, Freedman–Lane permutation). The variance inflation factor (VIF) is centered and reported for the wine revenue share; the specification maximum falls on the French share (2.69 in column 4, 2.84 in column 7). Condition numbers are Belsley–Kuh–Welsch, constant included. † the indicator for attested-producer cantons; ‡ the canton's share of verified 1902–06 purchases of monopoly spirit, both defined in §3.1.4. The two encodings never enter the same column; the raw purchase share is in Appendix Table A.7 and loose membership in Appendix Table A.8.

much wine, and dropping either raises the estimate from 0.677 to 0.801 without Fribourg and 0.797 without Basel-Stadt. Deleting each canton in turn leaves the coefficient positive and at least 0.598, and no single deletion costs significance at ten percent. Two deletions lose significance at five percent on the Bell–McCaffrey reading, for different reasons. The first is Vaud, which holds 32.1 percent of national wine revenue. Vaud also carries 34.5 percent of the wine coefficient's identification weight (Appendix Table A.17). Dropping Vaud raises the beta to 0.724, but the p-value rises to .058, a loss of precision rather than a sign of distortion (Hoaglin and Welsch, 1978, eq. 5.5). The other is Geneva, at .059, one of the two cantons that voted against the ban and, with Neuchâtel, the most French-speaking canton after Vaud (Appendix Table A.12).

5.1.1 Robustness

The wine coefficient moves from −0.196 in column (1) to 0.677 in column (4), with the controls added sequentially. We thus investigate how much of the coefficient movement each additional regressor term accounts for and whether the answer depends on the order. Table 3 answers both. The decomposition of Gelbach (2016, p. 522) splits the movement into one contribution per added covariate, and the split is invariant to the order in which they enter, so the fixed order of the cascade does not determine the attribution. Contributions are signed so that a positive entry moves the coefficient from its column (1) value toward its column (4) value, which is the reverse of the orientation in Gelbach's own equations; the shares are the same either way. Of the total movement of 0.873, the difference between the two coefficients, the French share accounts for 0.637, or 73 percent; the Protestant share contributes 0.090 (10 percent) and the producer indicator 0.147 (17 percent).[11] For column 7's specification with the continuous share of absinthe firm proxy-shares, the French share accounts for 89 percent of a smaller total movement, and log population density entering with the only negative contribution in the table, −0.014.

That language carries the movement is the design's premise, not a finding. The decomposition is an accounting identity, not a test: each contribution inherits the causal standing of its own coefficient and has none of its own (Gelbach, 2016, pp. 525–526). What it shows is the size of the confounding the design names: without language, the wine coefficient stands in for a linguistic pattern that runs the other way.

Table A.11 reports the selection bounds of Oster (2019) on two baselines, from column

[11]The split between the French share and the producer indicator is the least reliable part of the table. Attested producers sat in five cantons, four of them French-speaking, so the two carry much of the movement together, 90 percent of it between them; the decomposition reports that joint share more reliably than it divides it.

Table 3: Gelbach decomposition of the bivariate-to-adjusted wine coefficient shift

Added covariate	Column (4), preferred To col. (4)	Share	Column (7), reference To col. (7)	Share
French share	+0.637	73%	+0.565	89%
Protestant share	+0.090	10%	+0.034	5%
Absinthe producer (attested, binary)	+0.147	17%		
Milliet absinthe purchase share (corrected)			+0.051	8%
Log population density			–0.014	–2%
Total shift	+0.873	100%	+0.635	100%

Notes: Gelbach decomposition of the change in the wine revenue-share coefficient from the raw bivariate model to (i) the column-(4) preferred specification (French share, Protestant share, and the attested-producer binary; the wine coefficient moves from −0.196 to +0.677, a shift of +0.873) and (ii) the column-(7) continuous reference (French share, Protestant share, the corrected Milliet purchase share and log population density; −0.196 to +0.439, shift +0.635). Entries are signed adjusted-minus-raw contributions, rounded to three decimals; unrounded, they sum to each column's total. Shares are contributions divided by the total shift. This is the reverse of the orientation in Gelbach's (2016) equations (11) and (12); shares are unaffected. The two absinthe encodings are alternative measures of the same variable and never enter together. $N = 25$ cantons.

(1) to column (4) and, with the French share kept in both regressions, from column (2) to column (4). Appendix A.5.7 sets out the method, and Blum and Dimico (2018, pp. 386–387) present it as a check for economic history, where what was omitted is often unknowable. At her calibrated ceiling, unobserved variables that selected as strongly as the observed controls would raise the wine coefficient from 0.677 to 0.949 on the first baseline and to 1.008 on the second, by 1.26 and 1.53 standard errors, and both identified sets lie within 2.8 standard errors of the estimate, the half-width of a 99.5 percent confidence interval. On both baselines the controls move the coefficient away from zero, so no selection in the same direction as the observed controls produces a zero. At the limiting ceiling of one, which she shows over-adjusts, the set on the second baseline lies outside that interval. The exercise is a sensitivity analysis, not inference.

5.2 The petition

Table 4 sets the petition beside the vote, and the two arms load on different variables: the Protestant share, our proxy for organized temperance, predicts the signatures, and the wine share predicts the vote. Panels C and D of Figure 2 plot the two stages against the French-speaking share before any controls. The petition's Protestant share is positive and significant, at 0.170 (0.056) in the focal column. All four readings of the focal petition column reject at five percent: .007 on HC3, .006 under Bell–McCaffrey (on 13.35 effective degrees of

freedom), .007 on the wild bootstrap and .004 on the Freedman–Lane permutation test. The wine revenue share has no detectable effect at the petition stage: it enters at −0.142 (0.603), and none of its four readings is significant (Appendix Table A.23, which also shows no near-dependency in the focal or tavern column; condition numbers 4.54 and 6.24). Neither result turns on a single canton: deleting each in turn leaves the Protestant share between 0.146 and 0.201 and below five percent under Bell–McCaffrey every time (largest p .021), and leaves the wine share short of ten percent under every deletion (smallest p .152, without Vaud). The wine coefficient's standard error at the petition is almost three times the 0.216 it carries at the vote, and its Bell–McCaffrey reading rests on 2.50 effective degrees of freedom. The two outcomes are on different scales, signatures per hundred eligible voters and points of the yes-share, but their standard deviations are similar (Table 1), so the wine coefficient is estimated far less precisely at the petition than at the vote. The standardized panel states the cross-arm contrast in one place: the Protestant share stands at 0.624 standard deviations at the petition against −0.234 at the vote, and the wine revenue share runs the other way, 0.436 at the vote against −0.127 at the petition.

The stage-separation between the coefficients of the Protestant share and the wine revenue share is consistent with the campaign's course of events. The Blue Cross founded the standing delegation responsible for convening the Bern congress of December 17, 1905, after which the initiative committee formed. That same committee ran the initiative collection (Trechsel, 1990, pp. 33, 122, 151–153). The Catholic bodies followed: the Katholikentag, a general congress of Swiss Catholics rather than a temperance society, declared for a ban in 1906; and the Swiss bishops did so in 1908 (Trechsel, 1990, p. 107). The Catholic People's Association called for an emphatic yes on June 22, 1908, with the Methodist pastors endorsing unanimously (Journal de Genève, 1908, p. 3), nearly seventeen months after the signatures were submitted to the Federal Chancellery by January 31, 1907 (Schweizerischer Bundesrat, 1907b, p. 353). A reader who sets the two arms side by side can conclude that religion mattered only for getting the petition started, or that temperance had stopped mattering by the time the country voted. Neither is what the coefficients identify. What the Protestant share picks up at the petition is early confessional organization, and the Blue Cross tie is why: Protestants were already organized as a temperance movement when the collection opened, and Catholics, who did not want hard prohibition as uniformly, could unite against a drink with absinthe's reputation more easily. The Catholic Abstinent League backed the initiative, as it did other abstinence causes, but Trechsel's account shows the difference in depth: its activity was limited and seems to have leaned largely on the groundwork of the other abstinence societies, and its later proposals toward prohibition found hardly any support among Catholics (Trechsel, 1990, pp. 105,

107).

By 1908 both confessions stood on the same side, so at the vote the Protestant share no longer separates supporters of the ban from opponents; it measures the weight of one wing of a joint coalition. Trechsel credits the Catholic bishops' pastoral letter for the initiative with some influence, and by then the parties had come over or stood aside: only the Radicals of Neuchâtel and Geneva opposed the initiative, the other Radical cantonal parties abstained, and the rest, Conservatives and Socialists alike, supported the ban (Trechsel, 1990, p. 152). What is left of the confessional difference at the vote is small and runs the other way: holding wine, language and producers fixed, the Protestant share enters negative, short of five percent in every column. The late Catholic endorsements above are consistent with that, though nothing here tests it: they would leave the more Catholic cantons slightly ahead when the country voted. Trechsel adds that the vote came on a low turnout and that the yes votes were by far the poorest return on signatures of any federal initiative to that date, and he reads this as a sign that the emotion of the collection had already subsided somewhat (Trechsel, 1990, p. 153). Together, the two arms describe a mobilization in stages.

The tavern column of Table 4 replaces the wine share with tavern employment; wine and tavern never enter a column jointly, for the reason given with the specification. Its measure is the canton's share of male employment, net of cottage industry, in class 301 of the federal business census of August 9, 1905, restaurants and taverns not attached to an inn (BFS, 1908a, p. 32), and Appendix Table A.20 reports it among the petition-arm covariates. The coefficient is −3.762 (1.176), with .005 on HC3, .024 under Bell–McCaffrey (on 3.38 effective degrees of freedom), .039 on the wild bootstrap and .009 on the Freedman–Lane permutation test. The estimate therefore rests on a few cantons rather than on all twenty-five, and Appendix Table A.18 shows the details. Ticino carries 42.6 percent of its identifying weight (Appendix Table A.17), and deleting each canton in turn leaves the estimate between −4.388 and −3.251 without turning the sign. The column is consistent with a lower signature rate where the consumer-facing drink trade was a larger employer.[12]

[12]The broad companion that adds the lodging trades is where the measure is weakest, and Appendix Table A.15 sets out both; Appendix Table A.16 runs the same cascade against the 1908 vote.

Table 4: The 1906 anti-absinthe petition versus the 1908 ban vote

	Petition arm focal (4)	Tavern focal	Petition arm density-in (5)	Vote arm focal (4)	Vote arm share form (7)	Vote arm producer form (5)
Wine revenue share	–0.142		–0.169	0.677***	0.439	0.559*
	(0.603)		(0.653)	(0.216)	(0.315)	(0.305)
French share	0.134	0.103	0.126	–0.271	–0.240**	–0.298
	(0.126)	(0.074)	(0.173)	(0.158)	(0.113)	(0.258)
Protestant share	0.170***	0.170***	0.179**	–0.078	–0.030	–0.048
	(0.056)	(0.048)	(0.079)	(0.046)	(0.068)	(0.080)
Log density			–0.619		–2.084	–2.110
			(3.120)		(2.654)	(3.757)
Absinthe	–6.760	–5.418	–5.502	–4.974	–0.189	–0.514
	(10.020)	(6.265)	(13.907)	(11.844)	(0.199)	(20.235)
Tavern employment share		–3.762***				
		(1.176)				
Constant	12.84	20.41	15.32	70.48	78.32	79.05
	(2.54)	(3.67)	(12.63)	(2.80)	(10.96)	(14.67)
Standardized β:						
Wine revenue share	–0.127		–0.151	0.436	0.283	0.360
Protestant share	0.624	0.622	0.657	–0.234	–0.089	–0.145
Tavern employment share		–0.399				
R^2	0.41	0.56	0.42	0.58	0.62	0.60
Adjusted R^2	0.30	0.47	0.26	0.49	0.52	0.49
Coefficient of interest:	*Prot.*	*Tavern*	*Prot.*	*Prot.*	*Prot.*	*Prot.*
HC3 robust p	0.007	0.005	0.036	0.106	0.666	0.552
Bell–McCaffrey p	0.006	0.024	0.025	0.089	0.633	0.469
Wild-bootstrap-t p	0.007	0.039	0.021	0.088	0.630	0.499
Freedman–Lane p	0.004	0.009	0.020	0.082	0.626	0.488
Effective df	13.35	3.38	11.52	12.56	11.09	11.34

Notes: Ordinary least squares; $N = 25$ cantons. The outcome is signatures per 100 eligible voters in the petition-arm and tavern columns and the yes-share of July 5, 1908, in the vote-arm columns. Stars mark HC3 $p < 0.10$, 0.05 and 0.01. The four p-value rows and the effective degrees of freedom refer to the coefficient of interest named above them. Absinthe is an indicator for attested-producer cantons, except in the share-form column, where it is the canton's share of verified 1902–06 purchases of monopoly spirit. Standardized β: coefficients in standard-deviation units. Tithe-based temperance measures: Appendix Tables A.21–A.22.

5.3 Falsification and corroboration

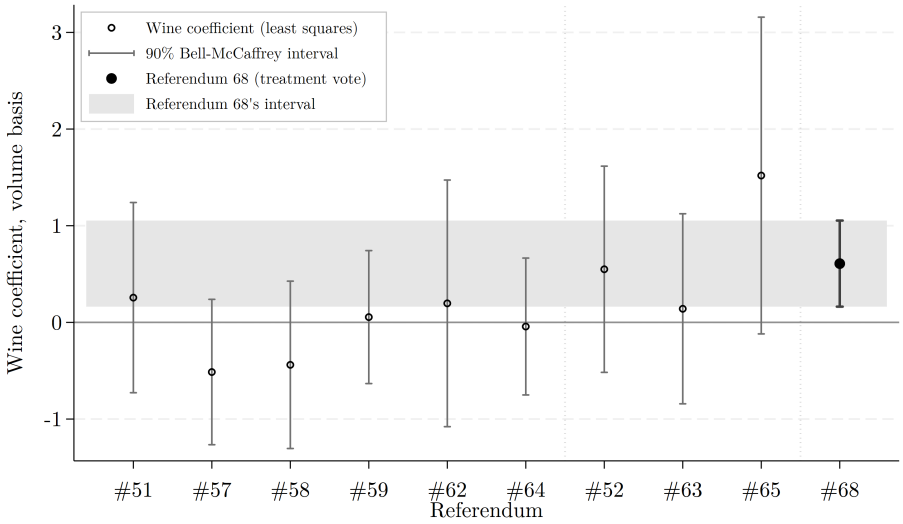


Figure 3: Wine coefficient by referendum, volume basis, at the column (3) specification (the wine, French and Protestant shares; Appendix Table A.6). Points are least-squares estimates with 90 percent Bell–McCaffrey intervals computed on each vote's own effective degrees of freedom. Dotted lines separate the falsification, corroborative and treatment votes. The shaded band is referendum 68's interval, drawn across the plot for comparison.

While no vote of this period is a true placebo for the absinthe ban, we use unrelated referenda votes to test whether the wine coefficient is significant only where a wine interest was at stake. Tables A.13 and A.14 report every estimate quoted below. The two arms and the criteria that fix them are given in §3.1.1, and were settled before any of these votes were estimated. Figure 3 and Table A.14 read them on the volume basis, which is the only measure on which all ten votes exist and is also the less favorable of the two for the treatment vote. Across the six falsification referenda the wine coefficients cluster near zero, the largest in absolute value being −0.514 on referendum #57, and the intervals are too wide to resolve any of them, the least-squares Freedman–Lane readings at the revenue basis running from .181 to .874.[13]

The coefficient appears where the design says it should. On the absinthe vote the wine revenue share returns 0.639 in column (3) of Table 2, the specification every vote in the set shares, at .016 on the least-squares Freedman–Lane test. Of the three corroborative votes, referenda #52 and #63 do not reach significance. Referendum #65 does, at the margin:

[13]These imprecise nulls cannot rule out a wine relationship on the falsification votes, but if the wine share's significance were an artifact of some omitted variable associated with a canton's share of the national wine revenue or volume, that variable would have to be silent on every falsification vote, and our results show no significance for it there.

on the specification the set shares, its least-squares Freedman–Lane reading is .083 for the revenue measure (Table A.13), just under the ten percent line, and its least-squares HC3 and Bell–McCaffrey readings sit at and just above that line, while the permutation reading of the fractional logit, our secondary estimator, falls below it (Table A.4). On that shared specification we read it as weakly corroborative. Its own coalition specification, conditioning on cattle and on whether a pre-existing canton-level food control institution existed, is more robust, though it rests on very few effective degrees of freedom: it reaches five percent on every reading but Bell–McCaffrey's, which falls just above it (Table A.19). The main text quotes the specification the set shares.

Referenda #52 and #65 are two related food-law referenda. Because wine was regulated as food and not as a spirit, they are the two food-law votes on which a wine interest might be at stake; the 1905 food act reserved federal legislation on distilled spirits and never touched absinthe (Schweizerische Bundesversammlung, 1905b, p. 18). Referendum #52, of July 1897, was the Federal Decree of March 26, 1897, that added Article 69bis to the Federal Constitution and empowered the Confederation to legislate on the trade in foodstuffs and stimulants (Schweizerische Bundesversammlung, 1897, p. 233); the article left execution to the cantons under federal supervision and with federal money and reserved control of imports at the national frontier to the Confederation (Schweizerische Bundesversammlung, 1897, p. 234). Referendum #65, of June 1906, was the act that turned that power into actionable regulatory policy (Schweizerische Bundesversammlung, 1905b, p. 1). The prima facie justification for the resulting food regulations was consumer protection. The message that carried the 1897 article argued that wines were very often adulterated with artificial wine and listed wines with butter, fats and oils for inspection at the frontier (Schweizerischer Bundesrat, 1895a, pp. 776 and 805). Appendix A.9.2 gives further detail on vote #52.

Much of the political support came from those who would benefit from producer protections. The act carried the demands of a pork-barreled agricultural coalition, containing wine, cattle and dairy in its substantive parts (Appendix A.9.1). The Farmers' Union called frontier control the cardinal point of the law, and the rapporteur said the frontier experts were added mainly at its demand and at that of the honest wine merchants and vineyard owners (Schweizerische Bundesversammlung, 1905a, National Council, June 9 and 10, 1903, pp. 418, 442). The Council of States committee's spokesperson called the clause on mixtures with substitutes "a measure of protection in favor of natural products" that "the circles of the farmers and of the wine-growers" counted on, and named "margarine and artificial wine" as its targets (Schweizerische Bundesversammlung, 1905a, pp. 892–893). By 1899 the government's own message named the protection of "the genuine producer (farmer

and manufacturer)" against "dishonest competition" as an aim co-equal with consumer protection (Schweizerischer Bundesrat, 1899b, p. 615). The Zurich and Swiss wine merchants' associations asked for artificial wine to be banned outright and were refused each time (Schweizerische Bundesversammlung, 1905a, pp. 393, 1050–1051). France's legislative history shows similar political pressure, albeit with differing outcomes. France enacted legislation in August 1905 and June 1907 that made it illegal to produce "adulterated" wine and prohibited, in vague terms, the production of wine that was "artificial" rather than "natural." Which wines counted as artificial was left to whoever enforced the laws, and associations of producers did much of the enforcing (Franck et al., 2014, p. 88). We presume it was a convenient coincidence that the vague laws left producers with a vested interest to decide which wines, and thereby which competitors, broke them. Franck et al. (2014, p. 88) read artificial wine as an issue the wine lobbies used to press the government, leading to laws that prevented adulteration and protected producers of cheap wine made from real grapes. They grant that adulteration was real and say they cannot determine how far interest drove the concern (Franck et al., 2014, p. 88). The Swiss record leaves the same question open, since the government that warned that wines were very often adulterated also counted the protection of the genuine producer among the law's aims.

The Swiss Farmers' Union, founded in 1897 and run by Ernst Laur (Baumann, 2006b,a), was an organized interest of the kind Olson (1965) describes, able to write producer protections into a consumer-protection law. In our reading of the results and the historical evidence, referendum #65 was backed by an agricultural coalition with wine-growers in its ranks. Although the act bears on absinthe only indirectly, through alcohol markets, the vote is consistent with our main measure capturing wine producers' interests.

6 Discussion

Across the two stages the evidence shows two constituencies: organized temperance is detectable at the petition, and the wine interest at the vote. The historical track record of Swiss wine interests, both in our own work and others' works, shows both attempts and successes in influencing alcohol regulation and taxation policies. While prior research hesitates to disambiguate the effects of culture from economic interests, our results suggest a monocausal cultural explanation, or a purely religious one, is insufficient. It would be equally reductive to claim the ban's success was purely driven by economic interests. Instead the ban's enactment was at least in some part, a product of both factors.

Referendum #65 isolates the wine interest from absinthe, because it was wine-salient

without being absinthe-salient: the food act regulated wine as a food, reserved the legislation on distilled spirits and nowhere names absinthe (Schweizerische Bundesversammlung, 1905b, p. 18), and where it bore on wine it bore on artificial wine (*Kunstwein*). On its own coalition specification, column (3) of Appendix Table A.19, the wine revenue share enters at 0.916 (0.221) and cattle per thousand inhabitants, for the meat and dairy side of the act, at 0.065 (0.017); a dummy for cantons that already had food control in 1895, three of which told the federal government their own authority was enough (Schweizerischer Bundesrat, 1896a, pp. 880–882), enters at −2.516 (6.792) and is not detectable. On the specification every vote in the set shares, we read #65 as weakly corroborative and no further.

Temperance carried the petition to the vote, lowering the cost of supporting the ban for voters with a stake in the wine market, particularly farmers. A deputy for the ban invoked farm opinion abroad, citing the Central Union of French Agriculturists' call of 1907 for a total ban on absinthe in France (Schweizerische Bundesversammlung, 1908, National Council, April 2, 1908, p. 47). The wine share is not detectable at the petition, and that is an absence of evidence that wine had a part in the collection, not evidence that it played none. The American record gives a contrast of venues: Munger and Schaller (1997, p. 156) find the drink trade's stake not detectable in the state legislatures' ratifications of 1919 but detectable in the popular votes on repeal in 1933, where the measure of the stake postdates the vote, and they take the change to suggest that producers had effectively organized by then. At the ballot, revealed preference could align cheaply with local economic interest. We read that revealed preference with a caution. Lewis could separate preference from mobilization by counting each side's votes against the eligible voters (Lewis, 2008, p. 383). We cannot use turnout that way, because some cantons fined non-voters in 1908 (Section 3), and compulsion pushes voters to a decision even where they hold no strong preference.

There are alternative explanations suggested by other scholars, which plausibly complement our observed results. For instance, Trechsel writes that, against a murdered family, a few possibly unemployed absinthe distillers and wormwood growers counted for little with the broad public, and the ban's supporters had popular emotion behind them, "the more clearly, the farther away from the absinthe-consuming regions" (Trechsel, 1990, p. 153). He judges that a ban on a spirit the great majority of voters had hardly drunk even once was no gauge of the popular mood about alcohol (Trechsel, 1990, p. 153). We read the gradient he describes as placing the moral catalyst of the petition in temperance areas with no stake in absinthe, and because the ban cost most voters so little, it left room for a canton's economic stake to show at the ballot. That premise rests on the contemporary record as well as on Trechsel's modern judgment: Müller, against the ban, relayed the cantons' reports that there was no appreciable consumption of absinthe liqueur in German-speaking

Switzerland, nor in Ticino and Graubünden (Schweizerische Bundesversammlung, 1908, National Council, April 2, 1908, p. 27). The French-speaking share turns between the stages: it is positive and not detectable at the petition, and at the vote it is negative in every column (Table 4). How precisely it is measured there depends on the absinthe measure beside it. Beside the purchase share its four readings run from .047 to .077: all clear ten percent, and only the HC3 reading clears five. Beside the producer indicator, which marks four of the five French-plurality cantons (correlation 0.77), none clears ten percent; in the focal column they run from .102 to .273. Our reading is that the French-speaking share leaned against the ban at the vote once the wine share is held fixed, but that with twenty-five cantons it cannot be separated precisely from absinthe presence where the two coincide.

Stigler names four main policies that an industry or occupation may seek of the state: a direct subsidy of money, control over entry by new rivals, powers over substitutes and complements, and price-fixing. Of the third he writes, "Crudely put, the butter producers wish to suppress margarine and encourage the production of bread" (Stigler, 1971, pp. 4–6). On our reading, referendum #65 comes close to that example, and on both of its goods. The act's power over mixtures of natural foodstuffs with substitutes reached margarine beside artificial wine (Section 5.3). The commission's other rapporteur said that what the producers aimed at was "in reality, to prevent the manufacture of margarine. They do not hide it" (Schweizerische Bundesversammlung, 1905a, Council of States, June 28, 1905, p. 894). The ban of 1908 is closer to the pairing that Yandle (1999, pp. 5, 7) adds as a footnote to the Stigler–Peltzman theory, in which durable social regulation "evolves when it is demanded by both of two distinctly different groups," one on moral grounds and one that expects to profit from it: here temperance demanded the ban, and the wine-growers were its predicted, indirect beneficiaries. It differs from that statement in two ways that we take up below: the record shows temperance confining its own initiatives to spirits, which left wine untouched, and wine need not have paid. On our reading the absinthe ban took Stigler's substitutes route.

The record explains why one spirit could be singled out as public enemy number one (Section 2.2), and why the temperance movement's concession on fermented drinks was strategic. The temperance movement, while not homogeneous, aimed for what was realistically achievable, yet Trechsel's account states most of the movement's societies strove for nothing less than a nationwide ban on alcohol (Trechsel, 1990, p. 171). Still, both of its popular initiatives, against absinthe in 1908 and for local option in 1921–29, were confined to spirits, although the social-hygiene wing stressed the danger of fermented drinks too (Trechsel, 1990, p. 171). We do not infer that temperance intended a gradual prohibition; we argue only that these attestations are consistent with Peltzman's coalition

theory, in which a majority-seeker spares the would-be losers who can fight back, so that "some losers will then be made winners when there is a rise in opposition" (Peltzman, 1976, p. 221). We have no evidence that wine interests funded the campaign for the ban, and our results do not attest financial collusion of the kind in Yandle's pairing. We point to the convenient shared goal, which served both the wine interests and the temperance actors' aims, gradualist or otherwise.

What counts is relative, as Becker's corollary has it: a group's political effectiveness "is mainly determined not by its absolute efficiency" at producing political pressure "but by its efficiency relative to the efficiency of other groups" (Becker, 1983, p. 380). On our application, the absinthe interest was weak relative to the other groups of the federal polity. However coarsely absinthe is measured, the arithmetic is plain: a handful of producer cantons could not defeat a double-majority vote on their own. A similar case points the same way: Poelmans et al. (2022, pp. 215, 221, 223–224) associate larger banned trades with less time under statewide prohibition across the American states, and read national prohibition's passage partly through brewing's concentration in a few states. Their bans were decided state by state; ours was imposed federally, over the cantons. That level of decision is the focal difference: Neuchâtel and Geneva, which rejected the ban, would presumably have preferred to decide the matter themselves.

Our analysis adds detail to why absinthe fit the scapegoat role that other scholars have posited (Room, 1984, fol. 3; Heimberg, 2000, p. 99; Ben Slimane et al., 2026, p. 751). On our evidence absinthe was vulnerable on two counts, economic and cultural. By contrast, France's petition had passed 400,000 signatures by the spring of 1907, only for the Chamber's hygiene commission to narrowly reject prohibition (Prestwich, 1979, pp. 311–312). Prestwich attributes this to the tax revenue at stake, fifty to sixty million francs a year, to indemnities and to fears of unrest in the Doubs (Prestwich, 1979, pp. 309, 312). France's ban succeeded only by wartime decree (Prestwich, 1979, p. 317) and law in 1915 (Adams, 2004, p. 214), and was then "relatively easy," Prestwich writes, because "the legislation affected only one area of France and compensation was assured" (Prestwich, 1979, p. 318). We read the two-country comparison as a difference in stakes, consistent with Peltzman's winning-coalition framing.

7 Conclusion

At the petition, cantons with more organized temperance (our proxy is the Protestant share) signed at a higher rate per hundred eligible voters, and the wine share is not detectable. At

the vote the two trade places. Among cantons equally French-speaking, a larger share of national wine revenue goes with more support for the ban. Of the other federal ballots, only the food-act referendum of 1906 shows a detectable wine relationship, and only weakly.

The record supplies the motive, the ballot the opportunity: temperance confined its initiative to spirits and spared wine, and once the ban was on the ballot, a vote could follow local economic interest at little cost. One limit is unavoidable: attested absinthe producers sat in five cantons, four of them French-speaking, too few for their coefficient to separate from language.

Our results provide a rare empirical test of the "culprits" of the green fairy's demise. The campaign record shows that temperance led the federal petition, and we find that the wine market predicted support at the vote. In Gantner's caricature the Blue Cross is the killer, and modern popular accounts, writing of France, name wine an accomplice (Hicks, 2010; Clifford, 2023; Adams, 2004, p. 197). The Swiss record fits the looser arrangement that Smith and Yandle (2014, pp. 14–18) call noncooperative, in which economic interests often come in as "latecomers to long-standing moral crusades" (p. 14), a description that matches the federal stage, where temperance organized the initiative and the wine share registers only at the vote. Our evidence, a canton-level predictor and not a confession, sustains the lesser charge, support at the vote, and is insufficient to convict on the graver one, collusion at the petition. That is not an acquittal. Coinciding interests are enough: the two groups in the coalition of Yandle (1983) back a common outcome "for decidedly different reasons" (Yandle, 2022, p. 24), and in its noncooperative form (Smith and Yandle, 2014, pp. 14–18) the coalition requires neither explicit collusion nor funding between wine producers and temperance activists. Others have posited that absinthe served as a scapegoat (Room, 1984, fol. 3; Heimberg, 2000, p. 99; Ben Slimane et al., 2026, p. 751). Scapegoating is not our thesis, but it is one reason the noncooperative arrangement could succeed: banning one spirit was a half-measure that placated a temperance movement whose radicals wanted abstinence rather than moderation, and it left wine, beer and cider untouched. The shared goals of temperance activists and the wine industry, combined with our findings, suggest the absinthe ban was aided by the support of both groups, albeit through different channels. The evidence in this paper points to both groups as responsible for absinthe's demise.

CRediT authorship contribution statement

Nicholas A. Jensen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jacob Smith:** Conceptualization, Data curation, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used AI tools (Claude and ChatGPT) to check Stata, R, Python, and LaTeX code; to locate spelling, translation, and grammar mistakes; and to help draft, revise, and translate text and check citations against their sources. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

We will share the replication package upon acceptance.

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A Appendix

A.1 The historical record

This subsection gives the full record behind Section 2, in that section's order.

The market for drink. The 1884 message also describes how drinking varied across the country:

> "In the zone we have already singled out with respect to alcohol consumption, schnapps holds the mastery as the household drink at the between-meals of the agricultural population; only there and then, where cheap wine or cider is abundantly present as a local product, does it share the dominion with it, or is it pushed into second place." (Schweizerischer Bundesrat, 1884, p. 385)

Industrial alcohol was itself born of a vine blight: when blight struck French vineyards in 1853, distillers turned to beets and cereals as the base for new drinks, and when phylloxera cut the wine crop two decades later that production received "further encouragement" (Marrus, 1974, p. 128). The insect was first found in Switzerland in 1874, in the vine greenhouses of an estate at Pregny near Geneva; the canton, which had taken the necessary powers from its Grand Council two years before, expropriated the owners and ordered the infected vineyards destroyed (Furrer, 1889, s.v. Reblaus, p. 646). By 1877–78 the federal government had undertaken by federal decree to meet not less than a third of cantonal eradication costs, and in September 1878 the international phylloxera convention was concluded at Bern (Furrer, 1889, pp. 653–654). Making up the shortfall in Swiss wine after 1876, the Council reckoned, would have taken "a large quantity of additional import and additional production of spirits" before one could say the population was drinking as much as it had in the good years (Schweizerischer Bundesrat, 1884, p. 382). Opening its section on distilled drinks, it put the same point as a matter of record: these circumstances "encouraged domestic manufacture to supply the increased demand itself as far as possible" (Schweizerischer Bundesrat, 1884, p. 382). The Council had put the fiscal half of the case earlier in the same message: if legislation treats a necessary foodstuff "like a luxury article" and "through corresponding taxes" makes it inaccessible to the worker, it asked, where then does the fault lie when the need for a fortifying drink drives him to the cheaper schnapps (Schweizerischer Bundesrat, 1884, p. 445).

The settlement of 1885–87. Revenue was a requirement of the design, not its measure of success: the yield had to suffice to relieve the undistilled drinks and, until 1890, to make good the entry duties the cantons were losing, while sparing the Confederation's own finances, and at least a tenth of what reached the cantons was reserved for combating alcoholism (Schweizerischer Bundesrat, 1886, pp. 423–424; Schweizerische Bundesversammlung, 1885, pp. 478–479). Two protections bounded the settlement. The distilling of wine, fruit and their residues, gentian and juniper stayed outside federal law on manufacture and taxation, on the Council's reasoning that such spirits were too dear to be much abused (Schweizerischer Bundesrat, 1886, pp. 422, 440), and from December 1887 only for domestic materials (Schweizerischer Bundesrat, 1889a, p. 185); fermented drinks could bear no special cantonal taxes and no restrictions beyond those needed against adulterated or unhealthy drink, the cantons keeping their powers over public houses and retail sales below two liters (Schweizerischer Bundesrat, 1886, p. 422). The 1886 message singled out the export trade in liqueur and absinthe, an industry working predominantly from foreign fine spirit, and its draft provided on export for refunding the tax and half the duty on that input (Schweizerischer Bundesrat, 1886, p. 468). By the 1884 message's first-stated aim, the reform set one of its own standards of success: whether it displaced the drinks it had named as most harmful, or reduced their harm. Before the reform not every canton levied an entry duty on spirits (Schweizerischer Bundesrat, 1886, p. 425). The Council answered the farmers' petition of March 1886 with what it called a first concession to agriculture: distilling in Switzerland would not be suppressed (Schweizerischer Bundesrat, 1886, p. 458). The statute had the Federal Council prefer domestic raw materials and agricultural cooperatives when awarding the distilling lots (Schweizerische Bundesversammlung, 1886, p. 1310), an arrangement the Federal Council still called the compromise of 1886 twelve years later (Schweizerischer Bundesrat, 1898b, pp. 622–623). The Council's draft would have taxed production and made a control meter central to licensing the ordinary taxable distilleries, provisions under which it expected the smallest household stills to disappear (Schweizerischer Bundesrat, 1886, pp. 444, 473). The meter moved into the contracts: the conditions of 1888 placed each contract still under official seal from the point where condensation begins and required a control reservoir under official lock or a control meter (Schweizerischer Bundesrat, 1888, p. 901).

The small distillers. The Council's survey counted 1,022 distilleries that made more than two hectoliters a year from starchy materials and sold the product; the 985 with returns reported about 50,000 hectoliters of absolute alcohol in 1885/86, 937 of them less than 100 hectoliters each, below the smallest lot the law would offer, and the Council treated

most returns as minima and added 15,000 hectoliters for the medium and small works (Schweizerischer Bundesrat, 1886, pp. 425, 428, 435). By 1898 the monopoly bought from seventy contracted suppliers (Schweizerischer Bundesrat, 1899a, p. 330). Most of that gap was closed by buying distillers out. In July 1887 the Federal Council had cantonal officials seal every still that had distilled the newly reserved materials, leaving a few dozen distilleries working under provisional contracts (Schweizerischer Bundesrat, 1889a, p. 123). By the spring of 1890 it had settled 1,171 of 1,376 claims for compensation, not all of them from surveyed distilleries, each owner undertaking, for successors too, never to distill again; and it bought such stills as it could and sold them for scrap (Schweizerischer Bundesrat, 1889a, pp. 185–187; Schweizerischer Bundesrat, 1890a, p. 56; Schweizerischer Bundesrat, 1892a, pp. 811–812). What remained formed a class of its own under the new law, the free distilleries, which might distill fruit, wine or fee-paid materials "without further involvement of the Confederation" (Schweizerischer Bundesrat, 1889a, p. 109). The record counts the distillers who took compensation, names the free distilleries without ever counting them, and sees illicit distilling only where a case was brought; the Council had admitted in 1886 that it could not count the exempt stills either (Schweizerischer Bundesrat, 1886, p. 425).

Supervision and enforcement. The decree of 1890 named ten controllers, charged them first with taking over and controlling the contract distilleries' output and last with penal reports, and drew their nine districts as lists of those distilleries (Schweizerischer Bundesrat, 1890b, pp. 161–164). The free distilleries were the cantons' to supervise (Schweizerische Bundesversammlung, 1886, p. 1312). The Federal Council had to ask the cantons, in 1889 and again in 1892, how they were doing it (Schweizerischer Bundesrat, 1889b, pp. 332–333; Schweizerischer Bundesrat, 1892b, pp. 233–234); parliament's committee found in 1897 that several had not properly carried it out (Nationalrätliche Alkoholkommission, 1897, p. 1016); and some cantonal offices had licensed what the federal law forbade, permits the Board called unlawful (Schweizerischer Bundesrat, 1891, p. 421). Becker and Stigler add that a rule usually tells no one where violations lie and gives no one a reason to find them, and propose paying enforcers by results (Becker and Stigler, 1974, p. 14). The Council's draft had expected taxed distillers to act as natural and unpaid watchers of the exempt ones (Schweizerischer Bundesrat, 1886, p. 481); the monopoly made them suppliers of the Confederation instead. Over 1887–98 the Board received 413 complaints and proceeded on 329, of which 265, about twenty-two a year, ended in a penal order; 241 of the 329 concerned unauthorized distilling of monopoly materials and 28 distilling on premises the Confederation had paid to close (Schweizerischer Bundesrat, 1899a, pp. 301–302,

460–461). Fines paid in fell from Fr. 13,883.45 in 1890 to Fr. 537.78 in 1897 (Schweizerischer Bundesrat, 1899a, p. 462); in 1898, when half the year's cases fell to Solothurn, sixteen penal orders brought in Fr. 3,634.05 (Schweizerischer Bundesrat, 1899a, p. 300; Nationalrätliche Alkoholkommission, 1899, p. 657), though the Council judged the statutory minimum of five times the sum defrauded too high in some cases (Schweizerischer Bundesrat, 1898b, p. 627). At the border, where every lawful import owed its fee, customs fined 553 evasions of the import fee over the same twelve years (Schweizerischer Bundesrat, 1899a, p. 303). The informer's bounty reached people outside the Board in the monopoly's first years and very little afterwards: informers were paid Fr. 3,682.35 directly in 1890 but Fr. 73.40 in 1898 (Schweizerischer Bundesrat, 1899a, p. 462), and at least one was a policeman (Schweizerischer Bundesrat, 1893, p. 341); the fund's occasional payments to police and other outsiders came to a few hundred francs in a year at most (Schweizerischer Bundesrat, 1896b, p. 366; Schweizerischer Bundesrat, 1897a, p. 312; Schweizerischer Bundesrat, 1898a, p. 99). The Board's informers' fund changed its meaning in between: founded in 1887/88 for thirds with no particular informer or refused by one (Schweizerischer Bundesrat, 1889a, p. 196), it also took from 1895 the third on every offense the Board's own officials reported (Schweizerischer Bundesrat, 1895b, p. 185), and the committee explained its receipts for 1896 by the complaints having come, with few exceptions, from the Board's employees (Nationalrätliche Alkoholkommission, 1897, p. 1018). Officials, gendarmes and police were bound to report every offense they found (Schweizerischer Bundesrat, 1888, p. 877), and the general rule gave a third of each collected fine to whoever reported it (Schweizerische Bundesversammlung, 1886, p. 1313). A contract distillery was visible before any offense occurred; production concealed outside the contracts had first to be found, and that was the weak step. The few cases on record are consistent with costly detection, weak incentives or offenders' adaptation; they cannot tell us whether illicit production was large or small. The accounts cannot separate the new rule from the falling number of cases; either way, by the late 1890s the bounty was bringing in few reports from outside. Taken together, the record suggests a low ex ante likelihood that production concealed outside the contracts would be detected and punished. A distiller once found was likely to be punished. The fines collected were small in francs, a total that cannot show the severity of individual penalties: the expected sanction was modest because detection was.

The surplus. A majority of the cantonal delegates expected a revision of the law to increase alcohol revenue rather than reduce it (Nationalrätliche Alkoholkommission, 1899, p. 654), and after the good harvest of 1899 the committee foresaw four or five years of reduced consumption of monopoly spirit and, with the larger population of the 1900

census, a lower quota per head and less money for cantons whose population had stayed level (Nationalrätliche Alkoholkommission, 1900, pp. 734–735). Parliament's committee counted a larger surplus as a success; it welcomed a modest fall in drinking spirit in 1899 on ethical grounds (Nationalrätliche Alkoholkommission, 1900, p. 732), but led with the surplus. "This favorable result," it wrote of the surplus of 1897, "is explained . . . also by increased consumption of spirits subject to the monopoly" (Nationalrätliche Alkoholkommission, 1898, p. 193), an increase it traced not to more drinking but to poor fruit and wine harvests, which had left less monopoly-free spirit, must and wine to drink (Nationalrätliche Alkoholkommission, 1898, p. 191).

The public case. Dr Pallard, whose Vernier speech the *Journal de Genève* reported (Section 2.3), also warned of harm to absinthe drinkers' descendants, a claim from degeneration theory, which Magnan and Fillassier took to the first eugenics congress in 1912 (Journal de Genève, 1908, July 3, p. 4; Prestwich, 1979, p. 306; Luauté, 2007, pp. 524, 530).

The campaign. Hercod's abstinence secretariat served as the initiative committee's general secretariat and took the name *Zentralstelle zur Bekämpfung des Alkoholismus* in 1913 (Trechsel, 1990, pp. 125, 152), the name under which Swissvotes records the initiator (Swissvotes, 1908). On April 7, 1908, in the Council of States, Federal Councilor Ruchet recalled the voters' rejection in 1903 of a revision that would have raised the floor for the free sale of wine and beer from two liters to ten (#63): "there was nothing to do but bow" (Schweizerische Bundesversammlung, 1908, Council of States, April 7, 1908, p. 64; for the date, Schweizerische Bundesversammlung, 1908, National Council, April 6, 1908, p. 85).

A.2 The yearbook wine table

The two wine shares differ by exactly one term. Let $\bar{P}_i = V_i/Q_i$ be the average price implied by canton i's row, and $\bar{P}_{\mathrm{CH}}$ its national counterpart. Then

$$X_{3i} \;=\; X_{2i} \times \frac{\bar{P}_i}{\bar{P}_{\mathrm{CH}}}.$$

The wine revenue share is the volume share scaled by relative price, and that ratio is the only thing separating the two measures. Comparing X_2 against X_3 therefore reads off how much the price term contributes. It is not a comparison of two independently constructed

variables. Wine revenue is concentrated: in the 1907 harvest the five largest wine cantons held 77.7 percent of national wine revenue, Vaud alone 32.1.

V_i records no transaction, and no harvest is sold entire. We nonetheless call it wine revenue throughout (§3.1.3). We avoid "value" because the word already carries its statistical sense in every table.

Alongside the totals, the yearbook reports harvest, value and average price separately for red, white and mixed wine. On the 1907 harvest the three class values sum to the printed total exactly for every canton and for the national row, and each class price equals that class's value divided by its quantity to within print rounding. The average-price column is therefore a quantity-weighted composite of the class prices and carries the canton's grape mix by construction. Grape class is too coarse to stand in for price, and we do not use it that way.

Two printed cells are internally inconsistent, both on negligible quantities: the Geneva mixed-wine price against a nineteen-hectoliter cell, and the Appenzell Outer Rhodes white-wine price against a half-hectoliter cell. Neither enters the shares, which are built from the total columns.

Three different years attach to every wine observation, and Table A.2 separates them. A referendum's measures are built from the last complete harvest preceding its ballot, and that harvest is published in a later volume, whose own imprint year is later again. The citation year is the imprint year, so vote #68 of July 1908 draws on the 1907 harvest, printed in the volume labeled 1908, which was issued in Bern in June 1909 and is cited as 1908. Because the ballot fell in mid-year and wine is the variable of interest, we take the nearest complete harvest rather than extrapolate across a partial year, which would import whatever revenue and yield trends the ban itself set in motion. The rule has one exception. No 1897 yearbook was ever published, so the 1896 harvest first appears in the 1898 edition, and votes #51 and #52 reach two years back rather than one.

One row departs from the one-harvest-per-referendum pattern: the petition arm is measured on the 1906 harvest, the last complete vintage before the signature drive closed, rather than on the vintage preceding the 1908 ballot.

A.3 Verifying the absinthe producers

The Milliet register (§3.1.4) lists forty houses that bought monopoly spirit, not forty absinthe distilleries. We therefore traced each house in the records a firm of the period would leave behind. These are the commercial-register notices of the *Schweizerisches Handelsamtsblatt*,

which state a firm's line of business at registration and at each change of ownership; the trade directory of the canton of Vaud; the federal trademark register; advertisements and court reports in the period press; the firm histories of the Zug and Rigi kirsch trade; and, for the Val-de-Travers, the local historiography of the absinthe industry. We call a house *confirmed* when the sources document it as an absinthe producer, whatever else it also made; *unconfirmed* when they document a working distillery but name no product; and *struck* when they document a house that produced primarily or exclusively another drink, or a merchant with no production of its own. Confirmed and unconfirmed houses stay in; struck houses leave both numerator and denominator. The classification is encoded house by house in the replication code, and the two series of §3.1.4 follow from it: the loose series strikes the five houses documented as kirsch or sparkling-wine producers, and the floor series also strikes the Fribourg merchant. The paragraphs below give, canton by canton, the decisive source for every house the code names and the page on which it appears.

Neuchâtel. Thirteen of the canton's seventeen houses are confirmed, and all thirteen lie in the Val-de-Travers: C. Berger, Couvet; Kübler & Romang, Travers; Giovenni, Bovet & Cie, Sandoz & fils and J. Borel-Pettavel, Môtiers; Dornier-Tüller & Cie, J.-F. von Almen & Cie, Th. Henny fils and J. Ammann, Fleurier; Ed. Pernod S.A., Legler-Pernod and A. Fraissard, Couvet; L.-A. Bolle, Les Verrières. The source is the history of the Kübler house in the Fête de l'absinthe cahier, which ranks the thirteen as the valley's sister distilleries by their 1902–1906 spirit purchases and prints three of the quantities (Fête de l'absinthe, Boveresse, 2002, p. 5). Its list rests on Couleru (1908, p. 244), the contemporary account of the absinthe industry, and on the Kübler family archives. The same chapter reproduces Kübler's partnership deed of December 27, 1875, whose stated object is "la fabrication et de la vente d'extrait d'absinthe, vermouth, liqueurs fines et autres produits analogues" (Fête de l'absinthe, Boveresse, 2002, pp. 4–5), so the Travers house is a documented absinthe producer with a mixed line. The four lakeside houses are not covered by that source. One of them is struck: Bouvier frères of Neuchâtel appears in the catalog of the Swiss section at the 1897 Brussels exhibition as a maker of "Champagne suisse; vins mousseux", exhibiting hors concours as a jury member (Exposition internationale de Bruxelles 1897, Section suisse, 1897, pp. 18, 57), consistent with its export ratio of 81 percent and not with an absinthe still. The Distillerie d'Auvernier, E. Haller fils of Neuchâtel and W. Dubois & Cie of Cormondrèche returned no registration or press notice in the archives searched and are unconfirmed.

Geneva. Eight of the canton's eleven houses are confirmed. Seven are confirmed by the Board's own document: the vermouth table of the Milliet annex cross-lists L. Guillemaud, Petit-Saconnex; Vve Mirault & Cie, Moillesulaz; A. Floquet, Chêne-Bourg; Muraour & Cie, J. Demont and Bouvier & Fechoz, Geneva; and Vve J. Sauter fils, Plainpalais, each with the note "vide Absinth" (Milliet, 1907, p. 363). The eighth, J. Laverrière & Cie of Chêne-Bourg, is not in that table; it registered federal trademark no. 2337 on July 11, 1888, as "distillateurs-liquoristes, Chêne-Bourg. Absinthe et vermouths" (Schweizerisches Handelsamtsblatt, 1888, Bd. 6, p. 653). Two of the seven also attest absinthe in their own registrations: Mirault et Cie registered in 1903 as "fabrique de vermouths et d'absinthe; exportation", and its successor Vve Mirault et Cie in 1906 as "fabrique de vermouth et d'absinthe et commerce de vins" (Feuille officielle suisse du commerce, 1903, 1906); J. Sauter fils was registered from 1902 as "distillerie d'absinthe, fabrique de vermouth, liqueurs et sirops" (Feuille officielle suisse du commerce, 1911). The Geneva police court corroborates two more: it tried Muraour and Bouvier & Fechoz on July 2, 1908, as manufacturers of "l'achinte", the substitute sold after the cantonal ban (Journal de Genève, 1908, p. 5). For L. Guillemaud, A. Floquet and J. Demont the Board's own record is the only attestation. They stay in under the rule that houses making both vermouth and absinthe count: economies of scope made the combination common, and three Geneva houses document both products in their own registrations or trademark. This is a judgment of plausibility, not verification by an independent outside source. J. Crépieux and J. Bonnard returned no registration or press notice and are unconfirmed; L. Magnin has no recorded purchases and carries no weight in any series.

Vaud. One house is confirmed and three are unconfirmed. C.-A. Braillard of Yverdon advertised "Absinthe . . . distillation lente" in 1901 (Le Conteur Vaudois, 1901, vol. 39, no. 31, p. 4). Weber frères and F. Dæppen-Pache of Lausanne and Vve E. Gamboni & Cie of Morges are registered distillers. The 1905 *Indicateur vaudois* lists Weber frères and Dæppen among the five Lausanne distillers and again among its liquoristes, and gives Dæppen-Pache's trade as distillateur-liquoriste in its alphabetical section (Indicateur vaudois, 1905, pp. 289, 313 and 65). It lists Vve Gamboni & Cie as one of the two Morges distillers (Indicateur vaudois, 1905, p. 125). The 1908 edition lists both Lausanne houses again, Dæppen at the same address and Weber frères still on the place Chauderon (Indicateur vaudois, 1908, p. 312). The directory prints a specialty line for distillers who declare one, and none of the three does, so their product stays unnamed and they are kept as unconfirmed.

Basel-Stadt. Weil frères is confirmed: its 1883 registration gives the business as "Distillerie und Weinhandel. (Spezialität in Kirschwasser und Absinth.)" at Hebelstrasse 21 (Schweizerisches Handelsamtsblatt, 1883b, Bd. 1, H. 2, S. 17). Vve Louis Mayer & Cie is unconfirmed: the firm registered in 1883 as "Sprit und Branntwein en gros", a spirits wholesaler (Schweizerisches Handelsamtsblatt, 1883a, Bd. 1), and re-registered in 1896 as "Wein und Spirituosen en gros. Dampfbrennerei", adding a steam distillery (Schweizerisches Handelsamtsblatt, 1896, Bd. 14, H. 253). Production is documented at the second date; the product never is.

Schwyz and Zug. We strike all four houses as kirsch distillers. The Eichhorn distillery at Arth, founded in 1840, is recorded as the first commercial cherry-brandy distillery in the canton of Schwyz, and Emil Eichhorn's postcard of 1901 advertises "Genuine Rigi Cherry Brandy" (Zuger Rigi Kirsch, n.d., entry 4). Franz Holzgang opened his Küssnacht distillery in 1868 after an apprenticeship at a Zug kirsch house, and the firm's line was Rigi kirsch and Rigi bitter (Zuger Rigi Kirsch, n.d., entry 16). C. Landtwing of Zug is the cherry-brandy distillery that Carl Landtwing-Moos took over from Schmidt in 1902 (Kleeb, 2017), and the Cherry Brandy Society is named for its product. The Historical Dictionary of Switzerland names Zug kirsch and Neuchâtel absinthe as the two spirits that became export industries in the nineteenth century (Dubler and Maurer, 1999).

Valais. Louis Morand of Martigny is confirmed. The period record holds the registration of Morand frères as a distillery in 1888 (Feuille officielle suisse du commerce, 1888, Bd. 6, p. 1004), the 1890 succession notice (Schweizerisches Handelsamtsblatt, 1890, Bd. 8, p. 353), and an advertisement of 1895 in which the house calls itself a "distillerie à vapeur d'extrait d'absinthe suisse et de liqueurs fines" (Indicateur vaudois, 1895, p. 50). Its showcase was a herbal liqueur, the Liqueur du Simplon: the official list of the 1896 national exhibition gave the house a bronze medal in the liqueurs class (Exposition nationale suisse, Genève 1896, 1896, pp. 196–198), which the *Journal de Genève* reported in 1897 to have been a clerical error for silver (Journal de Genève, 1897, p. 2). The firm's own history says its sales before the ban were mostly absinthe (Cornaz Bays, 2020, p. 6), and its pear brandy, the Williamine, was registered as a trademark in 1953 (Cornaz Bays, 2020, p. 26).

Fribourg. Jean Pellet of Murten is the canton's only listed house. The house registered on January 29, 1883, as "vins en gros", and the gazette entry of September 15, 1910, made on his death, strikes "J. Pellet, vins en gros" and registers his heirs' firm as "vins et spiritueux en gros", a wine and spirits wholesaler, with no distilling apparatus recorded at either

date (Schweizerisches Handelsamtsblatt, 1910, Bd. 28, H. 240, S. 1642). Because the sources document trade rather than production of another spirit, he falls under the merchant rule rather than the strike rule: he is retained in the loose series and struck at the floor, the only house on which the two series differ, at 1.2 percent of the loose denominator.

Construction. The register's purchases sum to 5,402,342 kg and its refund-eligible exports to 320,553 kg. The export column is not used as a regressor: one Couvet house accounts for 79 percent of it. The loose series strikes Holzgang, Eichhorn, Landtwing, the Cherry Brandy Society and Bouvier frères, 265,688 kg or 4.9 percent of the register, leaving 5,136,654 kg; the floor series also strikes Pellet, 327,594 kg or 6.1 percent in all, leaving 5,074,748 kg. For each series the continuous measure is

$$\text{Abs}_i \;=\; 100 \times \frac{\sum_{f \in i} \text{purchases}_f}{\sum_f \text{purchases}_f}, \qquad f \text{ over retained houses}$$

and the indicator $\mathbf{1}\{\text{Abs}_i > 0\}$ marks the cantons with a positive share. Under the floor series the shares are Neuchâtel 62.4, Geneva 19.8, Vaud 10.9, Basel-Stadt 6.0 and Valais 1.0 percent; the loose series moves these to 61.6, 19.5, 10.8, 5.9 and 1.0 and gives Fribourg 1.2, so the loose indicator adds Fribourg to the five floor cantons. The uncorrected register share enters only the appendix table that re-estimates the main specification on the register as printed (Appendix Table A.7). For houses that also made vermouth, kirsch or liqueurs, purchases overstate absinthe-specific input; every documented absinthe producer counts as an absinthe interest whatever its product mix, and the two series bracket what the sources leave open. The main tables report both encodings but never enter them in the same column.

The Board itself warned that the purchase figures give "keinen sichern Anhalt über die Grösse der Absinthproduktion" (Milliet, 1907, p. 361). The verification removes the houses that demonstrably were not absinthe producers. The purchase register is nonetheless the closest measure linking high-proof inputs to the distilled spirit of interest. It was also the evidence the Federal Council and the Federal Assembly had before them, so in using it we work from the same data the coalitions of 1908 had to read.

A.4 Data sources

Vote records. We extract from Swissvotes the canton-level yes counts, no counts, turnout, and eligible voters for vote #68 (the absinthe ban; Swissvotes 1908), along with the other ref-

erenda that serve as falsification and corroboration measures (Swissvotes 1897b,a, 1900a,b, 1902, 1903b,a, 1905, 1906). The Federal Chancellery's archive (Schweizerische Bundeskanzlei, 1908a,b), against which the canton-level outcome totals are cross-validated, paginates the official Erwahrungsbeschluss to *Bundesblatt* 1908 Band IV, p. 572 (Schweizerischer Bundesrat, 1908).

Compulsory voting. Zürich (law of 1869), St. Gallen (duty from 1835, fine from 1867), Thurgau (1870/71), Aargau (1871) and Schaffhausen (constitutional duty 1876, fine under the electoral law of 1904) fined non-voters in 1908, though enforcement was uneven; Vaud's better-known regime dates only from 1925. See Vutkovich (1906, pp. 21–28), Arnold (1929, pp. 27–32), Kley (1989, pp. 162–168) and Bechtel et al. (2018, App. Table 6). Schwegler (2009, pp. 5–8) dates Aargau from its 1937 statute, but Lambelet (1893), as cited in Kley (1989, p. 167, n. 2), already lists these same five cantons.

Legal forms. A popular initiative reaches the ballot once the required signatures are filed and validated; the Federal Council may recommend rejection, as it did for the absinthe initiative, but cannot decline to hold the vote. In an optional referendum only the ballot is optional: the law takes effect unless citizens demand a vote, and once they do, the vote binds exactly as the other forms do.

The petition. The petition measure has no counterpart elsewhere in the set: the popular initiative dated only from 1891, and vote #68 is the only one of these ballots for which a canton-level signature record survives. The canton figures of the petition numerator (Schweizerischer Bundesrat, 1907a, p. 984) reconcile against the printed source: each of the three columns sums exactly to its own printed national total.

The census shares. A two-language denominator for the French-speaking share would omit the language most of Ticino speaks and put the canton at 11.3 percent instead of 0.3, and it would move Graubünden with its Romansh population for reasons unrelated to how French the canton is. Graubünden's largest language group is German, at 46.7 percent. The two cultural controls are built on different denominators because their excluded categories differ in kind rather than in size. Ticino is predominantly Italian-speaking, so omitting Italian speakers from the language denominator would misdescribe the canton; no canton is predominantly Jewish, Eastern Orthodox or unaffiliated, so omitting those categories from the Protestant share misdescribes none. The largest share held by any other group

anywhere is 2.3 percent, in Geneva, and entering these groups separately would consume degrees of freedom already binding at $N = 25$.

The Protestant share. The Blue Cross's confessional pattern (§3.2.2) is not one organization's idiosyncrasy. The Good Templar lodges likewise stood exclusively in Protestant or religiously mixed areas (Trechsel, 1990, p. 68). Nor was the boundary merely observed: as late as 1911 the Catholic Abstinent League was advising Catholics against joining the Good Templars (Trechsel, 1990, p. 72). Catholic temperance was not absent. The Catholic Abstinent League campaigned for the absinthe initiative, the Katholikentag declared for a ban in 1906, and the Swiss bishops followed in 1908 (Trechsel, 1990, pp. 105, 107, 152).

Population density. Land area is recovered exactly, not approximately: the published density was formed as population over a whole number of square kilometers, so every canton returns an integer area. The land area excludes lakes: the twenty-five cantonal areas sum to 39,934 square kilometers against a national 41,285, the difference being lake surface.

Table A.1: Federal referenda in the analysis, 1897–1908

Vote	Date	Object	Topic	Legal form	*N*	Arm
#51	11.07.1897	Bundesbeschluss über die Revision des Art. 24 der Bundesverfassung *Federal Decree on the revision of Art. 24 of the Federal Constitution*	Forestry; environmental policy	Obligatorisch *Mandatory*	24	Falsification
#52	11.07.1897	Bundesbeschluss betreffend Bundesgesetzgebung über den Verkehr mit Nahrungs- und Genussmitteln *Federal Decree on federal legislation over trade in foodstuffs and consumables*	Consumer protection; federalism; health policy	Obligatorisch *Mandatory*	24	Corroborative
#57	04.11.1900	Volksinitiative 'für die Proporzwahl des Nationalrates' *Popular initiative for proportional election of the National Council*	Electoral system; parliament	Volksinitiative *Popular initiative*	23	Falsification
#58	04.11.1900	Volksinitiative 'für die Volkswahl des Bundesrates und die Vermehrung der Mitgliederzahl' *Popular initiative for popular election of the Federal Council and an increase in its membership*	Government; electoral system; popular rights	Volksinitiative *Popular initiative*	23	Falsification
#59	23.11.1902	Bundesbeschluss betreffend die Unterstützung der öffentlichen Primarschule durch den Bund *Federal Decree on federal support for public primary schools*	Schools; federalism	Obligatorisch *Mandatory*	25	Falsification
#62	25.10.1903	Volksinitiative 'für die Wahl des Nationalrates aufgrund der Schweizer Wohnbevölkerung' *Popular initiative to elect the National Council on the basis of the Swiss resident population*	Electoral system; parliament	Volksinitiative *Popular initiative*	25	Falsification
#63	25.10.1903	Bundesbeschluss betreffend Abänderung des Art. 32bis der Bundesverfassung *Federal Decree amending Art. 32bis of the Federal Constitution*	Addictive substances; hospitality; crop production	Obligatorisch *Mandatory*	25	Corroborative
#64	19.03.1905	Bundesbeschluss betreffend Revision des Art. 64 der Bundesverfassung (Ausdehnung des Erfinderschutzes) *Federal Decree revising Art. 64 of the Federal Constitution, extending inventor protection*	Competition policy	Obligatorisch *Mandatory*	25	Falsification
#65	10.06.1906	Bundesgesetz betreffend den Verkehr mit Lebensmitteln und Gebrauchsgegenständen *Federal Act on trade in foodstuffs and articles of use*	Consumer protection; federalism; health policy	Fakultativ *Optional*	25	Corroborative
#68	**05.07.1908**	**Volksinitiative 'für ein Absinthverbot'** *Popular initiative for a ban on absinthe*	Addictive substances	**Volksinitiative** *Popular initiative*	**25**	**Focal**

Notes. Vote numbers, dates, official titles, legal forms and topics are from Swissvotes (Année Politique Suisse, University of Bern). The Object column gives the official German title with an English rendering beneath it; the German is authoritative. *Legal form* is the official *Rechtsform*: an *obligatorisches Referendum* is a constitutional amendment originating in the Federal Assembly and a *Volksinitiative* one proposed by citizens, and both require a double majority of voters nationally and of cantons; a *fakultatives Referendum* concerns ordinary legislation and requires only the popular majority. N is the estimation sample. Vote data are complete for all twenty-five cantons throughout; coverage varies only where the wine series has no observation for the harvest vintage in question, namely Thurgau in 1897 and Thurgau with Basel-Landschaft in 1900.

Table A.2: Wine-data source for each referendum, 1897–1908

Referendum	Vote	Harvest	Source (edition, page)
#51	11.07.1897	1896	*Jahrbuch* 1898, pp. 84–85 (BFS, 1898)
#52	11.07.1897	1896	*Jahrbuch* 1898, pp. 84–85 (BFS, 1898)
#57	04.11.1900	1899	*Jahrbuch* 1900, pp. 52–53 (BFS, 1900)
#58	04.11.1900	1899	*Jahrbuch* 1900, pp. 52–53 (BFS, 1900)
#59	23.11.1902	1901	*Jahrbuch* 1902, p. 65 (BFS, 1902)
#62	25.10.1903	1902	*Jahrbuch* 1903, p. 69 (BFS, 1904)
#63	25.10.1903	1902	*Jahrbuch* 1903, p. 69 (BFS, 1904)
#64	19.03.1905	1904	*Jahrbuch* 1905, p. 72 (BFS, 1906)
#65	10.06.1906	1905	*Jahrbuch* 1906, p. 67 (BFS, 1907)
Petition	1906–07 drive	1906	*Jahrbuch* 1907, p. 55 (BFS, 1908b)
#68	**05.07.1908**	**1907**	*Jahrbuch* 1908, p. 51 (BFS, 1909)

Notes. Each row gives the source of the wine measures for one referendum. *Vote* is the ballot date, *harvest* the vintage the measure is computed from, and the source column names the yearbook edition carrying that harvest together with the printed folio of its canton wine-harvest table (*Darstellung der schweizerischen Weinernte*). Citation years are the volumes' imprint years and differ from the edition labels; see §A.2 for the construction and for the vintage rule.

Table A.3: Canton composition: language, religion, wine, absinthe and scale

	Language, 1900 census (percent of canton population)						Share of national total (percent)				Population, 1907	
Canton	French	German	Italian	Romansh	Other	Protestant	Wine revenue	Wine yield	Absinthe	Producer	Persons	Per km^2
Aargau	0.4	98.3	1.2	0.0	0.1	55.6	3.7	3.9	0.0	No	213,582	153
Appenzell AR	0.1	98.7	1.0	0.1	0.1	90.2	0.0	0.0	0.0	No	55,926	230
Appenzell IR	0.1	99.4	0.5	0.1	0.0	6.2	0.0	0.0	0.0	No	13,835	81
Basel-Land	0.9	96.9	2.1	0.0	0.0	77.2	1.2	1.1	0.0	No	72,092	168
Basel-Stadt	2.3	95.1	2.1	0.1	0.4	66.3	0.2	0.1	6.0	Yes	129,880	3510
Bern	16.6	82.0	1.2	0.0	0.2	86.3	2.2	2.3	0.0	No	620,321	92
Fribourg	68.3	30.3	1.3	0.0	0.1	15.1	0.8	0.8	0.0	No	132,777	83
Geneva	82.8	10.1	5.5	0.1	1.6	48.2	2.8	2.6	19.8	Yes	149,172	606
Glarus	0.2	98.3	1.1	0.4	0.1	75.5	0.0	0.0	0.0	No	31,539	47
Graubünden	0.5	46.7	16.8	34.9	1.2	52.9	1.2	1.0	0.0	No	108,951	15
Lucerne	0.5	97.8	1.5	0.0	0.1	8.3	0.0	0.0	0.0	No	152,641	107
Neuchâtel	82.8	14.0	2.9	0.0	0.3	85.8	5.4	5.6	62.4	Yes	134,360	188
Nidwalden	0.2	97.5	2.2	0.1	0.0	1.3	0.0	0.0	0.0	No	13,360	56
Obwalden	0.2	98.0	1.7	0.1	0.0	1.6	0.0	0.0	0.0	No	15,379	32
Schaffhausen	0.6	97.1	2.1	0.0	0.1	82.1	5.3	5.6	0.0	No	43,561	146
Schwyz	0.5	97.2	2.0	0.2	0.1	3.3	0.1	0.1	0.0	No	58,170	69
Solothurn	1.9	97.2	0.8	0.0	0.1	30.9	0.1	0.1	0.0	No	109,070	138
St. Gallen	0.3	97.2	2.1	0.2	0.2	39.7	2.4	1.6	0.0	No	262,418	134
Thurgau	0.3	97.9	1.6	0.1	0.1	68.3	2.3	2.4	0.0	No	117,908	137
Ticino	0.3	2.3	97.2	0.1	0.1	1.6	6.2	11.5	0.0	No	145,162	53
Uri	0.1	94.8	4.8	0.2	0.0	3.9	0.0	0.0	0.0	No	21,043	20
Valais	65.2	30.0	4.8	0.0	0.0	1.4	19.5	19.8	1.0	Yes	118,856	23
Vaud	86.5	8.7	3.8	0.0	1.0	86.8	32.1	25.6	10.9	Yes	303,139	108
Zug	0.6	95.8	3.3	0.1	0.2	6.8	0.0	0.0	0.0	No	26,225	127
Zürich	0.9	95.8	2.6	0.1	0.5	81.1	14.5	15.9	0.0	No	475,162	287

One row per canton, N = 25, the 1908 federal universe; Bern includes the Jura. The language columns are the five 1900 census counts (HSSO B.32) over the canton's 1900 resident population (HSSO B.01a), and sum to 100 across each row; plurality is the largest of the five. Ticino is the only Italian-plurality canton (97.2 percent Italian against 0.29 percent French), and Romansh is material only in Graubünden (34.9 percent); 19 cantons are German-plurality and 5 French-plurality. The Protestant column is Protestants as a percent of Protestants plus Catholics (HSSO B.27, 1900). Wine revenue and wine yield are the canton's shares of the national 1907 harvest by wine revenue and by physical output, the same objects as the corresponding rows of Table 1; absinthe is the canton's share of verified 1902–06 purchases of 95-degree spirit (Milliet 1907 annex) after the record-verified exclusion of non-absinthe firms, and Producer is the indicator for attested-producer cantons. Each of those three share columns sums to 100 down the table. Density is the 1907 population over the fixed 1900-census land area.

A.5 Supplementary results for the 1908 vote (#68)

A.5.1 Estimator diagnostics

Table A.4 reports one specification test for each estimator, and the two are not comparable with each other, since they add different powers and are referred to different distributions: the Ramsey RESET for the linear model (Ramsey, 1969) and its Lagrange-multiplier analog for fractional response (Ramalho et al., 2011). Neither rejects at any vote. At twenty-three to twenty-five observations that is weak evidence, so we settle functional form on two counts instead: no fitted value leaves the bounds on any referendum, and the two estimators agree throughout.

Table A.4: Estimator diagnostics and the fractional-logit inference record, by vote

Referendum	HC3 p	Bell–McCaffrey p	Freedman–Lane p (AME)	Finite draws	Out-of-bounds	RESET (linear)	RESET (fract.)
#51‡	0.611	0.576	0.628	553,997	0	0.193	0.244
#52‡	0.331	0.306	0.325	958,566	0	0.171	0.117
#57	0.243	0.239	0.211	1,000,000	0	0.832	0.390
#58	0.394	0.382	0.302	1,000,000	0	0.542	0.700
#59	0.901	0.883	0.966	276,947	0	0.849	0.389
#62	0.636	0.605	0.464	87,367	0	0.369	0.260
#63	0.712	0.636	0.512	1,000,000	0	0.697	0.279
#64	0.801	0.652	0.806	428,459	0	0.927	0.255
#65§	0.100	0.106	0.019	560,150	0	0.711	0.573
#68¶	0.006	0.025	0.027	1,000,000	0	0.492	0.640

Notes: Estimator diagnostics and the inference record, by vote. Referendum rows are estimated at the Table 2 column (3) specification. Coefficients are in Table A.13, on the same per-vote measure basis as these p-values; standard errors are in Table A.14, which is volume basis throughout and so matches these readings only for the two votes marked ‡. OOB counts out-of-bounds fitted values from the linear model (zero for every vote). RESET (linear) is the Ramsey test; RESET (fractional) is the Ramalho LM test with powers 2–3. The final columns report the fractional-logit average-marginal-effect Freedman–Lane permutation p-value and the number of completed draws, out of 1,000,000; a p computed on surviving draws is computed on a selected sample. ‡ #51 and #52 are wine volume share only (no value series exists). § #65 has its tailored coalition model in Table A.19, naive model here. ¶ #68, the treatment vote: inference columns at column (3) like the referendum rows, out-of-bounds and both RESETs at column (4). The absinthe conditioner rests on records covering 1902–06. To maximize comparability of the other referenda, the referenda set runs at column (3); #68's specification diagnostics are reported at column (4).

A.5.2 Methods comparison

Table A.5 sets the wine coefficient under both estimators and both wine measures side by side, for description only.

Table A.5: Functional-form and measure invariance: the wine coefficient under two estimators and two wine measures (descriptive)

	Raw	+ French share	+ Protestant share	+ producer (binary)	+ population density	+ purchase share	+ population density
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Ordinary least squares, revenue share							
Wine coefficient	–0.196	0.547	0.639	0.677	0.559	0.514	0.439
HC3 *p*	0.379	0.044	0.006	0.005	0.083	0.093	0.180
Bell–McCaffrey *p*	0.385	0.072	0.025	0.014	0.078	0.078	0.139
Wild bootstrap-*t* *p*	0.435	0.036	0.031	0.032	0.147	0.029	0.105
Freedman–Lane *p*	0.325	0.059	0.016	0.012	0.072	0.079	0.134
Effective D.O.F.	2.04	3.58	3.74	3.68	3.81	4.08	4.40
B. Ordinary least squares, volume share							
Wine coefficient	–0.165	0.548	0.608	0.644	0.515	0.481	0.408
HC3 *p*	0.553	0.037	0.029	0.026	0.173	0.093	0.193
Bell–McCaffrey *p*	0.537	0.053	0.040	0.028	0.131	0.083	0.156
Wild bootstrap-*t* *p*	0.498	0.053	0.047	0.037	0.199	0.040	0.128
Freedman–Lane *p*	0.520	0.055	0.041	0.032	0.138	0.081	0.150
Effective D.O.F.	3.17	4.96	5.00	4.91	4.63	4.87	5.04
C. Fractional-logit average marginal effect (pp)							
Revenue share	–0.192	0.513	0.595	0.634	0.514	0.486	0.408
Freedman–Lane *p*	0.532	0.060	0.027	0.023	0.095	0.099	0.172
Volume share	–0.163	0.528	0.585	0.623	0.487	0.468	0.389
Freedman–Lane *p*	0.640	0.078	0.049	0.043	0.143	0.131	0.215

Notes: This table reports each specification under two estimators and two wine measures: ordinary least squares, and fractional-logit average marginal effects, each on the revenue-share and volume-share measures. Stars unshown; each p-value row names its scheme, for comparison across calculation schemes. Effective degrees of freedom are reported for the least-squares rows; average-marginal-effect rows are descriptive. Permutation tests draw 1,000,000 replications throughout. A fractional-logit permutation p-value is shown only where the refit returned a finite average marginal effect on every draw; for referendum 68 it did so on all 1,000,000, in each column and on both wine measures. Per-vote completion across the battery is in Appendix Table A.4.

A.5.3 Volume (yield) results

Table A.6: Wine volume (yield) share and support for the 1908 absinthe ban

	Raw	+ French share	+ Protestant share	+ producer (binary)	+ population density	+ purchase share	+ population density
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Wine volume share	–0.165	0.548**	0.608**	0.644**	0.515	0.481*	0.408
	(0.273)	(0.247)	(0.260)	(0.268)	(0.364)	(0.272)	(0.302)
French share		–0.304***	–0.298***	–0.257*	–0.289	–0.227**	–0.221**
		(0.080)	(0.072)	(0.147)	(0.254)	(0.092)	(0.089)
Protestant share			–0.074	–0.070	–0.038	–0.052	–0.020
			(0.049)	(0.049)	(0.080)	(0.052)	(0.067)
Log density					–2.371		–2.166
					(3.711)		(2.696)
Absinthe (corrected share)						–0.233	–0.214
						(0.267)	(0.203)
Absinthe (attested producer)				–4.640	0.318		
				(11.036)	(20.142)		
Constant	65.00	67.17	70.04	69.93	79.68	69.35	78.21
	(2.64)	(2.17)	(2.97)	(2.99)	(14.56)	(2.85)	(11.25)
R^2	0.01	0.50	0.55	0.56	0.58	0.58	0.61
Adjusted R^2	–0.03	0.45	0.48	0.47	0.47	0.49	0.51
AIC	197	182	181	182	183	181	181
BIC	199	185	186	189	190	187	189
Small-sample diagnostics for wine coefficient (p-values)							
HC3 robust	0.553	0.037	0.029	0.026	0.173	0.093	0.193
Bell–McCaffrey	0.537	0.053	0.040	0.028	0.131	0.083	0.156
Wild bootstrap-t	0.498	0.053	0.047	0.037	0.199	0.040	0.128
Freedman–Lane	0.520	0.055	0.041	0.032	0.138	0.081	0.150
Bell–McCaffrey effective D.O.F.	3.17	4.96	5.00	4.91	4.63	4.87	5.04

Notes: Ordinary least squares; $N = 25$ cantons. Wine is the canton's share of national wine volume (yield, ×100), under the same vintage rule as Table 2 (1907 harvest, the last complete before the vote). This table establishes measurement robustness for Table 2: the headline cascade re-estimated on the volume measure. *, **, and *** mark $p < 0.10$, 0.05, and 0.01, on heteroskedasticity-robust (HC3) p-values; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell–McCaffrey, wild bootstrap-t HC3-studentized, Freedman–Lane permutation).

A.5.4 The uncorrected producer register

Table A.7: Wine interests and the 1908 absinthe ban — raw (uncorrected) purchase-share back matter

	Raw (1)	+ French (2)	+ Protestant (3)	+ Density (4)	+ Absinthe (5a)	+ Absinthe (5b)
Wine revenue share	–0.196	0.547**	0.639***	0.552**	0.438	0.539*
	(0.219)	(0.257)	(0.207)	(0.253)	(0.315)	(0.267)
French share		–0.316***	–0.315***	–0.302***	–0.241**	–0.346***
		(0.087)	(0.073)	(0.079)	(0.113)	(0.112)
Protestant share			–0.083*	–0.048	–0.031	–0.026
			(0.046)	(0.064)	(0.067)	(0.077)
Log density				–2.184	–2.073	–2.891
				(2.657)	(2.647)	(2.682)
Absinthe (share)					–0.197	
					(0.235)	
Absinthe (producer)						4.103
						(7.326)
Constant	65.13	67.38	70.56	79.36	78.39	81.21
	(2.58)	(2.08)	(2.77)	(11.03)	(10.95)	(10.57)
R^2	0.02	0.51	0.56	0.60	0.62	0.61
Adjusted R^2	–0.03	0.46	0.50	0.52	0.52	0.50
AIC	196	181	180	180	181	181
BIC	199	185	185	186	188	189
Small-sample diagnostics for wine coefficient						
HC3 robust p	0.379	0.044	0.006	0.041	0.181	0.058
Bell–McCaffrey p	0.385	0.072	0.025	0.059	0.141	0.070
Wild bootstrap-t p (HC3-studentized)	0.435	0.036	0.031	0.068	0.109	0.085
Freedman–Lane p	0.325	0.059	0.016	0.046	0.135	0.055
Bell–McCaffrey effective D.O.F.	2.04	3.58	3.74	4.10	4.42	4.15

Notes: Raw (uncorrected) spirits purchase share. This encoding sweeps in kirsch distillers, a sparkling-wine house and a wine and spirits wholesaler; the corrected producer memberships appear in Appendix Table A.8. Alternative wine-industry encodings are never entered together (see the Table 2 note). *, **, and *** mark p < 0.10, 0.05, and 0.01, on heteroskedasticity-robust (HC3) p-values.

Table A.8: Wine revenue and the 1908 absinthe ban — bounds appendix (loose producer membership)

	Raw (1)	+Fr. (2)	+Prot. (3)	+Disc. (4)^{L}	+Dens. (5)^{L}	+Cont. (6)^{L}	+Dens. (7)^{L}
Wine revenue share	–0.196	0.547**	0.639***	0.648***	0.506	0.513*	0.438
	(0.219)	(0.257)	(0.207)	(0.205)	(0.294)	(0.294)	(0.318)
French share		–0.316***	–0.315***	–0.360	–0.496*	–0.249**	–0.240**
		(0.087)	(0.073)	(0.446)	(0.246)	(0.118)	(0.114)
Protestant share			–0.083*	–0.083	–0.012	–0.062	–0.030
			(0.046)	(0.059)	(0.065)	(0.051)	(0.068)
Absinthe producer (attested, loose)				3.573	16.481		
				(34.196)	(18.689)		
Milliet absinthe purchase share (corrected, loose)						–0.202	–0.189
						(0.299)	(0.239)
Log population density					–4.397		–2.088
					(2.718)		(2.664)
Constant	65.13	67.38	70.56	70.41	87.60	69.89	78.36
	(2.58)	(2.08)	(2.77)	(2.93)	(11.25)	(2.72)	(11.00)
R^2	0.02	0.51	0.56	0.57	0.65	0.59	0.62
Adjusted R^2	–0.03	0.46	0.50	0.48	0.56	0.50	0.52
AIC	196	181	180	182	179	181	181
BIC	199	185	185	188	186	187	188
Small-sample diagnostics for wine coefficient							
HC3 robust p				0.005	0.101	0.097	0.184
Bell–McCaffrey p				0.022	0.099	0.081	0.143
Wild bootstrap-t p (HC3-studentized)				0.029	0.152	0.036	0.109
Freedman–Lane p				0.014	0.083	0.081	0.136
Bell–McCaffrey effective D.O.F.				3.72	3.96	4.12	4.44

Notes: ^{L} marks columns (4)–(7), which use the loose producer membership. Floor membership: Neuchâtel, Geneva, Vaud, Basel-Stadt and Valais, the cantons with houses attested in the purchase register as absinthe producers. The loose set adds Fribourg, whose one listed house, a wine and spirits wholesaler, is retained under the merchant rule and struck at the floor (Appendix A.3); the kirsch and sparkling-wine houses are struck in both series. *, **, and *** mark p < 0.10, 0.05, and 0.01, on heteroskedasticity-robust (HC3) p-values.

A.5.5 Subsample diagnostics

Table A.9: The wine revenue-share effect in the non-French subsample ($N = 20$ cantons, cov. French $< 50\%$)

	Bivariate (1)	+ Prot. (2)	+ Prot., dens. (3)
Wine revenue share	0.614	0.879	0.908
	(0.304)	(0.385)	(0.550)
N	20	20	20
R^2	0.06	0.14	0.21
OLS fitted-value range	[66.6, 75.5]	[62.3, 75.7]	[56.4, 75.0]
shares in $[0, 100]$	20/20	20/20	20/20
Wine-coefficient inference			
HC3 robust p	0.06	0.04	0.12
Bell–McCaffrey effective D.O.F.	1.90	2.00	2.02
Bell–McCaffrey p	0.17	0.13	0.15
Wild bootstrap-t p (HC3-studentized)	0.12	0.07	0.23
Freedman–Lane p (10^6)	0.08	0.06	0.12
Leave-one-canton-out range	[0.494, 0.759]	[0.768, 1.112]	[0.702, 1.263]
(widest swing when dropping)	NW	TI	BS

Notes: Non-French-speaking subsample (N = 20), ordinary least squares. This exhibit is read for sign and existence only: with effective degrees of freedom of 2.02 in the restricted columns, significance testing is deferred to the full-sample battery. The quoted cell is the bivariate estimate, 0.614, with no controls.

A.5.6 Pairwise correlations of the regressors

Table A.10: Pairwise correlations among the Table 2 regressors

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Wine revenue share	1					
(2) French share	0.56	1				
(3) Protestant share	0.25	0.15	1			
(4) Producer (attested)	0.54	0.77	0.21	1		
(5) Absinthe share (corrected)	0.16	0.63	0.31	0.63	1	
(6) Log density	–0.05	0.10	0.46	0.39	0.26	1

Notes: Pairwise Pearson correlations among the regressors of Table 2 columns (4) and (7). $N = 25$ cantons; lower triangle. The French share is French speakers as a share of the total canton population. At $N = 25$ a correlation's standard error is roughly 0.21, so the table ranks pairs and does not measure them; no value is meaningful to a third decimal, and no significance stars are shown. The two absinthe encodings, (4) and (5), are alternative measures of the same variable and never enter a specification together.

A.5.7 Oster bounds

Table 2 shows the wine coefficient moving from -0.196 in column (1) to 0.677 in column (4) as three observed controls enter. Oster (2019) asks what a fourth, unobserved control would do, reading the coefficient's movement together with the movement in R^2: a coefficient that sits still reassures only when the controls explain a good deal of the outcome. Her adjustment needs two assumptions. The first is a ceiling, $R^2_{\max}$, the share of the yes-share that all relevant variables together could explain. The second is proportional selection: the unobserved variables are tied to the wine share with δ times the strength of the observed controls. She reports two summaries, computed here with her routine `psacalc`: δ^*, the δ that would drive the coefficient to zero, and β^*, the adjusted coefficient at $\delta = 1$.

Both summaries depend on the ceiling. Oster calibrates it on sixty-five randomized results from five top journals and sets it at 1.3 times the controlled R^2, the value at which ninety percent of them survive (Oster, 2019, §5.2). On her first standard a result passes if the identified set, from the estimate to β^*, excludes zero. We also report, on the French-adjusted baseline, the limiting ceiling of one, which she shows over-adjusts: fewer than half of the randomized results in her own calibration survive it (Oster, 2019, §5.2, Table 4). We show it for transparency and do not treat it as the test.

Table A.11 reports two baselines. The naive baseline runs from column (1) to column (4) and treats all three controls as observed; her formal results cover it in full. The French-adjusted baseline keeps the French share in both regressions, as her §3.3.3 prescribes, and

runs from column (2), 0.547, to column (4), so that the controls it studies are the two that are modeling choices, the Protestant share and the producer indicator. Its ceiling is 1.3 times the R^2 of column (4), her controlled regression. The French share is held fixed rather than studied because every specification includes it: it is correlated with the producer indicator at 0.77 and accounts for 73 percent of the wine coefficient's movement (Table 3), so leaving it out would let the wine coefficient carry the language pattern. Her §3.3.3 assumes the fixed control is unrelated to the remaining controls; that premise fails here (correlations 0.15 and 0.77; Appendix Table A.10), so the French-adjusted rows are an approximation. The French-adjusted baseline asks the design's question, and the naive one rests on firmer footing. Both return the same verdict.

On both baselines the controls move the wine coefficient up, from −0.196 through zero to 0.677 on the naive baseline and from 0.547 to 0.677 on the French-adjusted one: the case Oster describes as controls moving the coefficient away from zero. Provided the unobserved variables select in the same direction as the observed controls, the assumption that bounds her δ below at zero (Oster, 2019, §3.4), unobserved variables that behave like the observed ones would raise the coefficient further, and only selection in the opposite direction could bring it to zero. The δ^* entries in Table A.11 are negative on both baselines and at every ceiling reported, which records exactly that. For her approximation this is algebra; for the exact estimator the table reports, it is what the numbers show at every ceiling we run, not a general result. Her bounding set takes δ in $[0, 1]$, and within it no selection produces a zero. Her cutoff of one presumes selection in the same direction, so these negative values are read for their sign and not compared with it. That is also why her first standard cannot fail a coefficient of this kind, and why the second, below, is the one that matters.

Oster applies her first standard only to coefficients the controls move toward zero, because a coefficient that grows with controls passes it automatically (Oster, 2019, §2, §5.1). Her second standard asks whether the identified set stays within 2.8 standard errors of the estimate, the bounds of a 99.5 percent interval (Oster, 2019, §3.4, §5.1). Unlike the first, it can be applied to coefficients the controls move away from zero, for which, she writes, the question is whether considering the full set of variables would lead to very different conclusions than the controlled estimate (Oster, 2019, §5.1, §5.2). This is the test we could have failed. At her calibrated ceiling the adjusted coefficient is 0.949 on the naive baseline and 1.008 on the French-adjusted one, both above 0.677, so under proportional selection the estimate is a floor. The band, 0.677 ± 0.604, is Oster's yardstick and not a small-sample interval: the HC3 standard error of Table 2 times her fixed 2.8, wide, as it must be with twenty-five observations. The identified set on the naive baseline, from 0.677 to 0.949, lies

within it, and the set on the French-adjusted baseline, from 0.677 to 1.008, lies within it; both verdicts, and the one at the limiting ceiling below, are the same with the classical or the Bell–McCaffrey standard error in place of HC3. The adjustments, 1.26 and 1.53 standard errors, are both smaller than the 2.8 standard errors that bound her band. At the limiting ceiling of one, the set on the French-adjusted baseline runs from 0.677 to 1.500 and lies outside the band. The movement is away from zero, so this ceiling bears on how large the coefficient is, not on its sign, and the estimate remains a floor.

Oster's bounds address the control we do not have, and say that, at her calibrated ceiling, a control behaving like the ones we do would strengthen the estimate rather than weaken it, and would not move it outside her band on either baseline. One thing the exercise cannot rule out should be said plainly. An omitted variable that is more present in wine cantons than elsewhere once language, the Protestant share and producer status are held fixed, and that raises support for the ban for reasons unconnected to the wine interest, would select against the pattern of every observed control. No arithmetic on the observed controls can exclude it. The two candidates that come first to mind, a canton's density of settlement and the strength of organized temperance, are already in the cascade as the population-density and Protestant-share controls, and the coefficient survives their inclusion. The bounds cannot settle whether a third exists. What they settle is what it would have to be: something that runs against the observed controls.

Four limits apply. Neither this adjustment nor the decomposition in Table 3 is an inference procedure; inference is the four readings in Table 2, and Oster's guarantees are asymptotic: her applications run to thousands of observations, and the adjustment's behavior at twenty-five is unexamined. The ceiling is assumed, never estimated, and its calibration is external. The proportionality rule always holds for some δ, so what is assumed is a bound on it: $\delta \leq 1$, which no in-sample diagnostic can check (Oster, 2019, §3, §4), and, for the sign result, $\delta \geq 0$, that unobserved variables select in the same direction as the observed ones (Oster, 2019, §3.4). And the method is linear: it says nothing about the fractional logit, functional form or the measurement of the wine share. The result is robustness to unobserved selection that runs with the observed controls, not a demonstration that the estimate is unbiased.

Table A.11: Oster (2019) δ^* bounds for the wine revenue-share coefficient

Selection bound	$R^2_{\max}$	δ^*	$\beta^*(R^2_{\max}, \delta{=}1)$
French-adjusted baseline (French share held in both regressions)			
$1.3\,\tilde{R}^2$	0.749	–0.880	
$R^2_{\max} = 1$	1.000	–0.474	1.500
Identified set = [0.677, 1.500], band 0.677 ± 0.604 (HC3): **outside**			
Bias-adjusted β ($\delta = 1$)	0.749		1.008
Identified set = [0.677, 1.008], band 0.677 ± 0.604 (HC3): **within**			
Naive bivariate baseline			
$1.3\,\tilde{R}^2$	0.749	–1.066	
Bias-adjusted β ($\delta = 1$)	0.749		0.949
Identified set = [0.677, 0.949], band 0.677 ± 0.604 (HC3): **within**			

Oster (2019) proportional-selection bound δ^*: the strength of selection on unobservables, relative to the observed controls, that would drive the wine revenue-share coefficient to zero; δ^* is read for its sign: Oster's cutoff of one presumes selection in the same direction as the observed controls, so these negative values are not compared with it. $N = 25$ cantons. The French share is a suppressor: the raw bivariate wine coefficient is -0.196 and turns positive, to +0.547, once the French share enters (Table 2, column (2); a Gelbach/Simpson configuration, not a defect). δ^* is negative in both baselines at the conventional bound $R^2_{\max} = 1.3\,\tilde{R}^2$, for a different reason in each: on the naive baseline the controls carry the coefficient from -0.196 through zero to +0.677; on the French-adjusted baseline the Protestant share and the producer indicator raise it from +0.547 to +0.677. In both, the observed controls move the coefficient away from zero, so no same-direction selection of any strength produces zero. Bias-adjusted β ($\delta = 1$) is the coefficient adjusted for unobservables selecting as strongly as the observed controls; values above the estimate mean the estimate is a floor (naive 0.949; French-adjusted 1.008). $\tilde{R}^2$ is the controlled R^2 of column (4) in both panels; the French-adjusted rows keep the French share in both regressions (Oster 2019, §3.3.3; the mcontrol option of psacalc). Partialling the French share out of every variable first and setting the ceiling at 1.3 times the resulting partial R^2 (0.347) is a tighter ceiling than her calibration; it gives $\beta^* = 0.764$ and the same verdict. At a ceiling of one the two conventions coincide. The memo row closing each panel reports Oster's magnitude test: the identified set runs from the controlled estimate to the bias-adjusted β at the $\min\{1.3\,\tilde{R}^2, 1\}$ ceiling with $\delta = 1$ (Oster 2019, §3.4); the band is that estimate ± 2.8 HC3 standard errors, her 99.5 percent interval (§5.1). **Within** means the set does not leave the band. The $R^2_{\max} = 1$ row and the identified set beneath it are reported for transparency; Oster's calibration uses $R^2_{\max} = 1.3\,\tilde{R}^2$.

A.6 Leave-one-out stability

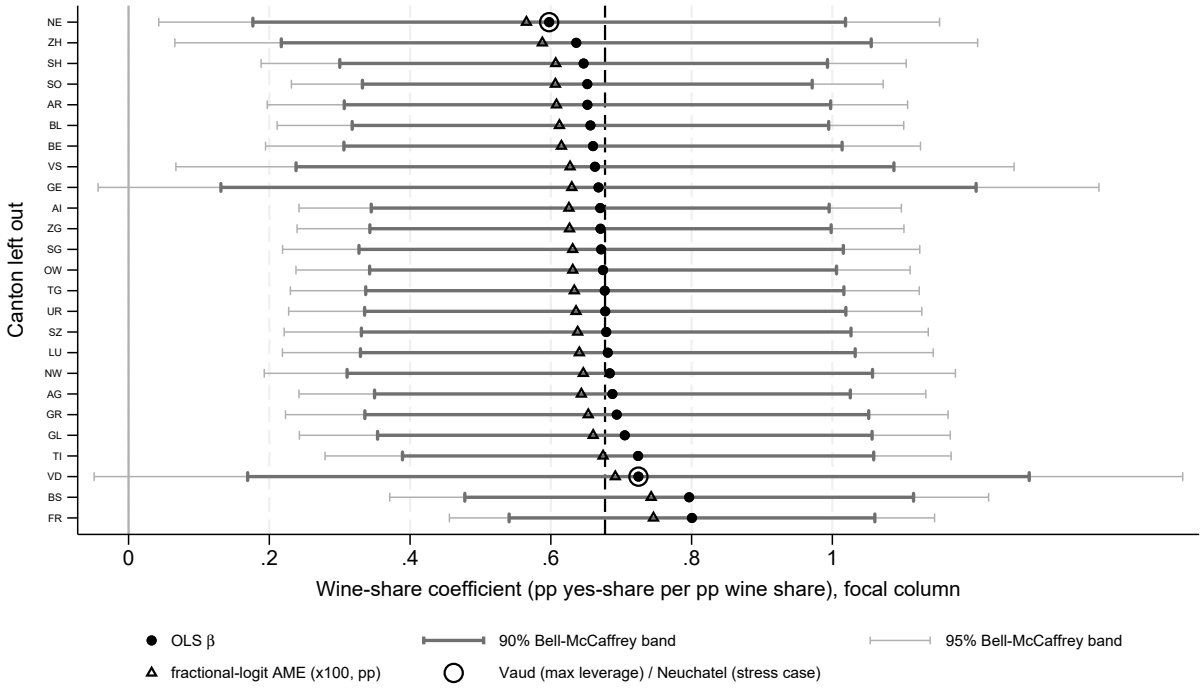


Figure A.1: Leave-one-canton-out stability of the wine coefficient, Table 2 column (4): all 25 single-canton deletions under both estimators on a shared percentage-point axis (A6 companion figure).

Table A.12: Leverage and influence anatomy of the wine coefficient, Table 2 column (4): where the influence sits, and survival under single-canton deletion

Canton	Wine rev. share	Producer (attested)	French share	h_{ii} (leverage)	DFBETA (influence, wine)	Drop-one β (p)
NE	5.43	Yes	82.8	0.364	+0.276	0.598 (0.041)
FR	0.76	–	68.3	0.534	-0.448	0.801 (0.004)
BS	0.17	Yes	2.3	0.652	-0.423	0.797 (0.008)
TI	6.16	–	0.3	0.142	-0.164	0.724 (0.012)
VD	32.11	Yes	86.5	0.631	-0.163	0.724 (0.058)
ZH	14.50	–	0.9	0.268	+0.141	0.636 (0.038)
SH	5.28	–	0.6	0.123	+0.109	0.647 (0.019)
GL	0.00	–	0.2	0.107	-0.099	0.705 (0.014)
SO	0.12	–	1.9	0.056	+0.095	0.652 (0.014)
AR	0.00	–	0.1	0.155	+0.087	0.652 (0.017)
BL	1.21	–	0.9	0.105	+0.074	0.656 (0.016)
GR	1.21	–	0.5	0.059	-0.064	0.694 (0.016)
BE	2.17	–	16.6	0.152	+0.059	0.660 (0.018)
VS	19.53	Yes	65.2	0.410	+0.049	0.663 (0.040)
AG	3.67	–	0.4	0.064	-0.036	0.687 (0.014)
GE	2.83	Yes	82.8	0.369	+0.032	0.668 (0.059)
NW	0.00	–	0.2	0.105	-0.025	0.684 (0.019)
AI	0.00	–	0.1	0.093	+0.024	0.670 (0.013)
ZG	0.00	–	0.6	0.091	+0.023	0.671 (0.013)
SG	2.44	–	0.3	0.053	+0.019	0.672 (0.016)
LU	0.03	–	0.5	0.087	-0.014	0.681 (0.016)
OW	0.00	–	0.2	0.104	+0.010	0.674 (0.014)
SZ	0.12	–	0.5	0.099	-0.006	0.679 (0.016)
TG	2.26	–	0.3	0.082	+0.001	0.677 (0.015)
UR	0.00	–	0.1	0.098	-0.001	0.677 (0.015)
Full sample (N=25)	100.0	5	16.5	5.00	–	0.677 (0.014)

Yes-vote share (vote #68) on national wine revenue share, the column-(4) preferred specification (+ French share + Protestant share + Absinthe producer (attested, binary)). Neuchâtel (NE) is listed first because dropping it gives the leave-one-out minimum; the remaining 24 cantons are sorted by absolute DFBETA on the wine coefficient. h_{ii} = OLS leverage, the canton's weight in its own fitted value (leverages sum to k = 5; the maximum-leverage canton on this design is Basel-Stadt). DFBETA = the canton's influence on the wine coefficient (BKW, dropped-one RMSE scaled): a positive value means the canton pulls the coefficient up, so deleting it lowers β – read each row against the drop-one column beside it. drop-one beta (p) re-estimates the wine coefficient omitting that canton; the parenthetical is the Bell–McCaffrey two-sided p (HC2 with Satterthwaite effective degrees of freedom), N=24 for the canton rows and N=25 for the full sample. The concentration of the wine coefficient's identification weight in Vaud (Table A.17), amplified by that canton's leverage, is why its Bell–McCaffrey effective degrees of freedom are 3.68. The final row reports full-sample aggregates, each appropriate to its column: wine revenue shares sum to 100.0 (they are national shares), attested producers are a count (5 cantons), French share is the sample mean, and leverages sum to 5.00 (= k, the identity noted above); DFBETA has no full-sample analog and is left blank. The wine effect stays positive and at least 0.598 under every single-canton deletion; the minimum comes from dropping Neuchâtel.

A.7 Other referenda

A.7.1 Full results: wine revenue

Table A.13: Referenda by role: least-squares and fractional-logit estimates per vote

Referendum	OLS coef.	Freedman-Lane p	AME	N
Falsification referenda				
#51	0.256	0.513	0.307	24
#57	–0.452	0.181	–0.462	23
#58	–0.377	0.289	–0.485	23
#59	–0.041	0.874	–0.020	25
#62	0.313	0.517	0.321	25
#64	–0.111	0.669	–0.116	25
Corroborative				
#52	0.550	0.261	0.572	24
#63	0.222	0.605	0.219	25
#65§	1.527	0.083	1.507	25
Treatment vote				
#68¶	0.639	0.016	0.595	25

Notes: Stars unshown. Columns report the least-squares coefficient, its Freedman–Lane permutation p-value, the fractional-logit average marginal effect (descriptive), and N. § marks #65, the wine-related corroborative candidate. Its row carries the universal battery specification. Referendum 65 also has a coalition specification of its own, adding the cattle and existing-regulation controls appropriate to a food-regulation vote, which the 1908 ballot's covariate set does not contain; it is reported in Table A.19, where the Bell–McCaffrey correction carries 1.72 effective degrees of freedom against 3.36 at the universal specification shown here. Panels group the votes by the role each plays in the argument. The falsification referenda concern proportional representation, primary-school funding and patent protection; their full official titles are in Table A.1. Measure basis, which Table A.2 does not carry: the wine regressor is the canton share of national wine revenue where a revenue series was published for the harvest behind the ballot, and the canton share of national wine volume where none was. Volume basis: #51, #52. Revenue basis: #57, #58, #59, #62, #63, #64, #65, #68. N is the complete-case count on the vote's own measure and is given in the final column. ¶ marks #68, the treatment vote, shown at the battery specification so that every row of the table is one regression. Its headline estimate is column (4) of Table 2, which adds the absinthe-producer indicator; that row and this one are different numbers. The battery carries no absinthe term because the producer register runs 1902–1906 while three of the six falsification ballots fall outside that window, so conditioning on it across the battery would hold the formula fixed while measurement quality varied from row to row. Per-vote HC3 and Bell–McCaffrey p-values, full estimator diagnostics and per-vote permutation completion appear in Appendix Table A.4.

A.7.2 Full results: wine volume

Table A.14: Volume-basis readings behind the quasi-falsification figure

Referendum	Wine coef.	HC3 (se)	Bell–McCaffrey (se)	Bell–McCaffrey p	Effective D.O.F.	N
Falsification referenda						
#51	0.256	(0.496)	(0.406)	0.576	2.79	24
#57	–0.514	(0.424)	(0.348)	0.217	3.81	23
#58	–0.439	(0.477)	(0.400)	0.337	3.81	23
#59	0.055	(0.397)	(0.298)	0.865	3.16	25
#62	0.197	(0.631)	(0.518)	0.732	2.70	25
#64	–0.043	(0.532)	(0.286)	0.892	2.68	25
Corroborative						
#52	0.550	(0.552)	(0.440)	0.306	2.79	24
#63	0.141	(0.562)	(0.399)	0.750	2.70	25
#65	1.519	(0.939)	(0.730)	0.118	3.40	25
Treatment vote						
#68†	0.608	(0.260)	(0.221)	0.040	5.00	25

Notes: Least squares on the volume basis at the Table A.6 column (3) specification – canton yes-share on the canton share of national wine volume (×100), the French share and the Protestant share. No absinthe term: the producer register averages 1902–1906 and three of the six falsification ballots fall outside that window, so conditioning on it across the referendum set would hold the formula fixed while measurement quality varied from row to row. The volume basis is used because the 1897 cohort publishes no revenue series, so it is the only measure on which the complete set of 10 referenda exists; it is also the less favorable of the two measures for the treatment vote. Standard errors are shown under both schemes because the gap between them is the small-sample correction: at these effective degrees of freedom, normal critical values would overstate precision. The Bell–McCaffrey p uses the effective degrees of freedom in the final column, never a normal approximation. Sample sizes are complete-case counts on the volume measure, which is missing at source for one canton in 1897 and two in 1900. Across the 6 falsification referenda the coefficients cluster near zero and the intervals are too wide to resolve; the design does not produce significance on any of them. † #68, the treatment vote, is the same fit shown in column (3) of Table A.6.

A.8 Secondary outcome: tavern employment share

The two cascades below regress each arm's outcome on the tavern employment share, defined in the note to Table 4, entering the controls in the order fixed in §4.1 and reporting the tavern coefficient of every column. At the petition the narrow measure runs −3.384 standing alone, then −3.518, −3.937, −3.762 and −3.783 as the controls enter, and in the focal column its standard error is 1.176, with .005 on HC3, .024 on Bell–McCaffrey, .039 on the wild bootstrap-t and .009 on the Freedman–Lane permutation. Its effective degrees of freedom in that column are 3.38, so the estimate rests on a few cantons rather than on all twenty-five, and deleting each canton in turn leaves it between −4.388 and −3.251 without a single change of sign. In the focal specification the broad measure, which adds the lodging

trades the census sets apart as class 300, inns, hotels and boarding houses, returns −1.539 with a standard error of 1.103, no reading below 0.10 and a standardized coefficient of −0.253, and log density does not rescue it: it moves to −2.028 with a standard error of 1.736, again with no reading below 0.10 (they run from .132 to .257), a leave-one-out range of [−3.28, −1.48] and no change of sign. The narrow measure is where the result lives, and the boundary is where it is weakest. At the vote the same coefficient is not detectable. The coefficient keeps its sign and loses magnitude as the controls enter, from −2.296 standing alone to −1.379 with log density entered, and the smallest p-value anywhere in that table's diagnostic block is .136, on the wild bootstrap-t in the column where the producer indicator enters.

The Federal Council's *Botschaft* of November 20, 1884, records that many taverns were run as a sideline by wine dealers, wine-growers or other farmers, or by grocers, butchers, bakers or boarding-house keepers (Schweizerischer Bundesrat, 1884, p. 403, n. 10), an overlap that these cantonal aggregates cannot see. Table A.17 sets out, canton by canton and for each coefficient, the squared Frisch–Waugh–Lovell residual that the canton contributes, normalized to sum to 100, which is the object underlying an added-variable plot (Belsley et al., 1980). The two coefficients rest on nearly separate sets of cantons. The tavern coefficient is identified largely in Ticino, which carries 42.6 percent of its weight, with Geneva at 19.3, Nidwalden 6.3, Obwalden 4.6 and Lucerne 4.3; the wine coefficient is identified in the wine cantons, Vaud 34.5, Zurich 15.5, Geneva 12.9, Valais 9.7 and Neuchâtel 9.6. Geneva is the one canton material to both, and it is also the second-largest attested absinthe producer, at 19.8 percent of the retained national total.

Ticino carries 42.6 percent of the tavern coefficient's identification weight and is its most influential canton (DFBETA 0.433), yet no single deletion moves the coefficient by more than about a fifth of its value or turns its sign (Table A.18). The measure stands for the consumer-facing side of the drink interest, a different thing from the producer interest the wine measures carry: a tavern-keeper's stake in a prohibition runs through the receipts he loses on a drink he may no longer sell, and a wine-grower's runs through what the removal of a competing drink does to the price of his own. We therefore read the tavern column of Table 4 as suggestive evidence that the petition drew fewer signatures where the consumer-facing drink trade employed more people, and we do not ask the column to carry more: it is one column, on a measure whose variation sits largely in two cantons, with no counterpart at the vote.

Table A.15: Tavern employment and the 1906 anti-absinthe petition (cascade)

	Class 301 (narrow)					Classes 300+301 (broad)	
	Tavern	+ French	+ Protestant	+ Producer	+ Density	Focal	+ Density
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Tavern employment share	–3.384*	–3.518**	–3.937***	–3.762***	–3.783***	–1.539	–2.028
	(1.849)	(1.439)	(1.209)	(1.176)	(1.305)	(1.103)	(1.736)
French share		0.076	0.050	0.103	0.107	0.099	0.064
		(0.053)	(0.043)	(0.074)	(0.098)	(0.083)	(0.148)
Protestant share			0.164***	0.170***	0.167**	0.132**	0.143*
			(0.047)	(0.048)	(0.065)	(0.063)	(0.076)
Absinthe (attested producer, = 1)				–5.418	–5.852	–5.706	–2.019
				(6.265)	(8.240)	(7.111)	(15.597)
Log population density					0.238		–1.707
					(2.389)		(3.596)
Constant	27.540	26.566	20.835	20.413	19.493	20.431	29.697
	(4.364)	(3.812)	(3.692)	(3.673)	(9.345)	(6.121)	(21.866)
Standardized β:							
Tavern employment share	–0.359	–0.373	–0.417	–0.399	–0.401	–0.253	–0.333
R^2	0.129	0.191	0.540	0.561	0.561	0.456	0.475
Adjusted R^2	0.091	0.117	0.474	0.473	0.446	0.347	0.337
AIC	183	183	171	172	174	177	179
BIC	186	187	176	178	181	184	186
Coefficient of interest (tavern employment share):							
HC3 robust p	0.080	0.023	0.004	0.005	0.009	0.178	0.257
Bell–McCaffrey p	0.097	0.045	0.023	0.024	0.027	0.133	0.158
Wild bootstrap-t p	0.082	0.083	0.046	0.039	0.034	0.139	0.132
Freedman–Lane p	0.103	0.023	0.010	0.009	0.012	0.127	0.167
Effective df	3.70	3.78	3.37	3.38	3.39	9.78	6.81
BKW condition no. (with constant)	4.45	4.85	5.69	6.24	15.77	9.33	21.63
VIF (tavern employment share)	1.00	1.00	1.01	1.03	1.04	1.28	1.62
N (cantons)	25	25	25	25	25	25	25

Notes: Ordinary least squares; $N = 25$ cantons in every column. Outcome is 1906 anti-absinthe petition signatures per 100 eligible voters. Columns (1)–(5) enter controls cumulatively with population density last; columns (6)–(7) are the broad-measure companion at the focal architecture. The tavern measure is class 301 of the federal business census of August 9, 1905 — restaurants and taverns not attached to an inn (BFS, 1908a); columns (6)–(7) add inns, hotels and boarding houses (classes 300+301). Density and, where applicable, other harvest-year inputs are 1906 vintage. *, **, and *** mark $p < 0.10$, 0.05, and 0.01, on every slope coefficient's heteroskedasticity-robust (HC3) p-value; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell–McCaffrey, wild bootstrap-t, Freedman–Lane permutation), for the tavern coefficient of each column; the effective degrees of freedom are Bell–McCaffrey's. The variance inflation factor (VIF) is centered and reported for the tavern employment share; the specification maximum falls on the Protestant share (1.03 in column 3) and on the producer indicator (2.58 in column 4, 3.32 in column 5, 2.63 in column 6, 3.94 in column 7); column 1 has a single regressor, and in column 2 the two regressors' VIFs coincide at 1.00. Condition numbers are Belsley–Kuh–Welsch, constant included. Leave-one-out on the column-(4) tavern coefficient: range [-4.39, -3.25], 0 sign changes (Appendix Table A.18 gives the anatomy); on the column-(7) broad-measure coefficient: range [-3.28, -1.48], 0 sign changes.

Table A.16: Tavern employment and the 1908 ban vote (cascade)

	Class 301 (narrow)				
	Tavern	+ French	+ Protestant	+ Producer	+ Density
	(1)	(2)	(3)	(4)	(5)
Tavern employment share	–2.296	–1.875	–1.731	–1.691	–1.379
	(3.593)	(1.858)	(1.532)	(1.542)	(1.652)
French share		–0.239***	–0.230***	–0.217***	–0.276
		(0.072)	(0.070)	(0.065)	(0.201)
Protestant share			–0.056	–0.055	–0.012
			(0.055)	(0.057)	(0.072)
Absinthe (attested producer, = 1)				–1.242	5.034
				(4.197)	(15.241)
Log population density					–3.428
					(2.945)
Constant	69.152	72.214	74.179	74.082	87.332
	(7.112)	(4.387)	(4.299)	(4.371)	(10.059)
Standardized β:					
Tavern employment share	–0.199	–0.162	–0.150	–0.146	–0.119
R^2	0.039	0.448	0.476	0.477	0.541
Adjusted R^2	–0.002	0.398	0.401	0.372	0.420
AIC	196	184	185	187	185
BIC	198	188	190	193	193
Coefficient of interest (tavern employment share):					
HC3 robust p	0.529	0.324	0.271	0.286	0.414
Bell–McCaffrey p	0.485	0.291	0.249	0.258	0.330
Wild bootstrap-t p	0.541	0.188	0.138	0.136	0.223
Freedman–Lane p	0.495	0.305	0.251	0.251	0.341
Effective df	3.70	3.78	3.37	3.38	3.39
BKW condition no. (with constant)	4.45	4.85	5.69	6.24	15.76
VIF (tavern employment share)	1.00	1.00	1.01	1.03	1.04
N (cantons)	25	25	25	25	25

Notes: Ordinary least squares; $N = 25$ cantons in every column. Outcome is the 1908 yes-share, Swissvotes #68, in percent. Columns (1)–(5) enter controls cumulatively with population density last. The tavern measure is class 301 of the federal business census of August 9, 1905 — restaurants and taverns not attached to an inn (BFS, 1908a). Density and, where applicable, other harvest-year inputs are 1907 vintage. *, **, and *** mark $p < 0.10$, 0.05, and 0.01, on every slope coefficient's heteroskedasticity-robust (HC3) p-value; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell–McCaffrey, wild bootstrap-t, Freedman–Lane permutation), for the tavern coefficient of each column; the effective degrees of freedom are Bell–McCaffrey's. The variance inflation factor (VIF) is centered and reported for the tavern employment share; the specification maximum falls on the Protestant share (1.03 in column 3) and on the producer indicator (2.58 in column 4, 3.33 in column 5); column 1 has a single regressor, and in column 2 the two regressors' VIFs coincide at 1.00. Condition numbers are Belsley–Kuh–Welsch, constant included. Leave-one-out on the column-(4) tavern coefficient: range [-2.60, -1.06], 0 sign changes.

A.8.1 Tavern and wine identification weights

Table A.17: Identification weights for the tavern and wine coefficients

	Identification weight (%)	
Canton	Tavern (petition arm)	Wine revenue (vote arm)
TI	42.6	3.8
GE	19.3	12.9
NW	6.3	0.0
OW	4.6	0.0
LU	4.3	0.0
AR	3.9	1.2
FR	2.6	4.3
BE	2.4	0.6
BS	2.3	5.2
TG	2.2	0.0
AI	2.2	0.0
AG	1.4	0.3
ZG	1.4	0.0
VD	1.2	34.5
VS	1.1	9.7
SZ	0.9	0.0
NE	0.5	9.6
ZH	0.2	15.5
BL	0.2	0.3
SO	0.2	0.2
SG	0.1	0.1
GL	0.0	0.9
SH	0.0	0.6
UR	0.0	0.0
GR	0.0	0.1

Identification weights: for each coefficient, the weight of a canton is the squared Frisch–Waugh–Lovell residual of the focal regressor on the other controls in its own arm's column-(4) architecture (tavern employment share: petition arm; wine revenue share, 1907 vintage: vote arm), normalized to sum to 100 — the object underlying an added-variable plot (Belsley et al., 1980), tabulated rather than plotted. All 25 cantons are shown, sorted by the tavern-arm weight, descending.

A.8.2 Leverage and influence in the tavern coefficient

Table A.18: Leverage and influence anatomy of the tavern coefficient, Table 4 tavern column: where the influence sits, and survival under single-canton deletion

Canton	Tavern emp. share	Producer (attested)	French share	h_{ii} (leverage)	DFBETA (influence, tavern)	Drop-one β (p)
TI	5.18	–	0.3	0.530	+0.433	-4.388 (0.043)
NW	0.75	–	0.2	0.168	-0.367	-3.251 (0.028)
AR	3.09	–	0.1	0.183	-0.329	-3.310 (0.029)
GE	4.42	Yes	82.8	0.433	+0.273	-4.156 (0.079)
ZG	1.41	–	0.6	0.104	+0.242	-4.083 (0.015)
BS	1.83	Yes	2.3	0.623	-0.182	-3.501 (0.029)
FR	0.98	–	68.3	0.517	-0.161	-3.531 (0.032)
TG	2.81	–	0.3	0.103	-0.139	-3.563 (0.030)
LU	0.98	–	0.5	0.130	+0.135	-3.956 (0.027)
VS	1.75	Yes	65.2	0.324	+0.108	-3.916 (0.017)
BE	1.31	–	16.6	0.169	+0.087	-3.888 (0.030)
GL	2.21	–	0.2	0.099	+0.066	-3.842 (0.021)
ZH	2.33	–	0.9	0.115	-0.062	-3.675 (0.028)
AI	2.71	–	0.1	0.114	+0.051	-3.836 (0.035)
AG	1.48	–	0.4	0.075	-0.039	-3.705 (0.026)
SH	2.01	–	0.6	0.117	-0.038	-3.710 (0.020)
VD	1.78	Yes	86.5	0.299	+0.031	-3.806 (0.025)
SO	1.83	–	1.9	0.055	-0.022	-3.729 (0.025)
SZ	2.45	–	0.5	0.108	+0.019	-3.789 (0.029)
OW	0.94	–	0.2	0.149	+0.012	-3.780 (0.032)
NE	1.99	Yes	82.8	0.272	-0.008	-3.749 (0.024)
GR	2.04	–	0.5	0.058	+0.005	-3.769 (0.023)
BL	1.91	–	0.9	0.104	-0.005	-3.755 (0.024)
SG	2.21	–	0.3	0.053	+0.000	-3.762 (0.025)
UR	1.94	–	0.1	0.098	-0.000	-3.761 (0.025)
Full sample (N=25)	2.09	5	16.5	5.00	–	-3.762 (0.024)

Petition signatures per 100 eligible voters on the class-301 tavern employment share, the tavern column of Table 4 and column (4) of Table A.15 (+ French share + Protestant share + Absinthe producer (attested, binary)). All 25 cantons are sorted by absolute DFBETA on the tavern coefficient. h_{ii} = OLS leverage, the canton's weight in its own fitted value (leverages sum to k = 5; the maximum-leverage canton on this design is Basel-Stadt). DFBETA = the canton's influence on the tavern coefficient (BKW, dropped-one RMSE scaled): a positive value means the canton pulls the coefficient up, so deleting it lowers β – read each row against the drop-one column beside it. drop-one beta (p) re-estimates the tavern coefficient omitting that canton; the parenthetical is the Bell–McCaffrey two-sided p (HC2 with Satterthwaite effective degrees of freedom), N=24 for the canton rows and N=25 for the full sample. The concentration of the tavern coefficient's identification weight in Ticino and Geneva (Table A.17), amplified by their leverage, is why its Bell–McCaffrey effective degrees of freedom are 3.38. The final row reports full-sample aggregates, each appropriate to its column: tavern employment share and French share are sample means, attested producers are a count (5 cantons), and leverages sum to 5.00 (= k, the identity noted above); DFBETA has no full-sample analog and is left blank. The tavern coefficient stays negative under every single-canton deletion, from -4.388 (drop-Ticino) to -3.251 (drop-Nidwalden).

A.9 Secondary outcome: the 1906 food act (#65)

A.9.1 The legislative record of the food act

Under the name of consumer protection, the act carried the demands of a mixed agricultural coalition; wine, cattle, and dairy were its substantive components.

The Farmers' Union declared control at the frontier the cardinal point of the law, and a law without it unacceptable, as the committee's rapporteur reported (Schweizerische Bundesversammlung, 1905a, National Council, June 9, 1903, p. 418); the same rapporteur said the customs experts at the frontier were added mainly at its demand, and also at that of other interested circles, the honest wine merchants and the vineyard owners among them (Schweizerische Bundesversammlung, 1905a, National Council, June 10, 1903, p. 442). The frontier laboratories the union had asked for were refused by the Council of States and by the committee, with those experts put in their place (Schweizerische Bundesversammlung, 1905a, National Council, June 9, 1903, p. 419), and the act made even the experts optional (Schweizerische Bundesversammlung, 1905b, p. 9).

A second clause, the power to forbid mixtures of natural foodstuffs with substitutes, drew the same protective reading. The Council of States committee's spokesperson, moving to strike that clause, told his chamber that it was "less a provision [. . .] to assure the sincerity of the goods" than "a measure of protection in favor of natural products," meant to support them "against an often ruinous competition," and that "the circles of the farmers and of the wine-growers" counted on it (Schweizerische Bundesversammlung, 1905a, p. 892), and he named "margarine and artificial wine" as the targets (Schweizerische Bundesversammlung, 1905a, p. 893).

The same committee's spokesperson said in October that the stated aim had been public health and the real target the intermediaries between producers and consumers (Schweizerische Bundesversammlung, 1905a, p. 1117).

In the first chamber to debate the bill, a member asked "who can blame the hard-pressed farmers" for the "ulterior motive" of protection against foreign produce, and warned that without it agriculture "would declare that it accepts no law" (Schweizerische Bundesversammlung, 1905a, p. 285).

The government's own statement of purpose openly named the protection of "the genuine producer (farmer and manufacturer)" against "dishonest competition" as an aim of the law (Schweizerischer Bundesrat, 1899b, p. 615). It was secondary to consumer protection in the 1895 message (Schweizerischer Bundesrat, 1895a, p. 803) but openly co-equal with it by 1899 (Schweizerischer Bundesrat, 1899b, p. 615). The 1895 message

Table A.19: Referendum #65 (Food Act 1906): wine-coalition specification ladder, wine revenue share, harvest 1905

	Wine (1)	+ Cattle (2)	+ Regime (3)	+ French (4)	+ Protestant (5)	+ Density (6)
Wine revenue share, harvest 1905	0.551	0.908***	0.916***	1.344**	0.979***	0.929**
	(0.388)	(0.191)	(0.221)	(0.602)	(0.332)	(0.386)
Cattle per 1,000 population (1901)		0.067***	0.065***	0.064***	0.054**	0.067
		(0.017)	(0.017)	(0.021)	(0.026)	(0.046)
Existing food-control regime, 1895 (=1)			–2.516	–0.456	–2.026	–2.592
			(6.792)	(7.046)	(7.790)	(7.830)
French share				–0.182		
				(0.232)		
Protestant share					–0.112	
					(0.183)	
Log population density						0.552
						(9.090)
Constant	59.83	28.52	30.78	31.31	39.88	27.19
	(4.87)	(10.12)	(10.70)	(12.13)	(19.00)	(63.59)
R^2	0.04	0.46	0.47	0.51	0.49	0.47
Adjusted R^2	–0.00	0.41	0.39	0.41	0.38	0.36
AIC	226	214	216	216	217	218
BIC	229	218	221	222	223	224
N	25	25	25	25	25	25
Small-sample diagnostics for the wine coefficient						
HC3 robust p	0.169	<0.001	<0.001	0.037	0.008	0.026
Bell–McCaffrey p	0.225	0.050	0.057	0.064	0.075	0.116
Wild bootstrap-t p (HC3-studentized)	0.371	0.020	0.020	0.053	0.051	0.094
Freedman–Lane p	0.161	0.006	0.010	0.037	0.027	0.042
Bell–McCaffrey effective D.O.F.	1.85	1.70	1.72	3.06	1.82	1.89

Notes: Ordinary least squares; $N = 25$ cantons. Dependent variable: the canton yes-share, in percent, in the referendum of June 10, 1906 on the Federal Act on trade in foodstuffs (Swissvotes #65). Wine: the canton's share of the national wine revenue total, harvest 1905, in percent. Cattle: head per 1,000 population — the 1901 federal-census stocks (HSSO Table I.9) over the 1900 population. The regime indicator marks the fifteen cantons that, in the Federal Council's survey of 1895, had organized food control by an enactment of their own and set up a central examination office (BBl 1895 I, printed 782–784); the other ten had no cantonal laboratory. Cantonal adoptions between 1896 and 1906 are unobserved, so misclassification attenuates the indicator. HC3 robust standard errors in parentheses. Stars use two-sided HC3 p: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The diagnostic rows read the wine coefficient only and are labeled by their own schemes; the Freedman–Lane and wild bootstrap-t p-values are computed in R with 10^6 draws (seed 20260630), and the wine regressor is never permuted. Column (3) is the coalition specification the text refers to.

relayed the food chemists' expectation that a federal law would protect domestic wine-growing, butter producers and beekeepers against dishonest competition (Schweizerischer Bundesrat, 1895a, pp. 794–795).

Meat was the one product given an obligatory inspection regime, and that regime was cut back to what the farm vote would bear: compulsory inspection was confined to meat destined for sale and home slaughter was left to the cantons (Schweizerische Bundesversammlung, 1905b, pp. 3–4).

The Council of States spokesperson called the exception "purely intended to save the law before the popular vote" (Schweizerische Bundesversammlung, 1905a, p. 65) and "a veritable right to poison" (Schweizerische Bundesversammlung, 1905a, p. 33); the National Council spokesperson called the meat clause "one of the most dangerous reefs for the little ship of the law" (Schweizerische Bundesversammlung, 1905a, p. 464); and the requirement that the inspector be a veterinarian was struck in the Council of States by twenty-six votes to four (Schweizerische Bundesversammlung, 1905a, p. 270).

Margarine and artificial wine, by contrast, were handled as one class of substitute from the 1895 message (Schweizerischer Bundesrat, 1895a, p. 791) to the National Council's final reading of the mixture clause, where its rapporteur gave mixed wine, natural with artificial, and butter mixed with margarine among his examples (Schweizerische Bundesversammlung, 1905a, National Council, September 29, 1905, p. 1050).

The Zurich wine merchants' association explicitly asked for the making of artificial wine to be banned outright (Schweizerische Bundesversammlung, 1905a, National Council, June 5, 1905, p. 393), and the Swiss association repeated the demand at nearly every stage of the deliberations, only for the commission to refuse each time (Schweizerische Bundesversammlung, 1905a, National Council, September 29, 1905, pp. 1050–1051). Instead, it left the power to prohibit deceptive mixtures to the Federal Council under Article 54 (Schweizerische Bundesversammlung, 1905b, p. 17).

A.9.2 The precursor vote: the 1897 food article (#52)

Referendum #52 was the vote on the Federal Decree of March 26, 1897, which added Article 69bis to the Federal Constitution of 1874; the people and the cantons accepted it on July 11, 1897 (Schweizerischer Bundesrat, 1897b, pp. 89–91). The article empowered the Confederation to legislate on the trade in foodstuffs and stimulants, and on the trade in other articles of use and consumption so far as these may endanger life or health (Schweizerische Bundesversammlung, 1897, p. 233). The health qualifier attached to the second of those alone, so the competence over foodstuffs, and therefore over wine, as

the main text sets out, did not depend on a danger to health at all. The message that carried the article opened with the Federal Assembly's instruction of 1889 to draft a federal law on the wine trade (Schweizerischer Bundesrat, 1895a, p. 767), said that wines were very often adulterated with artificial wine (Schweizerischer Bundesrat, 1895a, p. 776), called the extension of frontier control to butter, edible fats, oils and wines, articles often imported adulterated, highly desirable (Schweizerischer Bundesrat, 1895a, p. 805), and relayed the food chemists' expectation of protection for wine-growing, butter producers and beekeepers (Schweizerischer Bundesrat, 1895a, pp. 794–795). Two further paragraphs assigned the work: execution fell to the cantons, under federal supervision and with federal money, and control of imports at the national frontier rested with the Confederation (Schweizerische Bundesversammlung, 1897, p. 234). Neither stipulation existed in the Federal Council's earlier draft of March 8, 1895, which carried the two grants of competence and nothing further of the article (Schweizerischer Bundesrat, 1895a, p. 808). Parliament added the new stipulations, which the draft validation decree of August 1897 restates (Schweizerischer Bundesrat, 1897b, p. 91). In the recorded deliberations on the Food Act, the Farmers' Union declared frontier control the cardinal point of the law. In doing so, it asked for the use of a power the constitution had carried since July 1897 and not for a new one.

Four cantons voted against, Glarus, Fribourg, Valais and the half-canton Appenzell Innerrhoden, and of those four only Valais grows wine in quantity; Vaud, Neuchâtel and Ticino accepted, and so did Geneva, by a single vote (Schweizerischer Bundesrat, 1897b, p. 83). The rejection was not a wine bloc. The report records no complaint about the vote (Schweizerischer Bundesrat, 1897b, p. 83). The same electorate voted the same day on the forest-police article, referendum #51, and the two ballots did not divide the country the same way: Lucerne, Nidwalden, St. Gallen and Graubünden rejected the forest article and accepted the food article, Glarus did the reverse, and only Fribourg, Valais and Appenzell Innerrhoden rejected both (Schweizerischer Bundesrat, 1897b, pp. 82–85). What separates the two votes is the measure and not the canton's attitude to federal power, which is why referendum #51 sits in the falsification arm and referendum #52 in the corroborative arm without contradiction. Neuchâtel and Geneva, which accepted the competence in 1897, rejected the act that used it by wide margins in 1906 (Schweizerischer Bundesrat, 1906, p. 121) and were the only two cantons to reject the absinthe ban in 1908 (Schweizerische Bundeskanzlei, 1908b).

Bolliger's reading of the newspapers of the week of the 1897 vote finds little feeling about the new competence: the Free Democrats and the Grütli Association recommended acceptance, a joint appeal was signed by the main trade, industry and agricultural

associations, among them the Swiss Agricultural Association, which spoke for wine-growing with the other farm interests (Baumann and Moser, 2001), and the only organized opposition was a conservative committee in Geneva that charged the article with a "protectionist aftertaste" (Bolliger, 2010, pp. 2–3).

We estimate referendum #52 on the volume basis alone, because no wine revenue series exists for 1897, and on 24 cantons, because the wine series for the 1896 harvest lacks Thurgau. The gap is in the wine series and not in the vote: the result report returns Thurgau in full, with a cantonal yes (Schweizerischer Bundesrat, 1897b, p. 83). The wine coefficient is 0.550, and on a sample this size we do not rule out imprecision; what the two readings establish is that no wine relationship is detectable at conventional levels, on the Bell–McCaffrey reading or on the Freedman–Lane permutation reading. Tables A.14 and A.13 report its readings. Among the corroborative votes, referendum #52 comes nearest, on its volume basis, to the line referendum #65 reaches on the revenue basis, without reaching it (Tables A.14 and A.13).

A.10 Additional petition results

Table A.20: Descriptive statistics: the petition arm at its 1906 vintage

	All cantons (N = 25)					German-plurality (N = 19)		French-plurality (N = 5)	
Variable	Mean	SD	Median	Min	Max	Mean	SD	Mean	SD
Petition signatures (per 100 eligible)	20.5	9.5	17.5	3.4	41.3	20.4	9.1	23.9	9.4
Signatures submitted	6775	8500	4022	286	38337	6327	9001	9558	7135
Eligible voters, June 1906	31391	31682	24888	3021	134354	29386	35327	37318	18047
Wine revenue share, 1906	4.0	8.5	0.9	0.0	40.8	1.4	2.8	13.9	15.7
Wine volume share, 1906	4.0	7.6	0.9	0.0	34.4	1.4	3.0	13.5	12.7
Attested absinthe producer (0/1)	0.20	0.41	0.00	0.00	1.00	0.05	0.23	0.80	0.45
Absinthe-sector exposure, 1902–06	4.0	13.0	0.0	0.0	62.4	0.3	1.4	18.8	25.6
French share, 1900	16.5	31.3	0.5	0.1	86.5	1.4	3.7	77.1	9.7
Protestant share, 1900	43.1	35.0	48.2	1.3	90.2	44.1	34.4	47.5	39.3
Tavern employment share, 1905	2.09	1.01	1.94	0.75	5.18	1.91	0.64	2.18	1.31
Tavern and inn employment share, 1905	3.99	1.57	3.49	1.48	6.82	3.93	1.51	3.74	1.69
Population density, 1906 (per km^2)	260	671	106	15	3430	287	764	199	230
Log population density, 1906	4.66	1.12	4.67	2.73	8.14	4.67	1.15	4.76	1.19
Alkoholzehntel per capita, 1904 (Fr.)	0.30	0.47	0.19	0.15	2.56	0.33	0.54	0.19	0.01

Notes: N = 25; columns as in Table 1. Wine is the 1906 harvest and population the 1906 yearbook estimate over the fixed 1900 land area (the petition drive ran 1906–07; the 1907 harvest is post-outcome). Signatures are the submitted counts of the Federal Council's report of February 22, 1907 (169,377 nationally); eligible voters are the roll of the June 10, 1906 federal vote. The tavern employment shares are within-canton percentages of male employment, net of cottage industry, in class 301 of the federal business census of August 9, 1905 — restaurants and taverns not attached to an inn — and in classes 300+301, which add inns, hotels and boarding houses (BFS, 1908a). The Alkoholzehntel is 1904 spending per 1906 resident, shown for all 25 cantons; Table A.21 excludes Schaffhausen and Vaud.

Table A.21: Tithe-measure robustness (petition arm)

	Raw	Drop SH, VD	Genuine	Temperance
Alcohol-tithe per capita	0.270	–0.052	–0.072	–0.046
	[0.053]	[0.808]	[0.651]	[0.708]
Protestant share	0.609	0.636	0.666	0.651
	[0.025]	[0.023]	[0.031]	[0.016]
N (cantons)	25	23	23	23

Notes: Ordinary least squares. Cells are standardized coefficients with Freedman–Lane permutation p-values in brackets. Columns vary the tithe-based temperance measure: raw Alkoholzehntel spending per capita; the same excluding Schaffhausen and Vaud; the genuine-share measure; and temperance-rubric spending per capita. Schaffhausen and Vaud are excluded from tithe-only specifications on different grounds: Schaffhausen's recorded spending is inflated by welfare accounting unrelated to temperance effort, while Vaud's entire tithe was allocated to a single child-welfare institution, so the measure does not capture temperance effort there. Neither canton is dropped as an outlier. Column 1 retains all 25 cantons and is the pre-drop disclosure arm; columns 2–4 are estimated on the 23-canton tithe sample, so the exclusion is applied wherever a tithe quantity enters. The tithe measures are an alternative that does not displace the Protestant-share result.

Table A.22 is estimated on 23 cantons. Schaffhausen and Vaud are excluded from every column because the tithe measure is contaminated for each, on grounds that differ by canton: Schaffhausen's recorded spending is inflated by welfare accounting unrelated to temperance effort, and Vaud's entire tithe was allocated to a single child-welfare institution. Neither canton is dropped as an outlier. The columns without the tithe term are re-estimated on the 23-canton sample only so that each pair is a within-sample contrast; they are not the paper's estimates, which appear at $N = 25$ in Table 4. The pairs follow that table's order: the petition's focal and density-in specifications, then the vote's focal, share and producer forms. The tavern column has no tithe counterpart and remains in Table 4. The three inference schemes are reported side by side and are not compared with one another; the wild bootstrap-t is not reported, because it was not run on the 23-canton sample. Alternative tithe measures are in Table A.21.

Table A.22: Tithe pairs on a common sample (petition and vote arms, $N = 23$)

	Pet. focal (4)		Pet. density (5)		Vote focal (4)		Vote share (7)		Vote producer (5)	
	Without	With tithe	Without	With tithe	Without	With tithe	Without	With tithe	Without	With tithe
Wine revenue share	–0.833*	–0.839*	–0.905	–0.903	0.643*	0.641*	0.437	0.401	0.504	0.491
	(0.402)	(0.419)	(0.555)	(0.595)	(0.359)	(0.357)	(0.578)	(0.560)	(0.787)	(0.775)
French share	0.157	0.158	0.146	0.148	–0.264	–0.265	–0.231*	–0.225*	–0.292	–0.295
	(0.091)	(0.091)	(0.140)	(0.139)	(0.161)	(0.162)	(0.116)	(0.115)	(0.264)	(0.274)
Protestant share	0.149**	0.161**	0.163*	0.171*	–0.095*	–0.101	–0.047	–0.061	–0.065	–0.075
	(0.056)	(0.072)	(0.080)	(0.092)	(0.048)	(0.060)	(0.072)	(0.077)	(0.083)	(0.087)
Log density			–0.964	–0.873			–2.009	–2.223	–2.162	–2.300
			(2.824)	(2.839)			(2.946)	(3.067)	(4.256)	(4.392)
Absinthe	–4.393	–4.129	–2.367	–2.339	–4.194	–4.337	–0.178	–0.213	0.437	0.473
	(7.174)	(7.161)	(11.635)	(11.583)	(12.478)	(12.423)	(0.230)	(0.298)	(22.017)	(22.518)
Genuine-share tithe (fr. per capita)		–44.329		–36.839		26.807		84.730		48.768
		(99.427)		(97.834)		(112.695)		(116.492)		(131.221)
Constant	13.931	14.518	17.834	17.953	70.590	70.242	78.101	77.740	79.439	79.373
	(2.361)	(3.036)	(11.412)	(11.785)	(3.076)	(3.635)	(12.267)	(12.647)	(17.329)	(17.576)
Standardized β:										
Wine revenue share	–0.364	–0.367	–0.396	–0.395	0.269	0.268	0.182	0.168	0.210	0.205
Protestant share	0.570	0.613	0.624	0.654	–0.275	–0.294	–0.136	–0.178	–0.188	–0.218
R^2	0.43	0.43	0.44	0.44	0.58	0.58	0.62	0.63	0.60	0.61
Adjusted R^2	0.30	0.27	0.27	0.23	0.49	0.46	0.51	0.49	0.49	0.46
Coefficient of interest (Protestant share):										
HC3 robust p	0.016	0.040	0.057	0.081	0.063	0.111	0.525	0.438	0.448	0.401
Bell–McCaffrey p	0.013	0.044	0.041	0.066	0.048	0.097	0.475	0.378	0.349	0.311
Freedman–Lane p	0.008	0.023	0.029	0.042	0.040	0.082	0.471	0.365	0.370	0.317
Effective df	11.27	6.66	10.18	7.13	11.39	6.56	9.86	7.77	10.37	7.24
Wine revenue share (not the coefficient of interest):										
Bell–McCaffrey p	0.088	0.093	0.125	0.132	0.111	0.107	0.366	0.381	0.391	0.393
Freedman–Lane p	0.045	0.049	0.071	0.083	0.078	0.072	0.354	0.367	0.400	0.393
N (cantons)	23	23	23	23	23	23	23	23	23	23

Notes: Ordinary least squares on the 23 cantons with a usable tithe measure; Schaffhausen and Vaud are excluded throughout (see the text below Table A.21). Each column pair contrasts a specification without the tithe term, re-estimated on the same 23 cantons, with one that adds it; the paper's estimates without the tithe term, at $N = 25$, are in Table 4, whose column order the pairs follow. Stars mark $p < 0.10$, $p < 0.05$ and $p < 0.01$ on HC3 p-values. The diagnostic rows report the Protestant-share coefficient under HC3, Bell–McCaffrey and Freedman–Lane (permutation p-values at 10^6 draws), and the wine revenue share coefficient under Bell–McCaffrey and Freedman–Lane.

A.10.1 Petition-arm collinearity and wine readings

Table A.23: Design-matrix diagnostics and the wine revenue share's readings, petition arm

	Petition arm focal (4)	Petition arm density-in (5)	Tavern focal
Design matrix:			
BKW condition no. (with constant)	4.54	16.03	6.24
VIF (coefficient of interest)	1.11	1.50	1.03
VIF (wine revenue share)	1.82	2.03	
VIF, specification maximum	2.86	3.62	2.58
Regressor with the maximum	French share	Producer indicator	Producer indicator
Wine revenue share (not the coefficient of interest):			
HC3 robust p	0.816	0.798	
Bell–McCaffrey p	0.654	0.628	
Wild bootstrap-t p	0.575	0.590	
Freedman–Lane p	0.688	0.664	
Bell–McCaffrey effective df	2.50	2.59	

Notes: Diagnostics for three columns of Table 4, the petition-arm focal and density-in columns and the tavern column; ordinary least squares, $N = 25$ cantons in every column. The coefficient of interest is the Protestant share in the two petition columns and the tavern employment share in the tavern column. The variance inflation factor (VIF) is centered and reported for the coefficient of interest and for the wine revenue share; the specification maximum and the regressor that owns it are given for each column. Condition numbers are Belsley–Kuh–Welsch, constant included. The lower block reports the wine revenue share, which is not the coefficient of interest in any column of Table 4; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell–McCaffrey, wild bootstrap-t, Freedman–Lane permutation). The producer indicator is the Absinthe row of Table 4. The tavern column carries no wine regressor.